

A Braid Group Action on Category $\mathcal{O}$

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**Australian
National
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*To high school me, who thought mathematics was easy and boring. How very
wrong he was.*

Declaration

The work in this thesis is my own except where otherwise stated.

Yangda Bei

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Abstract

In this thesis, we investigate the relationship between the BGG category $\mathcal{O}$ and the 2-braid group defined by Rouquier. We show that there is a categorical braid group action induced by derived (co)shuffling functors on the bounded derived category of the principal block $D^b(\mathcal{O}_0)$ introduced by Mazorchuk–Stroppel. Via Soergel’s theorems, we transport the problem to the setting of Soergel bimodules, yielding a diagrammatic argument of Rouquier’s result.

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Notation and terminology

In the following $\mathfrak{g}$ is a complex semisimple Lie algebra and $\mathfrak{h} \subseteq \mathfrak{g}$ is a Cartan subalgebra.

Notation

Φ	root system
Δ	simple roots in Φ
s_α	reflection relative to $\alpha \in \Phi$
Φ^+	positive roots in Φ
W	Weyl group of Φ
H_α	hyperplane orthogonal to $\alpha \in \Phi$
$w_\circ$	longest element in W
Λ	integral weight lattice on $\mathfrak{h}^*$
ρ	half sum of roots in Φ^+
(W, S)	a Coxeter system
M_λ	λ -weight space of M
$U(\mathfrak{g})$	universal enveloping algebra of $\mathfrak{g}$
$\mathcal{O}, \mathcal{O}(\mathfrak{g})$	BGG category $\mathcal{O}$ associated to $\mathfrak{g}$
$[M : L]$	multiplicity of L in composition series of M
$\Delta(\lambda)$	Verma module associated to a weight $\lambda \in \mathfrak{h}^*$

$N(\lambda)$	unique maximal submodule of $\Delta(\lambda)$
$L(\lambda)$	unique simple quotient of $\Delta(\lambda)$
$(M : \Delta(\lambda))$	multiplicity of $\Delta(\lambda)$ in standard filtration of M
$K(\mathcal{O})$	Grothendieck group of $\mathcal{O}$
$M^\vee$	(BGG) dual of M in $\mathcal{O}$
$\nabla(\lambda)$	dual Verma module associated to a weight $\lambda \in \mathfrak{h}^*$
$Z(\mathfrak{g})$	centre of $U(\mathfrak{g})$
Sym	symmetric algebra
$\mathcal{O}_\lambda$	block in $\mathcal{O}$
T_λ^μ	translation functor from $\mathcal{O}_\lambda \rightarrow \mathcal{O}_\mu$
Θ_s	wall-crossing functor
$\text{Sh}_s, \text{CSh}_s$	shuffling and coshuffling functor
B_W	braid group associated to (W, S)
∂_s	Demazure operator
$\text{BS}(\underline{w})$	Bott–Samelson bimodule associated to expression $\underline{w}$
$[n]_{\textcolor{red}{s}}, [n]_{\textcolor{blue}{t}}$	$\textcolor{red}{s}$ - and $\textcolor{blue}{t}$ -coloured quantum numbers
JW	Jones–Wenzl projector
$F_{\underline{w}}$	Rouquier complex associated to braid word $\underline{w}$
$\mathbb{V}_\mu$	Soergel’s functor $\text{Hom}_{\mathfrak{g}}(P(w_\circ \cdot \mu), -)$
$A\text{-mod}, \text{mod-}A,$ $A\text{-mod-}B$	category of left A -modules, right A -modules, and (A, B) -bimodules
$A\text{-gmod}, \text{gmod-}A,$ $A\text{-gmod-}B$	category of graded left A -modules, graded right A -modules, and graded (A, B) -bimodules

Introduction

A *categorical group action* generalises the familiar notion of “a group acting on a set” to “a group acting on a category”. One can begin with a group of interest and look for categories on which it acts naturally, or begin with a category and investigate what group actions emerge from its internal structure. By passing from sets to categories, one often uncovers additional structure that upgrades familiar constructions into richer behaviour only visible at the categorical level. This is the principle of *categorification* [CF94]. In this thesis we adopt the second viewpoint from above: the category of interest is the *Bernstein–Gel’fand–Gel’fand (BGG) category* $\mathcal{O}$ of a complex semisimple Lie algebra $\mathfrak{g}$, and the group that arises is the *braid group* B_W associated to the Weyl group W of $\mathfrak{g}$.

Category $\mathcal{O}$ [BGG75, BGG76] is a natural enlargement of the category of finite-dimensional $\mathfrak{g}$ -modules containing certain well-behaved infinite-dimensional ones. A central problem was to understand the structure of Verma modules [Ver68] by computing their Jordan–Hölder multiplicities. This proved unexpectedly difficult and led to the Kazhdan–Lusztig conjecture [KL79], which asserted that these multiplicities are given by evaluating certain polynomials in the Hecke algebra at 1. The conjecture was proven two years later independently by Brylinski–Kashiwara [BK81] and Beilinson–Bernstein [BB81] using similar methods, which involved passing to a geometric categorification of the *Hecke algebra*.

In $\mathcal{O}$, we focus on the principal block $\mathcal{O}_0$ and its bounded derived category $D^b(\mathcal{O}_0)$. Mazorchuk–Stroppel [MS05] showed that certain derived functors induce an action of W on the Grothendieck group $K(\mathcal{O}_0)$. More strikingly, this upgrades to a categorical action of B_W on $D^b(\mathcal{O}_0)$. Whilst the images of functors related by braid relations were shown to be isomorphic, the isomorphisms were not specified. In this sense, their action is only a *weak* categorical action. Rouquier [Rou06] emphasised the role of morphisms between self-equivalences and made this weak action *strict*: to each braid word, he assigned a canonical object in the bounded homotopy category of certain bimodules, yielding explicit isomorphisms realising the relations. These complexes form what he called the *2-braid group*.

The bimodules appearing in the construction of the 2-braid group are the *Soergel bimodules* $\mathbb{S}\text{Bim}$. Soergel [Soe92, Soe90] constructed a functor from the full subcategory of projectives in $\mathcal{O}_0$ to a module category on which $\mathbb{S}\text{Bim}$ acts. Computations are typically clearer on the bimodule side, especially using a diagrammatic calculus to describe morphism spaces [EW16]. Rouquier’s proof used Soergel’s original formulation, but can be made more transparent with diagrammatic techniques [AMRW17].

Importantly, $\mathbb{S}\text{Bim}$ is an algebraic categorification of the Hecke algebra, and another incarnation of the *Hecke category*. Via Soergel’s bridge, the same category appears in the context of $\mathcal{O}$ as projective endofunctors. The diagrammatic presentation, which can be defined independently to Soergel bimodules, gives another model of the Hecke category. In this thesis, we explain the braid group action on $D^b(\mathcal{O}_0)$ through the relationship between these three incarnations.

Chapter 1 sets notation and provides a background of abstract root systems, complex semisimple Lie algebras, and their basic representation theory. Chapter 2 introduces category $\mathcal{O}$. In Chapter 3, we construct endofunctors of $\mathcal{O}$ that induce the action of B_W on $D^b(\mathcal{O}_0)$. Chapter 4 develops Soergel bimodules and their diagrammatic calculus. Chapter 5 introduces Rouquier complexes and shows that they give a weak categorification of the braid group diagrammatically. Finally, Chapter 6 links $\mathcal{O}$ with Soergel bimodules via Soergel’s theorems and the functor $\mathbb{V}$, proving the braid group action. For completeness, Appendix A records the statement of Gaussian elimination for complexes from [BN07].

We assume basic familiarity with representation theory, abelian and monoidal categories, and standard constructions in homological algebra (mapping cone, derived functors and categories). Knowledge of finite-dimensional $\mathfrak{sl}_2$ representation theory is also useful, but not required. The treatments in [EGH⁺11] and [Huy06, §2] provide sufficient background to read this thesis. The author gratefully acknowledges [Len] for clarifying aspects of [Rou06], as well as the lecture series [Maz] which greatly deepened his understanding of category $\mathcal{O}$. We give full proofs when pedagogically valuable and otherwise omit them with references.

Statement of originality

The exposition of these results are original. Examples 2.17, 2.21, 2.33, 2.45, 3.9, 3.10, 3.18, 3.24, 4.17, 5.2 and 6.7 were independently worked out. The proofs of Lemmas 3.6 and 5.8, Propositions 2.18, 2.19 and 3.5, and Theorems 2.39 and 6.8 were expanded upon from their source in detail. Lemmas 2.8, 3.20 and 4.5, Propositions 2.23 and 6.11, and Theorem 2.37 were results proven independently.

Chapter 1

Preliminaries

The aim of this chapter is to review the basics of complex semisimple Lie algebras and their representations and make clear the notation and conventions that we will be using. We begin with an axiomatic treatment of root systems and Weyl groups, which abstracts the structure theory of Lie algebras. This viewpoint will also motivate the definition of a realisation associated to a Coxeter system in Chapter 4. For a full treatment of Sections 1.1, 1.2 and 1.4, see [Hum80], and [Hum92] for Section 1.3. See also [Hum08, §0].

1.1 Root systems

We work with a finite-dimensional real vector space E . Let E^* be its dual and write $\langle \lambda, f \rangle := f(\lambda)$ for the canonical pairing $E \times E^* \rightarrow \mathbb{R}$.

Definition 1.1. An *abstract root system* is a finite set $\Phi \subset E \setminus \{0\}$ of *roots*, a finite set $\Phi^\vee \subset E^* \setminus \{0\}$ of *coroots*, and a bijection $\Phi \xrightarrow{\sim} \Phi^\vee : \alpha \mapsto \alpha^\vee$ satisfying the following conditions:

- (1) $\text{span}_{\mathbb{R}}(\Phi) = E$ and $\text{span}_{\mathbb{R}}(\Phi^\vee) = E^*$;
- (2) for each $\alpha \in \Phi$, $\mathbb{R}\alpha \cap \Phi = \{\pm\alpha\}$;
- (3) for each $\alpha \in \Phi$, define $s_\alpha(\beta) := \beta - \langle \beta, \alpha^\vee \rangle \alpha$ for $\beta \in E$ and $s_{\alpha^\vee}(f) := f - \langle \alpha, f \rangle \alpha^\vee$ for $f \in E^*$. Then $s_\alpha(\Phi) = \Phi$ and $s_{\alpha^\vee}(\Phi^\vee) = \Phi^\vee$.
- (4) $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$ for all $\alpha, \beta \in \Phi$, and $\langle \alpha, \alpha^\vee \rangle = 2$ for all $\alpha \in \Phi$.

For any root system Φ , there exists a set of *simple roots* $\Delta \subset \Phi$ such that $\Phi = \Phi^+ \cup (-\Phi^+)$, where $\Phi^+ = \mathbb{Z}_{\geq 0}\Delta \cap \Phi$ are the *positive roots*.

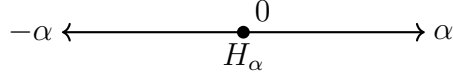


Figure 1.1: Roots and associated hyperplane for the A_1 root system. The chamber containing α lies above H_α and is its own upper closure. The chamber containing $-\alpha$ lies below H_α and its upper closure contains 0.

We call s_α and $s_{\alpha^\vee}$ from (3) *reflections*: each s_α (similarly $s_{\alpha^\vee}$) is a reflection through the hyperplane orthogonal to α . The *Weyl group* W of Φ is the subgroup of automorphisms of E generated by the reflections s_α . Since each s_α preserves Φ , one can prove that W is a finite group. Equip E with an inner product $(-, -)$, and identify $E \simeq E^*$. Then the coroot $\alpha^\vee \in E^*$ corresponds to the vector $2\alpha/(\alpha, \alpha)$ if and only if $(-, -)$ is W -invariant. Such a W -invariant inner product is unique up to scalar (if the root system is irreducible). Under this identification, we recover the familiar Euclidean definition of reflection $s_\alpha(v) = v - 2\frac{(v, \alpha)}{(\alpha, \alpha)}\alpha$.

A *chamber* is a connected component of $E \setminus \bigcup_{\alpha \in \Phi} H_\alpha$, where

$$H_\alpha := \{\lambda \in E \mid \langle \lambda, \alpha^\vee \rangle = 0\}$$

are the hyperplanes orthogonal to a root $\alpha \in \Phi$. We refer to H_α as the s_α -*wall*, or simply the s -*wall* if α is clear from context. A chamber C can be uniquely identified by a partition $\Phi_C^+ \sqcup \Phi_C^- = \Phi^+$ of positive roots such that

$$C := \{\lambda \in E \mid \langle \lambda, \alpha^\vee \rangle > 0 \text{ if } \alpha \in \Phi_C^+ \text{ and } \langle \lambda, \alpha^\vee \rangle < 0 \text{ if } \alpha \in \Phi_C^-\},$$

and we say that C lies *above* (resp. *below*) H_α if $\alpha \in \Phi_C^+$ (resp. $\alpha \in \Phi_C^-$). The *closure* $\overline{C}$ of a chamber C is defined by replacing the strict inequalities above with weak ones. The *upper closure* $\widehat{C}$ of a chamber C is defined as

$$\widehat{C} := C \cup \{\lambda \in E \mid \langle \lambda, \alpha^\vee \rangle \leq 0 \text{ if } \alpha \in \Phi_C^-\}.$$

The *dominant chamber* is the chamber $C_\circ := \{\lambda \in E \mid \langle \lambda, \alpha^\vee \rangle \geq 0 \text{ for all } \alpha \in \Delta\}$ and has the property that $\widehat{C}_\circ = C_\circ$. The *longest element* $w_\circ \in W$ is the unique element satisfying $w_\circ C_\circ = -C_\circ$; we call $-C_\circ$ the *antidominant chamber*.

The root systems for A_1 and A_2 are shown in Figure 1.1 and Figure 1.2, respectively. The Weyl groups associated to A_1 and A_2 are isomorphic to $S_2 = \langle s \mid s^2 = e \rangle$ and $S_3 = \langle s, t \mid s^2 = t^2 = (st)^3 = e \rangle$ respectively. These examples will serve as running models, and the reader is encouraged to keep them in mind throughout the thesis.

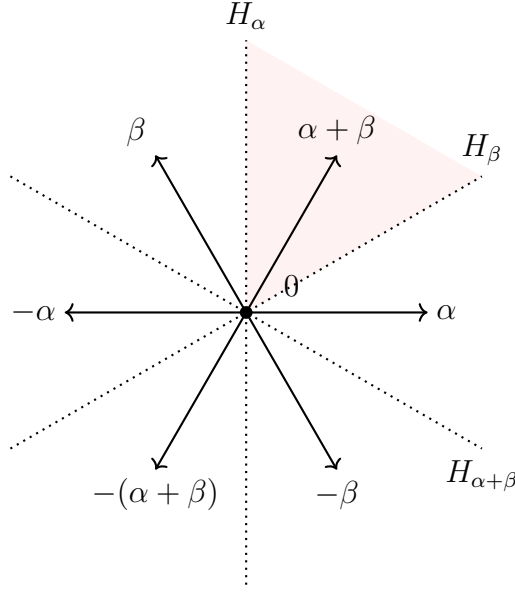


Figure 1.2: Roots and associated hyperplanes for the A_2 root system. The shaded region is the dominant chamber and lies above all hyperplanes.

1.2 Semisimple Lie algebras

Let $\mathfrak{g}$ be a complex semisimple Lie algebra and $\mathfrak{h} \subseteq \mathfrak{g}$ a Cartan subalgebra. We call elements of $\mathfrak{h}^*$ *weights*. The Lie algebra $\mathfrak{g}$ acts on itself via the adjoint action $\text{ad}: \mathfrak{g} \rightarrow \text{End } \mathfrak{g}$, where $\text{ad}_x(y) := [x, y]$. To each $\alpha \in \Phi$, associate the *root space*

$$\mathfrak{g}_\alpha := \{x \in \mathfrak{g} \mid [h, x] = \alpha(h)x \text{ for all } h \in \mathfrak{h}\}.$$

Then there is an abstract root system $\Phi \subset \mathfrak{h}^*$ associated to the real form of $\mathfrak{h}$. Picking a set of simple and positive roots $\Delta \subset \Phi^+ \subset \Phi$ defines a decomposition

$$\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+, \quad \mathfrak{n}^\pm := \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_{\pm\alpha},$$

called a *triangular decomposition*. We call $\mathfrak{b} := \mathfrak{h} \oplus \mathfrak{n}^+$ a *Borel subalgebra* of $\mathfrak{g}$.

An important example of a Lie algebra is $\mathfrak{sl}_2$, the algebra of all traceless 2×2 complex matrices, equipped with the commutator bracket $[x, y] = xy - yx$. It has a basis (e, h, f) given by the elements

$$e := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad h := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \text{and} \quad f := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

It is easy to check that these elements satisfy $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. There is a standard basis of $\mathfrak{g}$ containing $e_\alpha \in \mathfrak{g}_\alpha$, $f_\alpha \in \mathfrak{g}_{-\alpha}$ for $\alpha \in \Phi^+$, and

$h_\alpha := [e_\alpha, f_\alpha]$ for $\alpha \in \Delta$ such that $\alpha(h_\alpha) = 2$. We call the $\mathfrak{sl}_2$ -subalgebra of $\mathfrak{g}$ with basis $(e_\alpha, h_\alpha, f_\alpha)$ an $\mathfrak{sl}_2$ -triple. Another example we will touch upon is $\mathfrak{sl}_3$, the algebra of all traceless 3×3 complex matrices. The associated root systems of $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ are A_1 and A_2 respectively.

A weight $\lambda \in \mathfrak{h}^*$ is *integral* if $\langle \lambda, \alpha^\vee \rangle \in \mathbb{Z}$ for all $\alpha \in \Phi$. We say λ is *dominant integral* if $\langle \lambda, \alpha^\vee \rangle \in \mathbb{Z}_{\geq 0}$, and *antidominant integral* if $\langle \lambda, \alpha^\vee \rangle \in \mathbb{Z}_{\leq 0}$ for all $\alpha \in \Phi$. On $\mathfrak{h}^*$, define the *integral weight lattice* $\Lambda := \{\lambda \in \mathfrak{h}^* \mid \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \text{ for all } \alpha \in \Phi\}$, in which the *root lattice* $\mathbb{Z}\Phi \subset \mathfrak{h}^*$ is a sublattice. Define a partial order on $\mathfrak{h}^*$ in the following way: for $\lambda, \mu \in \mathfrak{h}^*$, we say $\mu \leq \lambda$ if $\lambda - \mu \in \mathbb{Z}_{\geq 0}\Phi$.

Let W be the Weyl group of Φ . We record a lemma to be used in Chapter 3.

Lemma 1.2 ([Hum80, Lemma 0.6]). *Let λ be an integral weight and fix a chamber C for the action of W corresponding to a set of simple roots Δ . Then the following are equivalent: (a) $\lambda \in \overline{C}$, (b) $\lambda \geq s_\alpha \lambda$ for all $\alpha \in \Delta$, (c) $\lambda \geq w\lambda$ for all $w \in W$.*

There is another action of W on $\mathfrak{h}^*$ called the *dot-action*: for $w \in W$ and $\lambda \in \mathfrak{h}^*$, define $w \cdot \lambda := w(\lambda + \rho) - \rho$, where $\rho := \frac{1}{2} \sum_{\alpha \in \Phi^+} \alpha$. We call λ *dot-dominant integral* if for all $\alpha \in \Phi$, $\langle \lambda + \rho, \alpha^\vee \rangle \in \mathbb{Z}_{\geq 0}$, and *dot-antidominant integral* if $\langle \lambda + \rho, \alpha^\vee \rangle \in \mathbb{Z}_{\leq 0}$. With respect to the dot-action, the reflecting hyperplanes are those for the linear W -action but shifted by $-\rho$. From now on, “dominant” will mean dot-dominant. When needed, we distinguish the actions by writing $(W, \cdot)$ -action for the dot action and W -action for the linear action. A weight λ is *regular* if $\langle \lambda + \rho, \alpha^\vee \rangle \neq 0$ for all $\alpha \in \Phi$, equivalently if its $(W, \cdot)$ -stabiliser is trivial.

1.3 Coxeter systems

Weyl groups are reflection groups arising from root systems. Coxeter systems generalise this notion to settings with no underlying Lie algebra.

Definition 1.3. A *Coxeter system* (W, S) is a group W and a finite set $S \subset W$ of generators for W with the presentation

$$W = \langle s \in S \mid (st)^{m_{st}} = \text{id for all } s, t \in S \rangle,$$

where $m_{ss} = 1$ for each $s \in S$, and $m_{st} \in \{2, 3, \dots\} \cup \{\infty\}$ for $s \neq t \in S$. When $m_{st} = \infty$, there is no relation between s and t .

In this thesis we will primarily work with the Coxeter system (W, S) coming from the Weyl group of our fixed Lie algebra, but we will define Soergel bimodules

for arbitrary Coxeter systems (see Chapter 4). For Weyl groups, the order m_{st} of st when $s \neq t$ satisfies $m_{st} \in \{2, 3, 4, 6\}$ when $s \neq t \in S$, that is Weyl groups are exactly the finite crystallographic Coxeter groups.

The elements of S are called *simple reflections*. Every $w \in W$ can be written as a product $w = s_1 \dots s_k$ with $s_i \in S$. We call $\underline{w} = (s_1, \dots, s_k)$ an *expression* for w . If k is minimal among all such expressions, $\underline{w}$ is called *reduced*. The *length* of w is denoted by $\ell(w) := k$.

The *geometric representation* of a Coxeter system (W, S) is a real vector space V with basis $\{\alpha_s \mid s \in S\}$ equipped with the symmetric bilinear form $(-, -)$ given by

$$(\alpha_s, \alpha_t) := -\cos\left(\frac{\pi}{m_{st}}\right).$$

When $m_{st} = \infty$, $(\alpha_s, \alpha_t) = -1$. The basis elements α_s are called *simple roots*. We can define a linear action of W on V where simple reflections act by

$$s(\alpha_t) = \alpha_t - 2(\alpha_s, \alpha_t)\alpha_s.$$

If W is a Weyl group of a Lie algebra, then the geometric representation is the real form $\mathfrak{h}^*$ equipped with the linear action of W . Note that the simple roots in this sense are not the same vectors as the Lie-theoretic simple roots: they determine the same angles, but here all simple roots are equal length, whereas in Lie theory (except in type A) simple roots can have unequal lengths.

1.4 Lie algebra representations

Our main concern in this thesis will be the study of Lie algebra representations. We will mostly use the language of modules rather than representations: a module is *simple* if it has no proper submodules; it is *indecomposable* if it is not isomorphic to a direct sum of two non-zero submodules.

We now briefly discuss an important class of $\mathfrak{g}$ -modules. For any $\lambda \in \mathfrak{h}^*$, the λ -*weight space* of a $\mathfrak{g}$ -module M is

$$M_\lambda := \{v \in M \mid h \cdot v = \lambda(h)v \text{ for all } h \in \mathfrak{h}\}.$$

These are the simultaneous $\mathfrak{h}$ -eigenspaces; in the adjoint representation, they recover the root spaces. The module M is called a *weight module* if $M = \bigoplus_{\lambda \in \mathfrak{h}^*} M_\lambda$. Non-zero vectors $v \in M_\lambda$ are called *weight vectors*. The (*contragredient*) *dual action* on $M^* := \text{Hom}_{\mathbb{C}}(M, \mathbb{C})$ is defined by $(x \cdot f)(v) = -f(v \cdot x)$ for $x \in$

$\mathfrak{g}, f \in M^*$. If M is a weight module, then so is M^* , with weight decomposition $M^* = \bigoplus_{\mu \in \mathfrak{h}^*} (M^*)_{\mu}$ where $(M^*)_{\mu} \simeq (M_{-\mu})^*$. If M is a simple finite-dimensional module of highest weight λ with respect to the partial order, then M^* has lowest weight $-\lambda$, hence highest weight $-w_o\lambda$. We will use this fact in the proof of Lemma 3.6.

We now turn $\mathfrak{g}$ into an associative algebra.

Definition 1.4. The *universal enveloping algebra* of $\mathfrak{g}$ is the associative algebra $U(\mathfrak{g}) := T(\mathfrak{g})/(x \otimes y - y \otimes x - [x, y])$, where $T(\mathfrak{g})$ is the tensor algebra of $\mathfrak{g}$. It satisfies the following universal property: if $\varphi: \mathfrak{g} \rightarrow A$ is a Lie algebra map to a unital associative algebra A , then there exists a unique unital algebra homomorphism $\psi: U(\mathfrak{g}) \rightarrow A$ such that $\varphi = \psi \circ i$, where $i: \mathfrak{g} \rightarrow U(\mathfrak{g})$ is the image of $\mathfrak{g} \simeq T^1\mathfrak{g} \hookrightarrow T\mathfrak{g}$.

Any $\mathfrak{g}$ -module may be regarded as a module over the universal enveloping algebra $U(\mathfrak{g})$, yielding an equivalence of categories. We will sometimes write $\text{Hom}_{\mathfrak{g}}(-, -)$ to mean $\text{Hom}_{U(\mathfrak{g})}(-, -)$.

The following theorem allows us to write an explicit basis for $U(\mathfrak{g})$.

Theorem 1.5 (Poincaré–Birkhoff–Witt (PBW) Theorem). *Given an ordered basis $(x_1, \dots, x_n)$ of $\mathfrak{g}$, the monomials $x_1^{t_1} \dots x_n^{t_n}$ for $t_i \in \mathbb{Z}_{\geq 0}$ form a basis for $U(\mathfrak{g})$. In the case that $\mathfrak{g}$ is semi-simple, we may write such monomials as*

$$f_1^{r_1} \dots f_m^{r_m} h_1^{s_1} \dots h_\ell^{s_\ell} e_1^{t_1} \dots e_m^{t_m},$$

where $f_i \in \mathfrak{g}_{-\alpha_i}$, $e_i \in \mathfrak{g}_{\alpha_i}$, and $h_i = [e_i, f_i]$ for simple roots α_i .

Lastly, we review induction and restriction. Let $\iota: \mathfrak{a} \rightarrow \mathfrak{g}$ be a Lie subalgebra inclusion, which induces an inclusion of their corresponding universal enveloping algebras. There are functors

$$\begin{array}{ll} \text{Res}_{\mathfrak{a}}^{\mathfrak{g}}: U(\mathfrak{g})\text{-mod} \rightarrow U(\mathfrak{a})\text{-mod} & \text{Ind}_{\mathfrak{a}}^{\mathfrak{g}}: U(\mathfrak{a})\text{-mod} \rightarrow U(\mathfrak{g})\text{-mod} \\ M \mapsto M & M \mapsto U(\mathfrak{g}) \otimes_{U(\mathfrak{a})} M \\ x \cdot m := \iota(x) \cdot m & x \cdot (y \otimes m) := (xy) \otimes m \end{array}$$

called *restriction* and *induction* respectively. Induction is left-adjoint to restriction, that is

$$\begin{aligned} \text{Hom}_{\mathfrak{g}}(\text{Ind}_{\mathfrak{a}}^{\mathfrak{g}} M, N) &\simeq \text{Hom}_{\mathfrak{a}}(M, \text{Res}_{\mathfrak{a}}^{\mathfrak{g}} N) \\ f &\mapsto (m \mapsto f(1 \otimes m)) \\ (x \otimes m \mapsto x \cdot g(m)) &\leftarrow g \end{aligned}$$

and we refer to this property as *Frobenius reciprocity*.

Chapter 2

“Basics” of Category $\mathcal{O}$

Introduced by Beilinson–Gel’fand–Gel’fand [BGG75, BGG76], the category $\mathcal{O}$ arose from the study of certain infinite-dimensional $U(\mathfrak{g})$ -modules initiated by Verma [Ver68] and, more generally, Harish-Chandra modules in the context of representation theory of complex Lie groups (see for example [BG80]). Category $\mathcal{O}$ enlarges the setting of finite-dimensional $\mathfrak{g}$ -modules with its axioms modelled on key properties of Verma’s modules, chosen to be minimal yet sufficient to admit a well-behaved family of objects. The structure of $\mathcal{O}$ allows for the study of certain well-behaved endofunctors (to be defined in Chapter 3) which play a key role in the braid group action.

We will work over the base field $\mathbb{C}$ throughout this chapter. Fix a Cartan subalgebra $\mathfrak{h}$ inside a chosen Borel $\mathfrak{b}$ and let $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{h} \oplus \mathfrak{n}^+$ be the corresponding triangular decomposition. Replacing finite-dimensionality by the weaker requirement of finite generation leads us to our first axiom:

$$M \text{ is a finitely generated } U(\mathfrak{g})\text{-module.} \quad (\mathcal{O}1)$$

This mild finiteness condition ensures our category remains large enough to contain every finite-dimensional module but still imposing noetherian-like properties to be discussed in the next section. A key property of finite-dimensional $\mathfrak{g}$ -modules is that M splits into simultaneous $\mathfrak{h}$ -eigenspaces, that is

$$M \text{ has a decomposition } M = \bigoplus_{\lambda \in \mathfrak{h}^*} M_\lambda, \quad (\mathcal{O}2)$$

where M_λ is the λ -weight space of M . Finally, we ask that

$$M \text{ is locally } \mathfrak{n}^+\text{-finite: } \dim U(\mathfrak{n}^+) \cdot v < \infty \text{ for each } v \in M. \quad (\mathcal{O}3)$$

While $(\mathcal{O}1)$ and $(\mathcal{O}2)$ model the finite-dimensional situation, $(\mathcal{O}3)$ allows us to restrict to a certain family of infinite-dimensional $\mathfrak{g}$ -modules, those that are “half infinite” or “finite towards the top” as we shall see.

We arrive at the following definition.

Definition 2.1. The *BGG category $\mathcal{O}$* , or just *category $\mathcal{O}$* , is defined as the full subcategory of $U(\mathfrak{g})\text{-mod}$ containing modules M satisfying $(\mathcal{O}1)$, $(\mathcal{O}2)$, and $(\mathcal{O}3)$.

We will write $\mathcal{O}(\mathfrak{g})$ for the category $\mathcal{O}$ whenever we wish to emphasise its associated Lie algebra $\mathfrak{g}$.

This chapter draws heavily from [Hum08]. The § section numbers cited below refer to that source. Our aim is to develop the two ingredients needed to make sense of the braid group action on $\mathcal{O}$, namely how $\mathcal{O}$ decomposes into blocks and how projective objects behave. For sake of pedagogy, we collect a number of general structural properties of $\mathcal{O}$ in Section 2.1. Next we introduce Verma modules in Section 2.2 whose study will provide insight into the structure of $\mathcal{O}$ (§1.2–1.7 and selected results of §4). The block decomposition of $\mathcal{O}$ will be explained in Section 2.5 (§1.7–1.13) and we discuss projective objects in Section 2.6 (§3). Along the way, we analyse $\mathcal{O}(\mathfrak{sl}_2)$ by classifying its indecomposable objects using the techniques listed above.

2.1 First properties of $\mathcal{O}$

We begin with a few preliminary observations about the kinds of objects that lie in $\mathcal{O}$. Let V be a finite-dimensional $U(\mathfrak{g})$ -module. Clearly V is finitely generated, so $(\mathcal{O}1)$ holds, and finiteness implies $U(\mathfrak{n}^+) \cdot v$ is finite-dimensional for each v , so $(\mathcal{O}3)$ holds. From the structure theory of finite-dimensional $\mathfrak{g}$ -modules the Cartan subalgebra $\mathfrak{h}$ acts semisimply on V , hence V decomposes as a finite direct sum of weight spaces and $(\mathcal{O}2)$ holds as well. We thus get the following corollary.

Corollary 2.2. *All finite-dimensional $U(\mathfrak{g})$ -modules are contained in $\mathcal{O}$.*

More generally, the structure of $\mathcal{O}$ enjoys good closure properties. Since $U(\mathfrak{g})$ is noetherian (see [MR87, §7.4]), every $M \in \mathcal{O}$ is a noetherian $U(\mathfrak{g})$ -module. In particular, $(\mathcal{O}1)$ is preserved under taking submodules, and $(\mathcal{O}2)$ and $(\mathcal{O}3)$ are trivially inherited, so $\mathcal{O}$ is closed under submodules. It is also straightforward to verify that $\mathcal{O}$ is closed under quotients and finite direct sums. Together with the fact that $U(\mathfrak{g})\text{-mod}$ is an abelian category, we get the following important structural result:

Theorem 2.3 ([Hum08, Theorem 1.1(c)]). *The category $\mathcal{O}$ is an abelian category.*

The braid action cannot be detected inside the abelian category $\mathcal{O}$. It becomes visible only after we pass to the homotopy category of projective objects, which is equivalent to the derived category of $\mathcal{O}$ (see Theorem 2.47). There, the relevant functors appear as complexes and satisfy the required braid relations.

In addition to being noetherian, $\mathcal{O}$ is *artinian*: every object satisfies the descending chain condition [Hum08, Theorem 1.11]. Hence each $M \in \mathcal{O}$ has *finite length*, that is M admits a finite filtration

$$0 = M_0 \subset M_1 \subset \cdots \subset M_n = M$$

with each quotient $L_i := M_i/M_{i-1}$ simple. We call this filtration a *composition series* for M . The multiplicity of a simple module L among the composition factors of M is denoted $[M : L]$. By the Jordan–Hölder theorem (see [EGH⁺11, Theorem 3.7.1]), this multiplicity is independent of the choice of composition series. To record composition factors compactly, we sometimes write $M_{i-1} \overset{L_i}{\subset} M_i$.

Remark 2.4. The finite length property is a crucial structural property of the category, but computing composition multiplicities is hard in practice. The search for systematic methods to find these multiplicities in the 1970s culminated in the Kazhdan–Lusztig conjecture [KL79], which provides a combinatorial formula for finding multiplicities of Verma modules (to be defined in the next section). See [Hum08, §8.1] for a summary of progress made towards computing multiplicities in special cases and [Kle07] for a historical account of the development of the geometric techniques used in the proof of the conjecture.

The last property of $\mathcal{O}$ that we state in this section applies to any abelian category satisfying the finite length condition.

Proposition 2.5 ([Kru25, Sch13, Rem23, Krull–Remak–Schmidt decomposition]). *Each $M \in \mathcal{O}$ decomposes uniquely as a direct sum of indecomposable modules, with summands unique up to isomorphism.*

We return to this statement in more detail in Section 2.5, where we explain how modules in $\mathcal{O}$ decompose in practice.

2.2 Highest weight and Verma modules

While the properties of $\mathcal{O}$ discussed so far have been largely formal, we now turn to a concrete and important family of modules in $\mathcal{O}$, the so-called *Verma modules*.

The idea behind their definition is to take an easily constructed family of modules for a subalgebra of $U(\mathfrak{g})$ and then induce it to a $U(\mathfrak{g})$ -module. Here, we shall see our first examples of infinite-dimensional modules in $\mathcal{O}$.

Definition 2.6. The *Verma module* corresponding to a weight $\lambda \in \mathfrak{h}^*$ is defined as

$$\Delta(\lambda) := \text{Ind}_{\mathfrak{b}}^{\mathfrak{g}} \mathbb{C}_{\lambda} = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_{\lambda},$$

where $\mathbb{C}_{\lambda}$ is the 1-dimensional $U(\mathfrak{b})$ -module such that $\mathfrak{h}$ acts by λ and $\mathfrak{n}^+$ acts trivially.

Verma modules are examples of *highest weight modules*. These are $U(\mathfrak{g})$ -modules M generated by a *highest weight vector* $v^+ \in M_{\lambda}$ in some weight space M_{λ} , such that $\mathfrak{n}^+ \cdot v^+ = 0$. If such a v^+ exists, then M is a *highest weight module of weight λ* . As their name suggests, all highest weight modules are weight modules, that is $\dim M_{\mu} < \infty$ for all $\mu \in \mathfrak{h}^*$, with $\dim M_{\lambda} = 1$. Conversely, if $M \in \mathcal{O}$ is non-zero then the weight decomposition together with local $\mathfrak{n}^+$ -finiteness implies the existence of a maximal weight λ and therefore of a highest weight vector in M_{λ} .

The following proposition gives a concrete description of a highest weight module generated by v^+ and shows that every weight of such a module is bounded above by the highest weight λ .

Proposition 2.7 ([Hum08, Theorem 1.2(a)-(b)]). *Let M be a highest weight module of weight $\lambda \in \mathfrak{h}^*$, generated by a highest weight vector v^+ . Fix an ordering of positive roots $\alpha_1, \dots, \alpha_m$ and choose corresponding vectors $f_i \in \mathfrak{g}_{-\alpha_i}$. Then M is spanned by the vectors $f_1^{i_1} \dots f_m^{i_m} \cdot v^+$ with $i_j \in \mathbb{Z}_{\geq 0}$, with respective weights $\lambda - \sum i_j \alpha_j$. Thus all weights μ of M satisfy $\mu \leq \lambda$.*

As such, all highest weight modules are in $\mathcal{O}$. Let us show that Verma modules are indeed in $\mathcal{O}$.

Lemma 2.8. *Verma modules are highest weight modules.*

Proof. By the PBW theorem (Theorem 1.5), we have a vector space isomorphism $U(\mathfrak{g}) \simeq U(\mathfrak{n}^-) \otimes U(\mathfrak{b})$, which implies that $\Delta(\lambda) \simeq U(\mathfrak{n}^-) \otimes \mathbb{C}_{\lambda}$ as left $U(\mathfrak{n}^-)$ -modules. This shows that $\Delta(\lambda)$ is a free $U(\mathfrak{n}^-)$ -module of rank one. Consider the vector $1 \otimes 1 \in \Delta(\lambda)$. Under the action of $\mathfrak{h}$, we get that $h \cdot (1 \otimes 1) = h \otimes 1 = 1 \otimes \lambda(h)1 = \lambda(h)(1 \otimes 1)$ for all $h \in \mathfrak{h}$, so $1 \otimes 1$ has weight λ . Under the action of $\mathfrak{n}^+$, we also have that $x \cdot (1 \otimes 1) = x \otimes 1 = 1 \otimes 0 \cdot 1 = 0$ for any $x \in \mathfrak{n}^+$. This implies that $1 \otimes 1$ is a highest weight vector generating $\Delta(\lambda)$. $\square$

If M is any highest weight module of weight λ generated by a highest weight vector $v^+ \in M_\lambda$, the assignment $1 \otimes 1 \mapsto v^+$ extends uniquely to a surjective homomorphism $\Delta(\lambda) \twoheadrightarrow M$. Thus $\Delta(\lambda)$ serves as a *universal* highest-weight module of weight λ . This property of highest weight modules can be generalised to all modules of $\mathcal{O}$ by the following proposition.

Proposition 2.9 ([Gai, Proposition 1.18]). *Every object in $\mathcal{O}$ is a quotient of a finite successive extension of Verma modules. In particular, every highest weight module of weight λ is a quotient of $\Delta(\lambda)$.*

Because every highest weight module of weight λ is a quotient of the Verma module $\Delta(\lambda)$, uniqueness statements about maximal submodules and simple quotients for $\Delta(\lambda)$ descend to all highest weight quotients.

Proposition 2.10 ([Hum08, Theorem 1.2(f)]). *If $M \in \mathcal{O}$ is a highest weight module of weight λ , then it has a unique maximal submodule, and thus a unique simple quotient. In particular, M is indecomposable.*

Proof. Let $v^+ \in M_\lambda$ be a highest-weight vector that generates M . Any proper submodule cannot contain v^+ , since v^+ generates M . Hence every proper submodule does not contain the weight space M_λ . In particular M cannot be written as a non-trivial direct sum, so it is indecomposable. Let N be the sum of all proper submodules, which is again proper and is the unique maximal submodule of M . It follows that M has a unique simple quotient, namely M/N . $\square$

For a Verma module $\Delta(\lambda)$, we will denote by $N(\lambda)$ its unique maximal submodule and $L(\lambda)$ the unique simple quotient. The previous proposition leads to a classification of simple modules in terms of Verma module quotients.

Theorem 2.11 ([Hum08, Theorem 1.3]). *Every simple object L is isomorphic to $L(\lambda)$ for some λ . Furthermore, $L(\lambda) \not\simeq L(\mu)$ if $\lambda \neq \mu$.*

Proof. Suppose $L \in \mathcal{O}$ is simple. By Proposition 2.9, there exists a homomorphism $\phi: \Delta(\lambda) \rightarrow L$ from a Verma module. The image of ϕ is a non-zero submodule of L , hence all of L . Thus $L \simeq \Delta(\lambda)/\ker \phi$ and $\ker \phi \simeq N(\lambda)$. Since every vector in $L(\lambda)$ has weight less than or equal to λ , then if $L(\lambda) \simeq L(\mu)$, $\lambda = \mu$ by symmetry. $\square$

As is customary in the study of $\mathcal{O}$, we will illustrate many concepts with $\mathfrak{g} = \mathfrak{sl}_2$, which serves as a simplified but instructive prototype for the general theory. Let us analyse the Verma modules in $\mathcal{O}(\mathfrak{sl}_2)$.

Example 2.12. Let v^+ be a highest weight vector generating $\Delta(\lambda)$. Since $\mathfrak{n}^-$ contains only f , $U(\mathfrak{n}^-) \simeq \mathbb{C}[f]$ is a polynomial ring in one variable. We take $\{\frac{1}{i!}f^i\}_{i \in \mathbb{Z}_{\geq 0}}$ as our preferred basis. Thus $\Delta(\lambda)$ has a $\mathbb{C}$ -basis $\{v_i\}_{i \in \mathbb{Z}_{\geq 0}}$ where $v_i := \frac{1}{i!}f^i \cdot v^+$ as it is a free $U(\mathfrak{n}^-)$ -module. Now $\mathfrak{g}$ acts on $\Delta(\lambda)$ as follows:

$$h \cdot v_i = (\lambda - 2i)v_i, \quad e \cdot v_i = (\lambda - i + 1)v_{i-1}, \quad f \cdot v_i = (i + 1)v_{i+1}.$$

These relations come from using the Lie bracket relations and iterating through the repeated powers of f in the definition of v_i . The factor of $\frac{1}{i!}$ in our basis tidies the action of e and f . From this description, we see that $\{v_i\}_{i \in \mathbb{Z}_{\geq 0}}$ is a weight basis for $\Delta(\lambda)$ with weights $\lambda, \lambda - 2, \lambda - 4, \dots$, each occurring with multiplicity one. The action of e raises the weight while the action of f lowers the weight. We can depict this information in the following diagram:

$$\begin{array}{ccccccc} & & \lambda-2i & & \lambda-4 & \lambda-2 & \lambda \\ & \nearrow^{i+1} & \text{loop} & \searrow^i & \nearrow^3 & \text{loop} & \searrow^2 & \text{loop} & \searrow^1 & \text{loop} \\ \cdots & \bullet & & \bullet & \cdots & \bullet & & \bullet & & \bullet & & \bullet \\ & \nwarrow_{\lambda-i} & & \nwarrow_{\lambda-i+1} & & \nwarrow_{\lambda-2} & & \nwarrow_{\lambda-1} & & \nwarrow_{\lambda} \end{array} \quad (2.1)$$

The dots represent the one-dimensional weight spaces, all of the form $\mathbb{C}v_i$. The loops represent the action of $\mathfrak{h}$ which is scalar multiplication by $\lambda - 2i$. The arrows going to the left represent the action of f and its scalar factor, which lowers the weight space. Similarly, the arrows going to the right represent the action of e . This pictorial description shows that Verma modules may be infinite-dimensional, and justifies why $(\mathcal{O}3)$ may be thought of as a “finite towards the top” condition.

If an arrow pointing to the right has coefficient zero, that is when $i = -\lambda$, then all the weight spaces to the left of that arrow form a proper submodule of $\Delta(\lambda)$. It is not hard to see that this happens precisely when λ is integral and $\lambda > -1$, that is when λ is dominant integral. Hence, the maximal submodule $N(\lambda)$ is isomorphic to $\Delta(-\lambda-2)$, and the irreducible quotient $L(\lambda)$ has dimension $\lambda + 1$. If λ is not dominant integral, then $\Delta(\lambda)$ has no proper submodules, hence is irreducible with $L(\lambda) \simeq \Delta(\lambda)$ being infinite-dimensional.

What we observed in the $\mathfrak{sl}_2$ picture is a special case of two general theorems. The first identifies which simples are finite-dimensional, while the next characterises when a Verma module is irreducible and explains how smaller Verma modules embed into larger ones.

Theorem 2.13 ([Hum08, Theorem 1.6]). *The simple module $L(\lambda)$ is finite-dimensional if and only if λ is dominant integral.*

Theorem 2.14 ([Hum08, Theorem 4.8, Theorem 4.6]). *For any $\lambda \in \mathfrak{h}^*$, there exists an embedding $\Delta(s_\alpha \cdot \lambda) \hookrightarrow \Delta(\lambda)$ if $s_\alpha \cdot \lambda \leq \lambda$ for positive roots α . In particular, the Verma module $\Delta(\lambda)$ is simple if and only if λ is antidominant.*

In the case of $\mathfrak{g} = \mathfrak{sl}_2$, $W \simeq S_2$ and $\rho = 1$. Since $s \cdot \lambda = s(\lambda + 1) - 1 = -\lambda - 2$, we have the embedding $\Delta(-\lambda - 2) \hookrightarrow \Delta(\lambda)$ when λ is a non-negative integer predicted by the theorem, which was confirmed by our analysis in Example 2.12.

Because Verma modules are as important to the structure of $\mathcal{O}$ as simple modules, one also studies filtrations whose successive quotients are Verma modules.

Definition 2.15. A module $M \in \mathcal{O}$ admits a *standard filtration*, or *Verma flag*, if it admits a finite filtration of the form

$$0 = M_0 \subset M_1 \subset \cdots \subset M_n = M,$$

where M_{i+1}/M_i is isomorphic to a Verma module $\Delta(\lambda)$. We denote by $(M : \Delta(\lambda))$ the multiplicity of $\Delta(\lambda)$ in the standard filtration of M .

Note that the filtration length and multiplicity with which each Verma module appears as a subquotient is well-defined (see [Hum08, §3.7]). We will investigate properties of Verma flags in subsequent sections.

We can further view simple and Verma modules as the building blocks of $\mathcal{O}$ in the following proposition. Let $K(\mathcal{O})$ denote the *Grothendieck group* of $\mathcal{O}$, which is the abelian group generated by the symbols $[M]$ for $M \in \mathcal{O}$ with the relation $[B] = [A] + [C]$ whenever $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is a short exact sequence in $\mathcal{O}$.

Proposition 2.16 ([Hum08, §1.11, 1.16]). *Both $\{[L(\lambda)] \mid \lambda \in \mathfrak{h}^*\}$ and $\{[\Delta(\lambda)] \mid \lambda \in \mathfrak{h}^*\}$ form a $\mathbb{Z}$ -basis for $K(\mathcal{O})$.*

2.3 Tensoring in $\mathcal{O}$

A natural way to build new modules is by tensoring known ones. Whilst the category of all $U(\mathfrak{g})$ -modules is closed under tensor products, the following example shows that this is not the case for $\mathcal{O}$.

Example 2.17. Let $\mathfrak{g} = \mathfrak{sl}_2$ and $\lambda, \mu \in \mathfrak{h}^*$ be dominant integral, that is $\lambda, \mu \in \mathbb{Z}_{\geq 0}$. Then we claim that $\Delta(\lambda) \otimes \Delta(\mu) \notin \mathcal{O}(\mathfrak{sl}_2)$. One might be tempted to think that $\Delta(\lambda) \otimes \Delta(\mu)$ trivially satisfies the finite generation axiom (O1), but this exactly the property that fails! Following the notation of Example 2.12, let $\{v_i\}$

and $\{w_i\}$ be weight bases of $\Delta(\lambda)$ and $\Delta(\mu)$ respectively. If $\Delta(\lambda) = \bigoplus_{i \geq 0} \Delta(\lambda)_{\lambda-2i}$ and $\Delta(\mu) = \bigoplus_{j \geq 0} \Delta(\mu)_{\mu-2j}$, then

$$\Delta(\lambda) \otimes \Delta(\mu) = \bigoplus_{n \geq 0} M_n, \quad M_n := \bigoplus_{\substack{i, j \geq 0 \\ i+j=n}} \Delta(\lambda)_{\lambda-2i} \otimes \Delta(\mu)_{\mu-2j}.$$

Now each M_n is generated by $\{v_i \otimes w_j\}_{i+j=n}$ with $\dim M_n = n + 1$, so $\dim M_n$ is unbounded as n increases.

Suppose instead that $\Delta(\lambda) \otimes \Delta(\mu)$ is a finitely generated $\mathfrak{sl}_2$ -module by a finite set $\{m_1, \dots, m_r\}$. Then if m_k has weight ν_k , any vector of weight $\lambda + \mu - 2n$ arising from m_k must be of the form $f^{n-d_k} m_k$ for a fixed d_k determined by ν_k , giving at most one vector in M_n . Hence $\dim M_n \leq r$ for all n , contradicting $\dim M_n = n + 1$. Thus $\Delta(\lambda) \otimes \Delta(\lambda)$ is not finitely generated.

This raises the question: for which modules V is tensoring with V an endofunctor of $\mathcal{O}$?

Proposition 2.18 ([Hum08, Theorem 1.2(d)]). *Let $V \in \mathcal{O}$ be finite-dimensional. Then $V \otimes -$ and $- \otimes V$ are exact functors on $\mathcal{O}$.*

Proof. We prove for $V \otimes -$, noting that the assertions for $- \otimes V$ follow similarly. Let $M \in \mathcal{O}$. Choose a basis $\{v_1, \dots, v_k\}$ of V and generators $\{w_1, \dots, w_\ell\}$ of M . Then the elements $\{v_i \otimes w_j\}$ generate $V \otimes M$ as a $U(\mathfrak{g})$ -module. For $v = \sum_{i=1}^n c_i v_i$ and $m = \sum_j x_j \cdot w_j$ for some $x_j \in U(\mathfrak{g})$, we have that

$$v \otimes m = \left(\sum_i c_i v_i \right) \otimes \left(\sum_j x_j \cdot w_j \right) = \sum_j x_j \cdot \left(\sum_{i,j} c_i v_i \otimes w_j \right) - \sum_{i,j} c_i x_j \cdot v_i \otimes w_j.$$

Writing $x_j \cdot v_i$ in terms of basis vectors gives us that $v \otimes m \in \langle v_i \otimes w_j \rangle$. The inclusion $\langle v_i \otimes w_j \rangle \subset V \otimes M$ is trivial, so we get the equality $V \otimes M = \langle v_i \otimes w_j \rangle$. The weight decomposition and local $\mathfrak{n}^+$ -finiteness are inherited from V and M , hence $V \otimes M \in \mathcal{O}$. Exactness follows as tensoring vector spaces over $\mathbb{C}$ is exact. $\square$

We defer further examples to Chapter 3 when we introduce translation functors. Below are two elementary but important properties of tensoring with a finite-dimensional module, the first being a formal adjointness property.

Proposition 2.19 ([Hum08, Lemma 7.2]). *For V finite-dimensional, the functor $V \otimes -$ (resp. $- \otimes V$) has a left and right adjoint both equal to $V^* \otimes -$ (resp. $- \otimes V^*$).*

Proof. We only prove for $V \otimes -$. Since V is finite-dimensional, $\text{Hom}_{\mathbb{C}}(V, M) \simeq V^* \otimes M \simeq M \otimes V^*$. On vector spaces, we get the isomorphism

$$\text{Hom}_{\mathbb{C}}(V \otimes M, N) \simeq \text{Hom}_{\mathbb{C}}(M, \text{Hom}_{\mathbb{C}}(V, N)) \simeq \text{Hom}_{\mathbb{C}}(M, V^* \otimes N),$$

where the first isomorphism comes from the tensor-hom adjunction. An explicit isomorphism $\Phi: \text{Hom}_{\mathbb{C}}(V \otimes M, N) \rightarrow \text{Hom}_{\mathbb{C}}(M, V^* \otimes N)$ is given by

$$f \mapsto \left(m \mapsto \sum_{i=1}^n v_i^* \otimes f(v_i \otimes m) \right)$$

with inverse $\Psi: \text{Hom}_{\mathbb{C}}(M, V^* \otimes N) \rightarrow \text{Hom}_{\mathbb{C}}(V \otimes M, N)$

$$g \mapsto (v \otimes m \mapsto (\text{ev}_v \otimes \text{id}_N)(g(m))),$$

where $\text{ev}_v: V^* \rightarrow \mathbb{C}$ denotes evaluation at v . We show that Φ (and similarly Ψ) is compatible with the action of $U(\mathfrak{g})$, that is $x \cdot \Phi(f)(m) = \Phi(f)(x \cdot m)$. We get the equalities

$$\begin{aligned} x \cdot \Phi(f)(m) &= x \cdot \left(\sum_{i=1}^n v_i^* \otimes f(v_i \otimes m) \right) \\ &= \sum_{i=1}^n x \cdot v_i^* \otimes f(v_i \otimes m) + v_i^* \otimes x \cdot f(v_i \otimes m) \\ &= \sum_{i=1}^n x \cdot v_i^* \otimes f(v_i \otimes m) + \sum_{i=1}^n v_i^* \otimes f(x \cdot v_i \otimes m) \\ &\quad + \sum_{i=1}^n v_i^* \otimes f(v_i \otimes x \cdot m) \quad (f \text{ is } U(\mathfrak{g})\text{-invariant}). \end{aligned}$$

The last term is $\Phi(f)(x \cdot m)$. We claim the first two terms cancel. Using the dual action $(x \cdot v_i^*)(v_j) = -v_i^*(x \cdot v_j)$ (recall from Section 1.2), write $x \cdot v_i^* = -\sum_{j=1}^n v_i^*(x \cdot v_j)v_j^*$ and $x \cdot v_i = \sum_{j=1}^n v_j^*(x \cdot v_i)v_j$. Thus the first two terms become

$$-\sum_{i,j=1}^n v_i^*(x \cdot v_j)v_j^* \otimes f(v_i \otimes m) + \sum_{i,j=1}^n v_j^*(x \cdot v_i)v_i^* \otimes f(v_j \otimes m) = 0$$

after re-indexing, proving the claim. $\square$

The following tensoring result produces a standard filtration on $V \otimes \Delta(\lambda)$, and will be an essential tool when we study projectives in Section 2.6 and translation functors.

Theorem 2.20 ([Hum08, Theorem 3.6]). *Let $V \in \mathcal{O}$ be a finite-dimensional module and $\lambda \in \mathfrak{h}^*$. Then $V \otimes \Delta(\lambda)$ admits a standard filtration such that each successive subquotient is isomorphic to $\Delta(\lambda + \mu)$, where μ runs the weights of V with $(V \otimes \Delta(\lambda) : \Delta(\lambda + \mu)) = \dim V_\mu$.*

Proof. Let Ind and Res denote the respective induction and restriction functors between $U(\mathfrak{g})$ and $U(\mathfrak{b})$. We show that $V \otimes \Delta(\lambda) \simeq U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} (\text{Res}(V) \otimes \mathbb{C}_\lambda)$, then construct a filtration of the right side satisfying the properties of the theorem.

Let $M \in \mathcal{O}$. Then we have the isomorphisms

$$\begin{aligned} \text{Hom}_{\mathfrak{g}}(V \otimes \Delta(\lambda), M) &\simeq \text{Hom}_{\mathfrak{g}}(\Delta(\lambda), V^* \otimes M) && \text{(Proposition 2.19)} \\ &\simeq \text{Hom}_{\mathfrak{b}}(\mathbb{C}_\lambda, \text{Res}(V^* \otimes M)) && \text{(Frobenius reciprocity)} \\ &\simeq \text{Hom}_{\mathfrak{b}}(\mathbb{C}_\lambda, \text{Res}(V)^* \otimes \text{Res}(M)) \\ &\simeq \text{Hom}_{\mathfrak{b}}(V \otimes \mathbb{C}_\lambda, \text{Res}(M)) && \text{(Proposition 2.19)} \\ &\simeq \text{Hom}_{\mathfrak{g}}(\text{Ind}(V \otimes \mathbb{C}_\lambda), M) && \text{(Frobenius reciprocity)} \end{aligned}$$

where $\text{Res}(V^*) \simeq \text{Res}(V)^*$ by forgetting the $\mathfrak{b}$ action. The Yoneda lemma gives the desired isomorphism.

As a $U(\mathfrak{b})$ -module, we can order a basis of weight vectors $\{v_1, \dots, v_n\}$ of $V \otimes \mathbb{C}_\lambda$, with v_i having weight ν_i , such that $i \leq j$ whenever $\nu_i \leq \nu_j$. Then $V \otimes \mathbb{C}_\lambda$ admits a filtration

$$0 = K_n \subset K_{n-1} \subset \dots \subset K_1 = V \otimes \mathbb{C}_\lambda,$$

where each K_i is a $U(\mathfrak{b})$ -module spanned by $\{v_i, \dots, v_n\}$. As $U(\mathfrak{g})$ is a free right $U(\mathfrak{b})$ -module, the functor $N \mapsto U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} N$ is exact whenever N is finite-dimensional. Then define $M_i := U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} K_i$. It follows that $M_i/M_{i+1} \simeq U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} K_i/K_{i+1} \simeq \Delta(\lambda + \nu_i)$. $\square$

2.4 Duality

A natural expectation is that $\mathcal{O}$ be closed under duals, but what does duality mean in $\mathcal{O}$? The naïve choice $M^* = \text{Hom}_{\mathbb{C}}(M, \mathbb{C})$ with the contragredient dual (see Section 1.2) fails: if M is infinite-dimensional, then M^* fails to be $\mathfrak{n}^+$ -locally finite and need not be a weight module since taking duals of vector spaces only commute with finite direct sums. Since weight spaces in $\mathcal{O}$ are finite-dimensional, one might instead try

$$M' := \bigoplus_{\lambda \in \mathfrak{h}^*} (M_\lambda)^*.$$

This gives a vector space, so it remains to check that M' is preserved by the action of $\mathfrak{g}$. Note that the parentheses above may be dropped. We can identify $(M_\lambda)^*$ with the set of $f \in M^*$ that vanish on M_μ for $\lambda \neq \mu$, which gives $(M_\lambda)^* \subset (M^*)_\lambda$. The other inclusion holds since if $f \in (M^*)_\lambda$, $v \in M_\mu$ and $h \in \mathfrak{h}$ with $\lambda(h) \neq \mu(h)$, then

$$\mu(h)f(v) = f(\mu(h)v) = f(h \cdot v) = -(h \cdot f)(v) = -\lambda(h)f(v),$$

so f vanishes outside of M_λ .

A quick computation shows that this definition is inadequate.

Example 2.21. Let $\mathfrak{g} = \mathfrak{sl}_2$ and consider $M = \Delta(2)$. Suppose that $v \in M$ is a weight vector. Following the notation introduced in Example 2.12, let $\{v_i^*\}$ be the dual basis vector to v_i . Consider the weight space M_{-4} , on which only v_3^* acts non-trivially. It follows that for $(e \cdot v_3^*)(v) = -v_3^*(e \cdot v)$ to be non-zero, we must have $v \in M_{-6}$ which is spanned by v_4 . In particular,

$$(e \cdot v_3^*)(v_4) = -v_3^*(e \cdot v_4) = -v_3^*(-v_3) = 1,$$

so $e \cdot v_3^* = v_4^*$. Iterating shows that $e^i \cdot v_2^* = i!v_{2+i}^*$ which is never zero for $i > 0$, so M' is thus not locally $\mathfrak{n}^+$ -finite.

The previous example hints at what the correct definition of duality should be: we should swap the actions of f and e .

Definition 2.22. For $f \in M^*$, $x \in \mathfrak{g}$ and $v \in M$, the *twisted action* of $\mathfrak{g}$ on M^* is defined by

$$(x \cdot_\tau f)(v) := f(\tau(x) \cdot v),$$

where $\tau: \mathfrak{g} \rightarrow \mathfrak{g}$ is the *Chevalley anti-involution*, sending $e_\alpha \mapsto f_\alpha$, $h_\alpha \mapsto h_\alpha$, and $f_\alpha \mapsto e_\alpha$ for each $\mathfrak{sl}_2$ -triple $(e_\alpha, h_\alpha, f_\alpha)$.

The Chevalley anti-involution is a well-defined Lie algebra *anti-homomorphism*, that is a map φ such that $\varphi([x, y]) = [\varphi(y), \varphi(x)] = -[\varphi(x), \varphi(y)]$. From now on, the action of $\mathfrak{g}$ on M^* will be this twisted action and we write $x \cdot f = x \cdot_\tau f$ for $x \in \mathfrak{g}$ and $f \in M^*$.

Proposition 2.23. Let $M \in \mathcal{O}$. Then the (BGG) dual $M^\vee$ of M in $\mathcal{O}$ is defined as

$$M^\vee := \bigoplus_{\lambda} M_\lambda^*,$$

with the $\mathfrak{g}$ -module structure on $M^\vee$ given by the twisted action. By construction, $M^{\vee\vee} \simeq M$. This duality operation preserves $\mathcal{O}$ and is an exact functor.

Proof. Exactness follows since taking duals of finite-dimensional weight spaces is exact. By construction, $M^\vee$ is a weight module, so it remains to check (O1) and (O3).

Take a composition series $0 = M_0 \subset M_1 \subset \cdots \subset M_n = M$ with simple quotients $L_i := M_i/M_{i-1}$. We have short exact sequences

$$0 \longrightarrow M_{i-1} \longrightarrow M_i \longrightarrow L_i \longrightarrow 0$$

for $1 \leq i \leq n$. Exactness of the duality functor yields the short exact sequence

$$0 \longrightarrow L_i^\vee \longrightarrow M_i^\vee \longrightarrow M_{i-1}^\vee \longrightarrow 0,$$

hence a filtration $M^\vee = M_n^\vee \twoheadrightarrow M_{n-1}^\vee \twoheadrightarrow \cdots \twoheadrightarrow M_0^\vee = 0$, with successive quotients $M_i^\vee/M_{i-1}^\vee \cong L_i^\vee$. To see that each $L_i^\vee$ is simple, note $L_i^{\vee\vee} \simeq L_i$. Let $0 \neq N \subseteq L_i^\vee$. Applying duality to the inclusion $N \hookrightarrow L_i^\vee$ yields a surjection $L_i^{\vee\vee} \twoheadrightarrow N^\vee$. Since L_i is simple, $N^\vee \simeq L_i$, hence $N \simeq L_i^\vee$. Thus $L_i^\vee$ has no proper non-zero submodules and is simple. Therefore, the filtration is a composition series for $M^\vee$. Hence $M^\vee$ has finite length. This implies that $M^\vee$ is finitely generated since we can choose an element $x_i \in M_i^\vee$ for each i whose image generates $L_i^\vee$. The set $\{x_1, \dots, x_n\}$ thus generates $M^\vee$.

If $f \in M_\mu^\vee$, then the twisted action of $\mathfrak{g}$ on $M^\vee$ means that $x_\alpha \cdot f \in M_{\mu+\alpha}^\vee$, where $x_\alpha \in \mathfrak{g}_\alpha$ for all a simple root α . This implies that $M^\vee$ is locally $\mathfrak{n}^+$ -finite. $\square$

We will denote the dual of a Verma module $\Delta(\lambda)$ as $\nabla(\lambda) := \Delta(\lambda)^\vee$.

Example 2.24. Let $\mathfrak{g} = \mathfrak{sl}_2$. The dual Verma module $\nabla(\lambda)$ has the same set of weights as $\Delta(\lambda)$. Using the same notation as in Example 2.21, $\{v_i^*\}$ is the dual weight basis for $\nabla(\lambda)$. The weight diagram for $\nabla(\lambda)$ is identical to (2.1) but with all arrows reversed:



For λ dominant integral, this means that $L(\lambda)$ is now a submodule of $\nabla(\lambda)$. The vector $v_0^* \in \nabla(\lambda)_\lambda$ is a maximal vector as it is annihilated by e , but it does not generate $\nabla(\lambda)$ as a $U(\mathfrak{g})$ -module since $f^{i+1} \cdot v_0^* = 0$. Thus, $\nabla(\lambda)$ is our first example of an indecomposable non-highest weight module in $\mathcal{O}$. From the diagram we obtain $\nabla(\lambda)/L(\lambda) \simeq \nabla(-\lambda-2) \simeq \Delta(-\lambda-2)$, so $L(-\lambda-2)$ is self-dual. Verifying these isomorphisms is straight-forward and follows the same calculations as in Example 2.12.

We list some properties of dual modules in the following theorem.

Theorem 2.25 ([Hum08, Theorem 3.3]). *Let $\lambda, \mu \in \mathfrak{h}^*$. Then*

- (a) $L(\lambda)^\vee \simeq L(\lambda)$,
- (b) $\nabla(\lambda)$ has $L(\lambda)$ as its unique irreducible submodule, and
- (c) $\dim \operatorname{Hom}_{\mathcal{O}}(\Delta(\lambda), \nabla(\lambda)) = 1$ containing the map $\Delta(\lambda) \twoheadrightarrow L(\lambda) \hookrightarrow \nabla(\lambda)$, while $\operatorname{Hom}_{\mathcal{O}}(\Delta(\mu), \nabla(\lambda)) = 0$ when $\mu \neq \lambda$.

2.5 Block decomposition

We now reach one of the chapter's principal goals, namely the block decomposition of $\mathcal{O}$, which breaks the category into smaller, more tractable pieces. So far, our arguments have used the root-space and weight-space structure built into the axioms of $\mathcal{O}$. However, to understand Verma modules better, we must also use the structure of $U(\mathfrak{g})$. A guiding principle in the representation theory of associative algebras is to study the centre. In finite-dimensional Lie theory, this viewpoint appears via the Casimir element, but here we work with the full centre $Z(\mathfrak{g}) \subset U(\mathfrak{g})$ and examine its action on objects of $\mathcal{O}$. The resulting central character decomposition yields a direct-sum decomposition of $\mathcal{O}$ into blocks, reducing many questions to statements inside a single block. We develop this decomposition in this section and study interactions between blocks in the next chapter.

To begin, we note an elementary but fundamental observation that makes the decomposition possible.

Lemma 2.26 ([Hum08, Theorem 1.1(e)]). *For any $M \in \mathcal{O}$, M is locally $Z(\mathfrak{g})$ -finite, that is $\dim Z(\mathfrak{g}) \cdot v < \infty$.*

Proof. Each $v \in M$ may be written as a finite sum of weight vectors, so assume that $v \in M_\lambda$ for some λ . Since the action of $\mathfrak{h}$ commutes with $Z(\mathfrak{g})$, it follows that $hz \cdot v = zh \cdot v = \lambda(h)z \cdot v$, so $z \cdot v \in M_\lambda$. The lemma follows since each weight space is finite-dimensional by Corollary 2.2. $\square$

Consider a highest weight module M generated by a highest weight vector $v^+ \in M_\lambda$ and fix $z \in Z(\mathfrak{g})$. We saw in the proof of Lemma 2.26 that $z \cdot v^+ \in M_\lambda$. In particular, z has to act by a scalar since $\dim M_\lambda = 1$, that is $z \cdot v^+ = \chi_\lambda(z)v^+$ for a central character χ_λ depending only on λ . Since $M = U(\mathfrak{n}^-) \cdot v^+$ and z

commutes with all of $U(\mathfrak{g})$, we see that z has to act by the same scalar on all of $\Delta(\lambda)$. This leads us to the following definition.

Definition 2.27. A *central character* is an algebra homomorphism $\chi: Z(\mathfrak{g}) \rightarrow \mathbb{C}$. A module M has *central character* χ if $z \cdot v = \chi(z)v$ for all $z \in Z(\mathfrak{g})$ and $v \in M$.

In particular, if $\Delta(\lambda)$ contains a highest weight submodule of highest weight $\mu < \lambda$, then $\chi_\mu = \chi_\lambda$. Moreover, since $\Delta(s_\alpha \cdot \lambda) \subset \Delta(\lambda)$ by Theorem 2.14, we also have $\chi_{s_\alpha \cdot \mu} = \chi_\lambda$ whenever λ is dominant integral. Because the Weyl group is generated by the simple reflections in this case, it follows that $\chi_\mu = \chi_\lambda$ whenever $\mu \in W \cdot \lambda$.

Definition 2.28. Two weights $\lambda, \mu \in \mathfrak{h}^*$ are *linked* if they lie in the same dot-orbit of W . The orbit $\{w \cdot \lambda \mid w \in W\}$ of λ is called the *linkage class* of λ .

Example 2.29. Let $\mathfrak{g} = \mathfrak{sl}_2$. The Weyl group for $\mathfrak{sl}_2$ is S_2 and $\rho = 1$. For any $\lambda \in \mathfrak{h}^*$, λ is linked to $s \cdot \lambda = -\lambda - 2$. When λ is dominant integral, we saw in Example 2.12 that $\Delta(-\lambda - 2)$ embeds into $\Delta(\lambda)$, so $\chi_{-\lambda-2} = \chi_\lambda$.

By the PBW theorem (Theorem 1.5), we may choose an ordered basis

$$f_1 < \cdots < f_m < h_1 < \cdots < h_\ell < e_1 < \cdots < e_m$$

of $\mathfrak{g}$ such that the monomials $f_1^{r_1} \cdots f_m^{r_m} h_1^{s_1} \cdots h_\ell^{s_\ell} e_1^{t_1} \cdots e_m^{t_m}$ form a basis for $U(\mathfrak{g})$, where $f_i \in \mathfrak{g}_{-\alpha_i}$ and $e_i \in \mathfrak{g}_{\alpha_i}$ for positive roots α_i , and $h_i = [e_i, f_i]$. Consider the subspace $U(\mathfrak{g})_0 \subset U(\mathfrak{g})$, the 0-weight space for the adjoint action of $\mathfrak{g}$ on $U(\mathfrak{g})$. Equivalently, $U(\mathfrak{g})_0$ is the centraliser of $\mathfrak{h}$ in $U(\mathfrak{g})$, hence a subalgebra containing both $Z(\mathfrak{g})$ and $U(\mathfrak{h})$ as unital subalgebras. In terms of the PBW basis, $U(\mathfrak{g})_0$ is spanned by those monomials for which $\sum_{i=1}^m (t_i - r_i)\alpha_i = 0$, as otherwise the monomial has non-zero weight under the adjoint $\mathfrak{h}$ -action.

Lemma 2.30 ([Hum08, Remark 1.7]). *As vector spaces, $U(\mathfrak{g})_0 = U(\mathfrak{h}) \oplus I$ for some two-sided ideal $I \subset U(\mathfrak{g})_0$.*

Proof. Define $L := U(\mathfrak{g})_0 \cap U(\mathfrak{g})\mathfrak{n}^+$ and $R := U(\mathfrak{g})_0 \cap \mathfrak{n}^-U(\mathfrak{g})$, which are left and right ideals of $U(\mathfrak{g})_0$ respectively. By the discussion in the previous paragraph, $L = R$, so define $I := L = R$ which is both a left and right ideal. Moreover, the PBW monomials which span I do not lie in $U(\mathfrak{h})$, so there is a direct sum decomposition $U(\mathfrak{g})_0 = U(\mathfrak{h}) \oplus I$. $\square$

We may then form the quotient $U(\mathfrak{g})_0/I$, which is an algebra isomorphic to $U(\mathfrak{h})$. The next definition and theorem shows why the dot-action of the Weyl group appears when working with $\mathcal{O}$.

Definition 2.31. The *Harish-Chandra homomorphism* is the algebra homomorphism $\gamma: Z(\mathfrak{g}) \rightarrow U(\mathfrak{h})$ defined as the composition of the projection $\pi: U(\mathfrak{g})_0 \rightarrow U(\mathfrak{h})$ restricted to $Z(\mathfrak{g})$ with the twist automorphism of $U(\mathfrak{h}) \simeq \text{Sym}(\mathfrak{h}) = \mathbb{C}[\mathfrak{h}^*]$ induced by the substitution $p(\lambda) \mapsto p(\lambda - \rho)$.

Theorem 2.32 (Harish-Chandra Theorem [HC51], [Hum08, §1.8-1.10]). *Consider the Harish-Chandra homomorphism $\gamma: Z(\mathfrak{g}) \rightarrow U(\mathfrak{h})$.*

- (a) *The homomorphism γ gives an isomorphism $Z(\mathfrak{g}) \simeq \mathbb{C}[\mathfrak{h}^*]^{(W, \cdot)}$.*
- (b) *For every $\lambda, \mu \in \mathfrak{h}^*$, $\chi_\lambda = \chi_\mu$ if and only if $\lambda = W \cdot \mu$.*
- (c) *Every central character is of the form χ_λ for some $\lambda \in \mathfrak{h}^*$.*

We illustrate this isomorphism by considering the image of the Casimir element of $\mathfrak{g} = \mathfrak{sl}_2$.

Example 2.33. Let $\mathfrak{g} = \mathfrak{sl}_2$ and e, h, f the standard basis. Set $z = ef + fe + \frac{1}{2}h^2 \in U(\mathfrak{sl}_2)$. A straightforward check shows that $[z, e] = [z, h] = [z, f] = 0$, so $z \in Z(\mathfrak{g})$.

Identify $\mathbb{C}[h] = \text{Sym}(\mathfrak{h}) \simeq \mathbb{C}[\mathfrak{h}^*]$. By the PBW theorem (Theorem 1.5), write

$$z = ef + fe + \frac{1}{2}h^2 = [e, f] + 2fe + \frac{1}{2}h^2 = 2fe + h + \frac{1}{2}h^2.$$

The projection π kills terms involving e or f , hence $\pi(z) = h + \frac{1}{2}h^2$. Denote $p(h) := h + \frac{1}{2}h^2 \in \mathbb{C}[h]$. The twist homomorphism with $\rho(h) = 1$ in $\mathfrak{sl}_2$ gives

$$p(h) \mapsto p(h - \rho(h)) = (h - 1)^2 + 2(h - 1) = \frac{1}{2}(h^2 - 1).$$

For $\mathfrak{sl}_2$, the Weyl group $W \simeq S_2$ acts by $s \cdot \lambda = s(\lambda + 1) - 1$. Now s fixes $\frac{1}{2}(h^2 - 1)$ since $s \cdot \frac{1}{2}(h^2 - 1) = \frac{1}{2}(s(h^2 - 1 + 1) - 1) = \frac{1}{2}(h^2 - 1)$, so $\gamma(z) \in \mathbb{C}[h]^{(W, \cdot)}$. Here, $\mathbb{C}[h]^{(W, \cdot)}$ is just $\mathbb{C}[h^2]$. Since $\frac{1}{2}(h^2 - 1)$ generates $\mathbb{C}[h^2]$, the Casimir element generates $Z(\mathfrak{sl}_2)$ by the Harish-Chandra theorem.

By Theorem 2.32, $\mathcal{O}$ can be decomposed as a direct sum of subcategories.

Definition 2.34. An abelian category $\mathcal{A}$ admits a *block decomposition* if $\mathcal{A} = \bigoplus_{i \in I} \mathcal{A}_i$ for full indecomposable abelian subcategories $\mathcal{A}_i$ and there are no non-trivial morphisms or extensions between objects in different summands. We call $\mathcal{A}_i$ a *block* of $\mathcal{A}$.

Since a highest-weight module is acted on by $Z(\mathfrak{g})$ through a central character, it is natural to ask for an analogous description for arbitrary $M \in \mathcal{O}$. For a character $\chi: Z(\mathfrak{g}) \rightarrow \mathbb{C}$, set

$$M^\chi := \{v \in M \mid z - \chi(z) \text{ acts nilpotently on } v \text{ for all } z \in Z(\mathfrak{g})\}.$$

Because $Z(\mathfrak{g})$ is central, M^χ is a $U(\mathfrak{g})$ -submodule of M as $(z - \chi(z))^n u \cdot v = u(z - \chi(z))^n \cdot v = 0$ for all $u \in U(\mathfrak{g})$ as $z \in Z(\mathfrak{g})$. We can decompose each (finite-dimensional) weight space M_λ as $M_\lambda = \bigoplus_\chi M_\lambda \cap M^\chi$ since $z - \chi(z)$ commute with each other and preserves weight spaces. Therefore, $M = \bigoplus_\chi M^\chi$ is a finite direct sum of non-zero submodules M^χ .

Write $\mathcal{O}_\lambda := \mathcal{O}_{\chi_\lambda}$ for the full subcategory of $\mathcal{O}$ containing modules M such that $M = M^{\chi_\lambda}$. Thanks to the Harish-Chandra theorem, we obtain a decomposition of $\mathcal{O}$ by central characters as follows.

Theorem 2.35 ([Hum80, Proposition 1.13]). *There is a decomposition*

$$\mathcal{O} = \bigoplus_{\lambda \in \mathfrak{h}^*/(W \cdot \cdot)} \mathcal{O}_\lambda$$

for which $\mathcal{O}_\lambda$ is a block of $\mathcal{O}$ if λ is integral.

Importantly, this is *not* a block decomposition! If λ is not integral, $\mathcal{O}_\lambda$ may further split. Nevertheless, for $\lambda \notin W \cdot \mu$, there are no non-trivial extensions and homomorphisms between $\mathcal{O}_\lambda$ and $\mathcal{O}_\mu$. In particular, $\mathcal{O}_\lambda$ is a *Serre subcategory* of $\mathcal{O}$, that is $\mathcal{O}_\lambda$ is closed under extensions in $\mathcal{O}$.

In this thesis we restrict to integral weights, making it convenient to compare integral blocks via translation functors (see Chapter 3), allowing us to concentrate on the *principal block* $\mathcal{O}_0$. Since all simple modules in $\mathcal{O}_0$ have central character χ_0 , Theorem 2.32 gives the following theorem.

Theorem 2.36. *The simples of $\mathcal{O}_0$ are $\{L(w \cdot 0) \mid w \in W\}$ up to isomorphism.*

Now we turn to the classification of indecomposable objects in $\mathcal{O}_0$.

2.6 Projectives in $\mathcal{O}$

Recall that an object P in an abelian category $\mathcal{A}$ is called *projective* if $\text{Hom}_{\mathcal{A}}(P, -)$ is exact. Equivalently, P satisfies the lifting criterion: given any epimorphism $\pi: M \rightarrow N$ and any morphism $\varphi: P \rightarrow N$, there exists $\psi: P \rightarrow M$ such that $\pi \circ \psi = \varphi$.

Over a ring R , projectives in $R\text{-mod}$ are the direct summands of free R -modules. Category $\mathcal{O}$ is a subcategory of $U(\mathfrak{g})$ -modules not containing $U(\mathfrak{g})$, so one cannot easily build projectives in $\mathcal{O}$ by taking direct sums of $U(\mathfrak{g})$. A natural substitute is given by Verma modules. The next theorem provides a precise criterion for when a Verma module is projective in $\mathcal{O}$.

Theorem 2.37. *The Verma module $\Delta(\lambda)$ is projective if and only if $\lambda \in \mathfrak{h}^*$ is dominant.*

Proof. We will only prove the “only if” direction, which is [Hum08, Exercise 3.11]. The converse is [Hum08, Proposition 3.8(a)].

Suppose λ is not dominant. Then there exists a positive root α such that $\lambda < s_\alpha \cdot \lambda$. By Theorem 2.14, $\Delta(\lambda) \hookrightarrow \Delta(s_\alpha \cdot \lambda)$. Dualising yields a surjection of dual Vermas $\nabla(s_\alpha \cdot \lambda) \twoheadrightarrow \nabla(\lambda)$. If $\Delta(\lambda)$ were projective, then $\text{Hom}_{\mathcal{O}}(\Delta(\lambda), -)$ would be exact, so we get a surjection

$$\text{Hom}_{\mathcal{O}}(\Delta(\lambda), \Delta(s_\alpha \cdot \lambda)) \longrightarrow \text{Hom}_{\mathcal{O}}(\Delta(\lambda), \nabla(\lambda)) \longrightarrow 0.$$

However, Theorem 2.25 says that the left term is zero while the dimension of the middle term is $\dim \text{Hom}_{\mathcal{O}}(\Delta(\lambda), \nabla(\lambda)) = 1$, which is impossible. $\square$

If we start off with a known projective, it is easy to construct new ones.

Proposition 2.38 ([Hum08, Proposition 3.8(b)]). *If $V \in \mathcal{O}$ is finite-dimensional and $P \in \mathcal{O}$ is projective, then $V \otimes P$ and $P \otimes V$ are projective in $\mathcal{O}$.*

Proof. We show that $\text{Hom}_{\mathcal{O}}(V \otimes P, -)$ is exact. By Proposition 2.19,

$$\text{Hom}_{\mathcal{O}}(V \otimes P, M) \simeq \text{Hom}_{\mathcal{O}}(P, V^* \otimes M).$$

Since P is projective and tensoring with the finite-dimensional V^* is exact, this shows that $V \otimes P$ is projective. The same argument applies for $P \otimes V$. $\square$

Category $\mathcal{O}$ has *enough projectives* if for every $M \in \mathcal{O}$, there is a projective object $P \in \mathcal{O}$ and an epimorphism $P \rightarrow M$. If $\mathcal{O}$ has enough projectives, standard homological methods apply which places $\mathcal{O}$ close in behaviour to a module category. Indeed this is true.

Theorem 2.39 ([Hum08, Theorem 3.8]). *There are enough projectives in $\mathcal{O}$.*

Proof. We proceed by inducting on the length of $M \in \mathcal{O}$.

When M has length 1, $M \simeq L(\lambda)$ for some $\lambda \in \mathfrak{h}^*$. Pick $n \in \mathbb{Z}$ large enough so that $\mu := \lambda + n\rho$ is dominant. Then $\Delta(\mu)$ is projective by Theorem 2.37. Since $n\rho$ is dominant integral $L(n\rho)$ is finite-dimensional by Theorem 2.13, so $P := L(n\rho) \otimes \Delta(\mu)$ is projective by Proposition 2.38. By Theorem 2.20, $P \twoheadrightarrow \Delta(\mu - n\rho) = \Delta(\lambda) \twoheadrightarrow L(\lambda)$. Thus $M \simeq L(\lambda)$ is a quotient of a projective module.

Now suppose M has length greater than 1. Then M has a simple submodule $L(\lambda)$ for some λ , yielding a short exact sequence

$$0 \longrightarrow L(\lambda) \hookrightarrow M \twoheadrightarrow N \longrightarrow 0.$$

By induction, there is a surjection $g: Q \rightarrow N$ for some projective $Q \in \mathcal{O}$. Since Q is projective, g lifts to a map $f: Q \rightarrow M$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & L(\lambda) & \hookrightarrow & M & \xrightarrow{\pi} & N \longrightarrow 0 \\ & & & & & \nwarrow f & \uparrow g \\ & & & & & & Q \end{array}$$

We thus have $\pi \operatorname{im}(f) = \operatorname{im}(g) = N$ as $g = \pi \circ f$, so $\pi|_{\operatorname{im}(f)}$ is surjective with kernel $L(\lambda) \cap \operatorname{im}(f)$. Since $L(\lambda)$ is simple, we have that $L(\lambda) \cap \operatorname{im}(f) \in \{0, L\}$.

If $L(\lambda) \cap \operatorname{im}(f) = 0$, then $\ker \pi|_{\operatorname{im}(f)} = 0$ so π is both injective and surjective on $\operatorname{im}(f)$. Thus $\operatorname{im}(f) \simeq N$, giving a splitting $M \simeq L(\lambda) \oplus N$. As both $L(\lambda)$ and N are mapped onto by projectives, $M \simeq L(\lambda) \oplus N$ will be mapped onto by the direct sum of those projectives, which is also projective.

If $L(\lambda) \cap \operatorname{im}(f) = L$, we proceed by a diagram chase to show that f is surjective. Let $m \in M$. Since g is surjective, there exists $q \in Q$ such that $\pi(m) = g(q)$. As $g = \pi \circ f$, $\pi \circ f(q) = g(q) = \pi(m)$. Then $\pi(m - f(q)) = 0$, so $m - f(q) \in \ker \pi$. By exactness, $m - f(q) \in L$. As $L \subseteq \operatorname{im}(f)$, $m - f(q) \in \operatorname{im}(f)$, so $m = f(q) + (m - f(q)) \in \operatorname{im}(f)$. Therefore, $Q \rightarrow M$ is surjective. $\square$

Remark 2.40. The dual notion of a projective object is an *injective object*. In $\mathcal{O}$, the exact contravariant duality $(-)^{\vee}$ sends projectives to injectives (and vice versa). Since $\mathcal{O}$ has enough projectives, it also has enough injectives.

As $\mathcal{O}$ is artinian and has enough projectives, we can associate a projective to each object in the following sense.

Definition 2.41. A *projective cover* of $M \in \mathcal{O}$ is an epimorphism $\pi: P_M \rightarrow M$, where P_M is the unique projective (up to isomorphism) such that no proper submodule of P_M is mapped onto M .

In the case that $M \simeq L(\lambda)$, write $P(\lambda) := P_{L(\lambda)}$ for its projective cover. Without using special properties of $\mathcal{O}$, all indecomposable projectives are isomorphic to $P(\lambda)$ for some λ (see [Hum08, §3.9] or [CR90, §6C]). By projectivity of $P(\lambda)$, the proof of Theorem 2.39 can be mimicked to yield epimorphisms

$$P(\lambda) \twoheadrightarrow \Delta(\lambda) \twoheadrightarrow L(\lambda), \quad (2.2)$$

so $P(\lambda)$ is also a projective cover of $\Delta(\lambda)$.

Example 2.42. As the linkage class of $-\rho$ contains only itself, here are no proper submodules of $\Delta(-\rho)$ by Theorem 2.14, so $\Delta(-\rho) \simeq L(-\rho)$. As $-\rho$ is dominant, $P(-\rho) \simeq \Delta(-\rho) \simeq L(-\rho)$ by Theorem 2.37.

Although we have provided some motivation for studying Verma modules, their construction can still seem arbitrary and lack an obvious categorical interpretation. However, (2.2) places them between simples and projectives. The following remarkable theorem makes this intermediate role precise, which also justifies the Verma-like axioms of $\mathcal{O}$.

Theorem 2.43 ([Hum08, Theorem 3.10], BGG Reciprocity [BGG76]). *Each projective module in $\mathcal{O}$ has a standard filtration. For $\lambda, \mu \in \mathfrak{h}^*$,*

$$(P(\lambda) : \Delta(\mu)) = [\Delta(\mu) : L(\lambda)] = [\nabla(\mu) : L(\lambda)].$$

Since any summand of a projective is itself projective, it is natural to ask which indecomposable projectives appear in its decomposition. The next lemma records a basic structural fact about projective modules.

Lemma 2.44 ([Hum08, Theorem 3.9(b)]). *Let $P \in \mathcal{O}$ be projective. The number of its indecomposable summands isomorphic to $P(\lambda)$ is $\dim \operatorname{Hom}_{\mathcal{O}}(P, L(\lambda))$.*

Proof. First note that $\dim \operatorname{End}_{\mathcal{O}} L(\lambda) = 1$ as all endomorphisms of a simple highest weight module correspond to scaling a highest weight vector. Write $P \simeq \bigoplus_{\mu} P(\mu)^{\oplus m_{\mu}}$. As $P(\mu)$ is the unique projective cover of $L(\lambda)$ (up to isomorphism), we have $\dim \operatorname{Hom}_{\mathcal{O}}(P(\mu), L(\lambda)) = 1$ if $\mu = \lambda$ and 0 otherwise. Therefore,

$$\operatorname{Hom}_{\mathcal{O}}(P, L(\lambda)) \simeq \bigoplus_{\mu} \operatorname{Hom}_{\mathcal{O}}(P(\mu)^{\oplus m_{\mu}}, L(\lambda)) \simeq \operatorname{Hom}_{\mathcal{O}}(P(\lambda), L(\lambda))^{\oplus m_{\lambda}} \simeq \mathbb{C}^{m_{\lambda}},$$

so $\dim \operatorname{Hom}_{\mathcal{O}}(P, L(\lambda)) = m_{\lambda}$. □

Let us now investigate the projective objects in $\mathcal{O}_0(\mathfrak{sl}_2)$.

Example 2.45. The dot-orbit of 0 is $\{0, 2\}$, so $L(0)$ and $L(-2)$ are the only simples in the principal block. As -2 is minimal, $\Delta(-2) \simeq L(-2)$.

Let $P(0), P(-2)$ be the indecomposable projective covers of $L(0), L(-2)$. There are surjections $P(0) \twoheadrightarrow L(0)$ and $P(-2) \twoheadrightarrow L(-2)$. The Verma $\Delta(0)$ has unique maximal submodule $\Delta(-2)$, giving the composition series

$$0 \xrightarrow{L(-2)} \Delta(-2) \xrightarrow{L(0)} \Delta(0),$$

so $[\Delta(0) : L(0)] = [\Delta(0) : L(-2)] = 1$. Theorem 2.43 gives $(P(0) : \Delta(0)) = (P(-2) : \Delta(0)) = 1$. As $[\Delta(-2) : L(-2)] = 1$, we also have $(P(-2) : \Delta(-2)) = 1$.

By Theorem 2.37, $\Delta(0)$ is projective so $P(0) \simeq \Delta(0)$. Now consider $P(-2)$. Since $(P(-2) : \Delta(-2)) = 1$ and $L(-2) \simeq \Delta(-2)$, the top of $P(-2)$ is $L(-2)$. As the multiplicities of $\Delta(0)$ and $\Delta(-2)$ in $P(-2)$ are both 1, we obtain the short exact sequence

$$0 \longrightarrow \Delta(0) \longrightarrow P(-2) \longrightarrow \Delta(-2) \longrightarrow 0.$$

We can thus read off the standard filtration for $P(-2)$ as

$$0 \xrightarrow{\Delta(0)} \Delta(0) \xrightarrow{\Delta(-2)} P(-2). \quad (2.3)$$

This implies the composition series

$$0 \xrightarrow{L(-2)} \Delta(-2) \xrightarrow{L(0)} \Delta(0) \xrightarrow{L(-2)} P(-2).$$

We have found five indecomposable objects of $\mathcal{O}_0(\mathfrak{sl}_2)$: $L(0)$, $L(-2) \simeq \Delta(-2)$, $\Delta(0) \simeq P(0)$, $\nabla(0)$, and $P(-2)$. In fact, these are the only indecomposable objects in $\mathcal{O}_0(\mathfrak{sl}_2)$ (see [Hum08, Proposition 3.12] for details).

Since $\mathcal{O}$ has enough projectives, every object in $\mathcal{O}$ has a projective resolution.

Theorem 2.46 ([Hum08, Theorem 6.9]). *All objects in $\mathcal{O}_0$ have projective resolutions with finite length.*

In any abelian category, if there are enough projectives and each object has a projective resolution of finite length, we have the following result.

Theorem 2.47. *There is an equivalence $D^b(\mathcal{O}_0) \simeq K^b(\mathcal{O}_0\text{-Proj})$.*

Proof. See [Wei94, Corollary 10.4.7, Theorem 10.4.8]. □

In the next chapter, we will define certain functors which do not see the braid group action at the abelian level, making it necessary to pass to the derived category.

Chapter 3

Functors Between Blocks

In many cases, the study of $\mathcal{O}$ can be reduced to the study of its block subcategories $\mathcal{O}_\lambda$. This naturally leads to an important question: when are two such subcategories equivalent? In this chapter, we introduce translation functors, which allows us to relate blocks with different central characters. Composing translations into and out of the principal block $\mathcal{O}_0$ yields wall-crossing functors, and their (co)unit give rise to (co)shuffling functors from which the braid action ultimately arises. Many properties follow from the adjunctions between the underlying translations.

This chapter is based on [Hum08, §7], [EMTW20, §15.2], and [MS05, §5.1]. Throughout this chapter, all weights considered will be integral. The results hold also for non-integral weights but requires extra technical details that would hinder the exposition (see [Hum08, §7.4] for the general case). We will compute the images of the indecomposable objects in $\mathcal{O}_0(\mathfrak{sl}_2)$ under these functors to illustrate these results.

3.1 Translation functors

For two integral weights $\lambda, \mu \in \Lambda$, we seek a functor between blocks $\mathcal{O}_\lambda$ and $\mathcal{O}_\mu$. Tensoring by a finite-dimensional $V \in \mathcal{O}$ preserves $\mathcal{O}$, but changes weight spaces and typically have summands over several blocks. Composing with the projection $\mathrm{pr}_\mu: \mathcal{O} \rightarrow \mathcal{O}_\mu$ remedies this and yields a functor $\mathrm{pr}_\mu(V \otimes -): \mathcal{O}_\lambda \rightarrow \mathcal{O}_\mu$.

If V is arbitrary, one cannot expect this construction to yield useful information (see Example 3.3). The idea of choosing V is as follows. If $M \in \mathcal{O}_\lambda$ has highest weight λ , we want V of highest weight $\mu - \lambda$ so that $V \otimes M$ has highest weight μ . Choosing $w \in W$ such that $\nu := w(\mu - \lambda)$ is dominant makes $V = L(\nu)$

finite-dimensional. Then $L(\nu) \otimes -$ best captures the desired shift from λ to μ . This leads to the following definition.

Definition 3.1. Let $\lambda, \mu \in \Lambda$ and ν be as above. The *translation functor* $T_\lambda^\mu: \mathcal{O}_\lambda \rightarrow \mathcal{O}_\mu$ is the composition

$$T_\lambda^\mu := \text{pr}_\mu (L(\nu) \otimes -),$$

where $\text{pr}_\mu: \mathcal{O} \rightarrow \mathcal{O}_\mu$ denotes the projection on the respective summand of the decomposition in Theorem 2.35.

Remark 3.2. The translation functors are examples of projective functors, first defined in [BG80]. We can consider the category of endofunctors of $U(\mathfrak{g})\text{-mod}$ (or its subcategories), where objects are functors and morphisms are natural transformations, so it makes sense to define direct sums (and hence direct summands) of functors formed pointwise. A *projective functor* is a direct summand of the functor $V \otimes_{\mathbb{C}} -$ for a finite-dimensional V . Projective endofunctors on $\mathcal{O}_\lambda$ will correspond to Soergel bimodules over the coinvariant algebra under the $\mathbb{V}$ functor, all to be defined in Chapters 4 and 6. We focus on translation functors, and pass through Soergel bimodules and their diagrammatics rather than developing projective functors abstractly, as this connection provides powerful computational tools that make the algebraic picture more concrete.

The translation functor can also be considered an endofunctor of $\mathcal{O}$ by composing with the projection and inclusion functors $\text{pr}_\lambda: \mathcal{O} \rightarrow \mathcal{O}_\lambda$ and $\iota_\mu: \mathcal{O}_\mu \rightarrow \mathcal{O}$ appropriately. The next two simple examples illustrate that even when λ and μ lie in the same block, there can be multiple translation endofunctors.

Example 3.3. When $\mu = \lambda$, we have $T_\lambda^\lambda M \simeq M$. Here $L(0)$ is the 1-dimensional module with trivial $\mathfrak{g}$ -action, which serves as the monoidal unit in the tensor category of $U(\mathfrak{g})$ -modules. Hence $T_\lambda^\lambda = \mathbb{1}_{\mathcal{O}_\lambda}$.

Example 3.4. When λ and μ are two different representatives of the same linkage class, T_λ^μ is an endofunctor of $\mathcal{O}_\lambda = \mathcal{O}_\mu$. For simplicity, assume that μ is also dominant and $\lambda := w_\circ \cdot \mu$, where $w_\circ$ is the longest element of W . Then $\mu - \lambda$ is also dominant, which means that $T_\lambda^\mu = \text{pr}_\mu (L(\mu - \lambda) \otimes -)$. Hence T_λ^μ is a non-trivial endofunctor of the block.

Several properties of translation functors follow readily from their definition. In particular, translation preserves projective objects. This is important because, at the abelian level, the braid group action appears only on projective objects.

Proposition 3.5 ([Hum08, Proposition 7.1]). *The translation functor $T_\lambda^\mu = \text{pr}_\mu(L(\nu) \otimes -)$ is exact, commutes with taking duals, and sends projectives to projectives.*

Proof. As $L(\nu)$ is finite-dimensional, tensoring with $L(\nu)$ is exact by Proposition 2.18. Projection to the $\mathcal{O}_\mu$ block is also exact as an exact sequence of $U(\mathfrak{g})$ -modules preserves weight spaces, so T_λ^μ is an exact functor.

Duality clearly commutes with block projection, so it suffices to show that $L(\nu) \otimes -$ and $(-)^\vee$ commute. We show that $(L(\nu) \otimes M)^\vee \simeq L(\nu)^\vee \otimes M^\vee$, from where the claim follows as $L(\nu)^\vee \simeq L(\nu)$ by Theorem 2.25.

As $\mathbb{C}$ -vector spaces, we have isomorphisms

$$\begin{aligned} L(\nu)^\vee \otimes M^\vee &= \text{Hom}(L(\nu), \mathbb{C}) \otimes \left(\bigoplus_{\lambda \in \mathfrak{h}^*} \text{Hom}(M_\lambda, \mathbb{C}) \right) \simeq \bigoplus_{\lambda \in \mathfrak{h}^*} \text{Hom}(L(\nu) \otimes M_\lambda, \mathbb{C}) \\ &\simeq \bigoplus_{\mu, \lambda \in \mathfrak{h}^*} \text{Hom}(L(\nu)_\mu \otimes M_\lambda, \mathbb{C}) \simeq \bigoplus_{\lambda \in \mathfrak{h}^*} \text{Hom}((L(\nu) \otimes M)_\lambda, \mathbb{C}) \\ &= (L(\nu) \otimes M)^\vee, \end{aligned}$$

where we used the fact that $L(\nu)$ is finite-dimensional and tensor products distribute over direct sums. Acting by $x \in \mathfrak{g}$ on $f \otimes g$ yields

$$x \cdot (f(u) \otimes g(v)) = f(\tau(x) \cdot u) \otimes g(v) + f(u) \otimes g(\tau(x) \cdot v),$$

which matches the action of $\mathfrak{g}$ on both $L(\nu)^\vee \otimes M^\vee$ and $(L(\nu) \otimes M)^\vee$. Therefore, $L(\nu)^\vee \otimes M^\vee \simeq (L(\nu) \otimes M)^\vee$ as $\mathfrak{g}$ -modules.

The last statement follows from Proposition 2.38 and projecting to a direct summand of the central character decomposition being an exact functor. $\square$

We also have an adjunction property of translation functors.

Lemma 3.6 ([Hum08, Proposition 7.1]). *The pair T_λ^μ and T_μ^λ are biadjoint.*

Proof. It suffices to prove that $\text{Hom}_{\mathcal{O}_\mu}(T_\lambda^\mu M, N) \simeq \text{Hom}_{\mathcal{O}_\lambda}(M, T_\mu^\lambda N)$ by symmetry. Write $\nu = w(\mu - \lambda)$. Then $L(\nu)^* \simeq L(-w_\circ \nu)$ (see Section 1.2). Note that $-w_\circ \nu$ is still dominant integral, and that $-w_\circ \nu = -w_\circ w(\mu - \lambda) = w_\circ w(\lambda - \mu) \in W(\lambda - \mu)$. Therefore, the functors $L(\nu) \otimes -$ and $L(-w_\circ \nu) \otimes -$ are biadjoint by Proposition 2.19. Furthermore, there is an adjunction $\text{pr}_\lambda \dashv \iota_\lambda$, where $\iota_\lambda: \mathcal{O}_\lambda \rightarrow \mathcal{O}$ is the inclusion of $\mathcal{O}_\lambda$ into $\mathcal{O}$, as $\text{Hom}_{\mathcal{O}_\lambda}(\text{pr}_\lambda M, N) \simeq \text{Hom}_{\mathcal{O}}(M, \iota_\lambda N)$ due to the fact that there are no non-trivial morphisms between objects in different blocks. This proves the claim. $\square$

Let us look at the effect of translation functors on Verma modules. Recall from Theorem 2.20 that $L(\nu) \otimes \Delta(\lambda)$ has a filtration whose subquotients are isomorphic to $\Delta(\lambda + \nu')$ where ν' runs over the weights of $L(\nu)$. Applying the projection functor pr_μ to this filtration kills Verma modules whose highest weight is not linked to μ . This yields the following corollary.

Corollary 3.7. *The module $T_\lambda^\mu(\Delta(\lambda))$ admits a standard filtration with successive subquotients isomorphic to $\Delta(\lambda + \nu')$, where ν' runs over the weights of $L(\nu)$ such that $\lambda + \nu' \in W \cdot \mu$ with multiplicity $\dim L(\nu)_{\nu'}$.*

The following lemma describes the objects that appear in the standard filtration of a Verma module after such a translation.

Lemma 3.8. *Assume that μ is maximal in its dot-orbit and that $\lambda - \mu$ is dominant integral. Then for any $w \in W$, $T_\mu^\lambda(\Delta(w \cdot \mu))$ admits a filtration with successive subquotients isomorphic to $\Delta(w w' \cdot \lambda)$, where w' runs over the stabiliser of μ under the dot-action, with each w' appearing once.*

Proof. By Corollary 3.7, we want to find weights ν' of $L(\lambda - \mu)$ such that

$$w \cdot \mu + \nu' = x \cdot \lambda \quad (3.1)$$

for some $x \in W$. This can be rewritten as $x^{-1}w(\mu + \rho) + x^{-1}\nu' = \lambda + \rho$. By [Hum80, Theorem 21.2], $x^{-1}\nu'$ is a weight of $L(\lambda - \mu)$. Since $\lambda - \mu$ is the highest weight of $L(\lambda - \mu)$, we have $\lambda - \mu \geq x^{-1}\nu'$. This yields the inequality

$$\lambda + \rho = x^{-1}w(\mu + \rho) + x^{-1}\nu' \leq x^{-1}w(\mu + \rho) + \lambda - \mu,$$

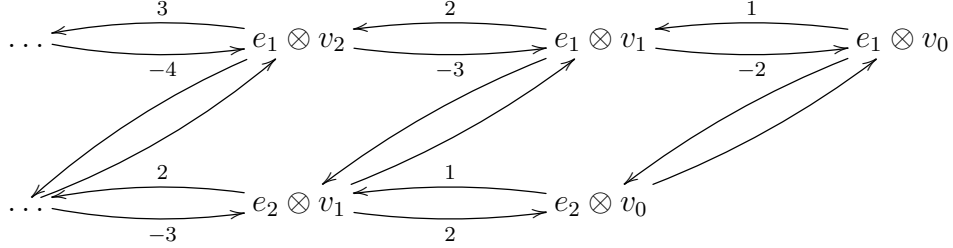
so $\mu \leq x^{-1}w(\mu + \rho) - \rho = x^{-1}w \cdot \mu$, implying that $x \cdot \mu \leq w \cdot \mu$. However, since μ is maximal in its dot-orbit, we must have the equality $x \cdot \mu = w \cdot \mu$. Substituting into (3.1) gives $\lambda - \mu = x^{-1}\nu'$, which implies that $\nu' = x(\lambda - \mu)$. Again, [Hum80, Theorem 21.2] implies that $\dim L(\lambda - \mu)_{\nu'} = 1$. Therefore, $x = ww'$ for some w' fixing μ and $\Delta(ww' \cdot \lambda)$ occurs as a subquotient exactly once. $\square$

These results allows us to determine how filtrations of Verma modules under translation split.

Example 3.9. We compute the images of $L(0)$ and $L(-2)$ under T_0^{-1} . The dominant weight ν in the linear W -orbit of $-1 - 0 = -1$ is $s(-1) = 1$. Thus $T_0^{-1} = \text{pr}_{-1}(L(1) \otimes -)$.

For $L(0)$, we have $T_0^{-1}L(0) = \text{pr}_{-1}(L(1) \otimes L(0))$. By Example 3.3, $L(1) \otimes L(0) \simeq L(1)$ and projecting to $\mathcal{O}_{-1}$ gives 0 as $1 \notin W \cdot (-1)$.

For $L(-2) \simeq \Delta(-2)$, we have $T_0^{-1}L(-2) = \text{pr}_{-1}(L(1) \otimes L(-2))$. As a $U(\mathfrak{sl}_2)$ -module, $L(1)$ has basis $\{e_1, e_2\}$ with highest weight 1. Using the same weight basis $\{v_i\}_{i \geq 0}$ for $L(-2)$ as in Example 2.12, a weight basis for $L(1) \otimes L(-2)$ is $\{e_i \otimes v_j\}_{i \in \{1,2\}, j \geq 0}$. We can visualise the weight spaces of $L(1) \otimes L(-2)$ as follows:



The arrows represent f (lowering) and e (raising) actions, and numbers denote the corresponding scalars with blank arrows meaning 1. The highest-weight vector $e_1 \otimes v_0$ has weight -1 , and weights decrease in steps of 2.

The top row is isomorphic to $\Delta(-1)$, generated by $e_1 \otimes v_0$. By Corollary 3.7, $T_0^{-1}L(-2)$ has $\Delta(-1) \simeq L(-1)$ and $\Delta(-3) \simeq L(-3)$ as subquotients in its standard filtration, so we get the short exact sequence

$$0 \longrightarrow \Delta(-1) \longrightarrow L(0) \otimes \Delta(-2) \longrightarrow \Delta(-3) \longrightarrow 0.$$

We expect that this short exact sequence splits since $\Delta(-1)$ and $\Delta(-3)$ are contained entirely within different blocks. Indeed, the vector $e_1 \otimes v_1 + 2e_2 \otimes v_0$ spans the space of all highest weight vectors of weight -3 , and is not a multiple of $f \cdot (e_1 \otimes v_0) = e_2 \otimes v_0 + e_1 \otimes v_1$. Thus there is a map $\Delta(-3) \hookrightarrow L(0) \otimes \Delta(-2)$ which splits the sequence above. Then $T_0^{-1}L(-2) \simeq L(-1)$ as the subquotient $\Delta(-3)$ is killed when projecting to the block $\mathcal{O}_{-1}$.

Example 3.10. Translating the other way, we compute $T_{-1}^0 L(-1) = \text{pr}_0(L(1) \otimes L(-1))$. Similar to the previous example, $L(1) \otimes L(-1)$ fits into the sequence

$$0 \longrightarrow \Delta(0) \longrightarrow L(1) \otimes L(-1) \longrightarrow \Delta(-2) \longrightarrow 0.$$

Both $\Delta(0)$ and $\Delta(-2)$ appear in the standard filtration of $L(1) \otimes L(-1)$ as $0, -2 \in W \cdot 0$, so $T_{-1}^0 L(-1)$ fits into the same sequence.

As translation preserves projectives and $L(-1)$ is projective in $\mathcal{O}_{-1}$, $T_{-1}^0 L(-1)$ is projective in $\mathcal{O}_0$. Comparing the standard filtration of $T_{-1}^0 L(-1)$ with indecomposable projectives in $\mathcal{O}_0$ (see Example 2.45) means that $T_{-1}^0 L(-1) \simeq P(-2)$.

3.2 Properties of translation functors

We now study how the translation functors T_λ^μ act within Weyl chambers (see Section 1.2). It is helpful to picture $\mathcal{O}_\lambda$ as “living over” the weight $\lambda \in \mathfrak{h}^*$. For $\mathfrak{sl}_2$, Example 3.9 illustrates translation *to* the s -wall and Example 3.10 translation *from* the s -wall. This perspective makes clear the difference between translations involving regular weights in the interior of a chamber and those involving weights on a wall. Below we consider two cases that exhibit these different behaviours. Throughout this section, fix a chamber C (see Section 1.2).

3.2.1 Translation to chamber closure

We continue to examine how translation functors act on Verma modules. The theorem below shows that translation within a chamber preserves Verma modules if the starting weight is regular. We first record a useful lemma, whose proof we omit.

Lemma 3.11 ([Hum08, Lemma 7.5]). *Suppose $\lambda \in C$ and $\mu \in \overline{C}$. Let $\gamma = \mu - \lambda$. Then for all weights $\nu' \neq \gamma$ of $L(\nu)$, the weight $\lambda + \nu'$ is not linked by W to $\lambda + \gamma = \mu$.*

Theorem 3.12 ([Hum08, Theorem 7.6]). *Let $\lambda \in C$ and $\mu \in \overline{C}$. Then $T_\lambda^\mu \Delta(w \cdot \lambda) \simeq \Delta(w \cdot \mu)$ for all $w \in W$. Furthermore, $T_\lambda^\mu \nabla(w \cdot \lambda) \simeq \nabla(w \cdot \mu)$.*

Proof. By Corollary 3.7, $T_\lambda^\mu \Delta(w \cdot \lambda)$ admits a standard filtration with subquotients $\Delta(w \cdot \lambda + \nu')$, where ν' appears with multiplicity $\dim L(\nu)_{\nu'}$. Since $\Delta(w \cdot \mu) = \Delta(w \cdot \lambda + (w \cdot \mu - w \cdot \lambda))$ and $w \cdot \mu - w \cdot \lambda = w(\mu - \lambda)$, the relevant shift is $w(\mu - \lambda)$, which lies in the W -orbit of ν . The Weyl group preserves the dimension of weight spaces (see [Hum80, Theorem 21.2]), so $\dim L(\nu)_{w(\mu - \lambda)} = \dim L(\nu)_\nu = 1$. By Lemma 3.11, no other ν' occurs, so the filtration is a single term and $T_\lambda^\mu \Delta(w \cdot \lambda) \simeq \Delta(w \cdot \mu)$. By Proposition 3.5, translation preserves duality. $\square$

Verma modules are also preserved under translation if λ and μ lie on the same wall (see [Hum08, Theorem 7.6]). However, the same cannot be said if λ is singular while μ is regular, evident in Example 3.10.

A bit more care is required when translating simple modules.

Theorem 3.13 ([Hum08, Proposition 7.7]). *Let $\lambda \in C$ and $\mu \in \overline{C}$. Then*

$$T_\lambda^\mu L(w \cdot \lambda) \simeq \begin{cases} L(w \cdot \mu) & \text{if } w \cdot \mu \in \widehat{w \cdot C} \\ 0 & \text{otherwise.} \end{cases}$$

Proof. First, $T_\lambda^\mu L(w \cdot \lambda)$ is either $L(w \cdot \mu)$ or zero. Indeed from the maps

$$\Delta(w \cdot \lambda) \twoheadrightarrow L(w \cdot \lambda) \hookrightarrow \nabla(w \cdot \lambda) \quad (3.2)$$

in Theorem 2.25, exactness of T_λ^μ gives $\Delta(w \cdot \mu) \twoheadrightarrow T_\lambda^\mu L(w \cdot \lambda) \hookrightarrow \nabla(w \cdot \mu)$. By Proposition 2.9, $T_\lambda^\mu L(w \cdot \lambda)$ is a highest weight module of either weight $w \cdot \mu$ or is zero. If non-zero, the injection implies that $T_\lambda^\mu L(w \cdot \lambda) \simeq L(w \cdot \mu)$ by Theorem 2.25.

From the maps (3.2), some composition factor $L(w'w \cdot \lambda)$ of $\Delta(w \cdot \lambda)$ maps to $L(w \cdot \mu)$ under T_λ^μ . Exactness forces this factor to be unique, and we must have $T_\lambda^\mu L(w'w \cdot \lambda) \simeq L(w'w \cdot \mu) \simeq L(w \cdot \mu)$ if $T_\lambda^\mu L(w \cdot \lambda) \neq 0$. Hence $w'w \cdot \mu = w \cdot \mu$, so w' lies in the subgroup $W' := \langle s_\alpha \mid \alpha \in \Phi_\mu \rangle$, where $\Phi_\mu := \{\alpha \in \Phi \mid s_\alpha \cdot (w \cdot \mu) = w \cdot \mu\}$.

If $w \cdot \mu \in \widehat{w \cdot C}$, we claim that $w'w \cdot \lambda = w \cdot \lambda$. It suffices to show that $s_\alpha w \cdot \lambda \geq w \cdot \lambda$ for all $\alpha \in \Phi_\mu$ as the opposite inequality was established in the previous paragraph. For such α , $\langle w \cdot \mu + \rho, \alpha^\vee \rangle = \langle w(\mu + \rho), \alpha^\vee \rangle = 0$ as μ lies on H_α . Since $w \cdot \mu \in \widehat{w \cdot C}$, it follows that $\langle w \cdot \lambda + \rho, \alpha^\vee \rangle = \langle w(\lambda + \rho), \alpha^\vee \rangle \leq 0$, hence $s_\alpha w \cdot \lambda \geq w \cdot \lambda$. By Lemma 1.2, $w'w \cdot \lambda \geq w \cdot \lambda$ for all $w' \in W'$, so $L(w \cdot \lambda)$ is the composition factor mapping to $L(w \cdot \mu)$. Thus $T_\lambda^\mu L(w \cdot \lambda) \simeq L(w \cdot \mu)$.

Conversely, if $w \cdot \mu \notin \widehat{w \cdot C}$, there exists a simple reflection s_α with $s_\alpha w \cdot \lambda \leq w \cdot \lambda$. By Theorem 2.14, there is an embedding $\Delta(s_\alpha w \cdot \lambda) \subset \Delta(w \cdot \lambda)$. Since $s_\alpha w \cdot \mu = w \cdot \mu$, both $\Delta(s_\alpha w \cdot \lambda)$ and $\Delta(w \cdot \lambda)$ are mapped by T_λ^μ into $\Delta(w \cdot \mu)$. Therefore, T_λ^μ maps the quotient $Q := \Delta(w \cdot \lambda) / \Delta(s_\alpha w \cdot \lambda)$ to zero, so the quotient $T_\lambda^\mu L(w \cdot \lambda)$ of $T_\lambda^\mu Q$ is zero. $\square$

3.2.2 Translation from chamber closure

Now let us see how projectives act under translation.

Theorem 3.14 ([Hum08, Theorem 7.11]). *Let $\lambda \in C$ and $\mu \in \overline{C}$. If $w \cdot \mu \in \widehat{w \cdot C}$ for some $w \in W$, then $T_\mu^\lambda P(w \cdot \mu) \simeq P(w \cdot \lambda)$.*

Proof. By Proposition 3.5, $T_\mu^\lambda P(w \cdot \mu)$ is projective. For any $w' \in W$, Theorem 3.13 gives

$$\begin{aligned} \operatorname{Hom}_{\mathcal{O}}(T_\mu^\lambda P(w \cdot \mu), L(w' \cdot \lambda)) &\simeq \operatorname{Hom}_{\mathcal{O}}(P(w \cdot \mu), T_\lambda^\mu L(w' \cdot \lambda)) \quad (\text{Lemma 3.6}) \\ &\simeq \operatorname{Hom}_{\mathcal{O}}(P(w \cdot \mu), L(w' \cdot \mu)) \end{aligned}$$

when $w \cdot \mu \in \widehat{w \cdot C}$. As $P(w \cdot \mu)$ is the projective cover of $L(w \cdot \mu)$,

$$\dim \operatorname{Hom}_{\mathcal{O}}(P(w \cdot \mu), L(w' \cdot \mu)) = \begin{cases} 1 & w' = w \\ 0 & w' \neq w, \end{cases}$$

so $\dim \operatorname{Hom}_{\mathcal{O}}(T_{\mu}^{\lambda}P(w \cdot \mu), L(w' \cdot \lambda)) = 1$ when $w' = w$ and 0 else. Thus, $P(w \cdot \lambda)$ appears exactly once as a summand of $T_{\mu}^{\lambda}P(w \cdot \mu)$ by Lemma 2.44, and no other summand can occur. Therefore, $T_{\mu}^{\lambda}P(w \cdot \mu) \simeq P(w \cdot \lambda)$. $\square$

3.2.3 Category equivalences

As a corollary of Theorem 3.12 and Theorem 3.13, we have an equivalence of categories between $\mathcal{O}_{\lambda}$ and $\mathcal{O}_{\mu}$ whenever λ and μ are in the same chamber. We first prove that $\mathcal{O}_{\lambda}$ and $\mathcal{O}_{\mu}$ are isomorphic on the level of Grothendieck groups.

Proposition 3.15 ([Hum08, Proposition 7.8]). *If $\lambda, \mu \in C$, then T_{λ}^{μ} and T_{μ}^{λ} induce inverse isomorphisms $K(\mathcal{O}_{\lambda}) \xrightarrow{\sim} K(\mathcal{O}_{\mu})$, sending $[\Delta(w \cdot \lambda)] \mapsto [\Delta(w \cdot \mu)]$ and $[L(w \cdot \lambda)] \mapsto [L(w \cdot \mu)]$.*

Proof. By Proposition 2.16, $\{[\Delta(w \cdot \lambda)] \mid w \in W\}$ and $\{[\Delta(w \cdot \mu)] \mid w \in W\}$ form $\mathbb{Z}$ -bases for $K(\mathcal{O}_{\lambda})$ and $K(\mathcal{O}_{\mu})$ respectively. By Theorem 3.12, $T_{\lambda}^{\mu}\Delta(w \cdot \lambda) \simeq \Delta(w \cdot \mu)$ and $T_{\mu}^{\lambda}\Delta(w \cdot \mu) \simeq \Delta(w \cdot \lambda)$. For an arbitrary $M \in \mathcal{O}_{\lambda}$, $[M]$ is written uniquely as a $\mathbb{Z}$ -linear combination of $[\Delta(w \cdot \lambda)]$. Therefore, the map induced by $T_{\mu}^{\lambda}T_{\lambda}^{\mu}$ sends $[M]$ to itself, and similarly for the maps in reverse order on $K(\mathcal{O}_{\mu})$. Hence T_{μ}^{λ} and T_{λ}^{μ} induce mutually inverse isomorphisms on Grothendieck groups.

For simples, Theorem 3.13 implies $T_{\mu}^{\lambda}T_{\lambda}^{\mu}$ sends $[L(w \cdot \lambda)]$ to itself or 0. The latter is impossible by the above isomorphism of Grothendieck groups. Theorem 3.13 then forces $T_{\lambda}^{\mu}L(w \cdot \lambda) \simeq L(w \cdot \mu)$, so T_{λ}^{μ} sends $[L(w \cdot \lambda)]$ to $[L(w \cdot \mu)]$. $\square$

In fact, in an artinian abelian category, a pair of mutually adjoint functors that induce isomorphisms on the Grothendieck group is enough to yield an equivalence of categories (see [Gai, Lemma 4.27]). We will prove this directly for $\mathcal{O}$. The argument is easily generalisable to an arbitrary finite-length abelian category.

Theorem 3.16 ([Hum08, Theorem 7.8]). *If $\mu, \lambda \in C$, then T_{λ}^{μ} and T_{μ}^{λ} are mutually inverse equivalences of categories $\mathcal{O}_{\lambda}$ and $\mathcal{O}_{\mu}$.*

Proof. We wish to show that $T_{\mu}^{\lambda}T_{\lambda}^{\mu}M \cong M$ (with the other order being similar). By Lemma 3.6, we have $\operatorname{Hom}_{\mathcal{O}_{\lambda}}(T_{\mu}^{\lambda}T_{\lambda}^{\mu}M, M) \cong \operatorname{Hom}_{\mathcal{O}_{\mu}}(T_{\lambda}^{\mu}M, T_{\lambda}^{\mu}M)$. Denote by $\varphi_M: T_{\mu}^{\lambda}T_{\lambda}^{\mu}M \rightarrow M$ the map corresponding to identity map $\operatorname{id}_{T_{\lambda}^{\mu}M}$ on the right. We will proceed by induction on the length of M in $\mathcal{O}_{\lambda}$ to show that φ_M is an isomorphism. For all short exact sequences

$$0 \longrightarrow N \longrightarrow M \longrightarrow L \longrightarrow 0$$

where L is simple, the sequence

$$0 \longrightarrow T_\mu^\lambda T_\lambda^\mu N \longrightarrow T_\mu^\lambda T_\lambda^\mu M \longrightarrow T_\mu^\lambda T_\lambda^\mu L \longrightarrow 0$$

is exact. The diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & T_\mu^\lambda T_\lambda^\mu N & \longrightarrow & T_\mu^\lambda T_\lambda^\mu M & \longrightarrow & T_\mu^\lambda T_\lambda^\mu L \longrightarrow 0 \\ & & \downarrow \varepsilon_N & & \downarrow \varepsilon_M & & \downarrow \varepsilon_L \\ 0 & \longrightarrow & N & \longrightarrow & M & \longrightarrow & L \longrightarrow 0 \end{array}$$

commutes by naturality. For the base case where M has length 1, Proposition 3.15 gives that ε_M is an isomorphism. Similarly, ε is an isomorphism and ε_N is also isomorphism by induction, so ε_M is an isomorphism by the five lemma. $\square$

Thanks to the previous theorem, to study a block $\mathcal{O}_\lambda$ for λ integral, it suffices to study $\mathcal{O}_0$ and its image under various translation functors.

3.3 Wall-crossing functors

A natural thing to consider now is the composition of different translation functors. Let C be the dominant Weyl chamber. Fix $\lambda \in C$ and let μ lie solely on the s -wall in $\overline{C}$, where s is a simple reflection. Denote by $T_{\text{on}} := T_\lambda^\mu$ the *translation onto the s -wall*, and by $T_{\text{off}} := T_\mu^\lambda$ the *translation from the s -wall*.

Definition 3.17. The functor $\Theta_s := T_{\text{off}} T_{\text{on}}$ is called a *wall-crossing functor*.

In the next example, we calculate the effect of Θ_s on the indecomposable objects of $\mathcal{O}_0(\mathfrak{sl}_2)$. Note that by Proposition 3.5, Θ_s preserves projectives, although not necessarily indecomposable projectives. This phenomena will appear in Example 3.24.

Example 3.18. There is a single wall at the weight $-\rho = -1$. Call this the s -wall. By Example 2.42, we have $P(-1) \simeq \Delta(-1) \simeq L(-1)$.

Write $T_{\text{on}} = T_0^{-1}$ and $T_{\text{off}} = T_{-1}^0$. For $\Delta(0)$,

$$\Theta_s \Delta(0) \simeq T_{\text{off}} \Delta(-1) \simeq T_{\text{off}} P(-1)$$

by Theorem 3.12. Using Theorem 3.14,

$$T_{\text{off}} P(-1) = T_{\text{off}} P(s \cdot (-1)) \simeq P(s \cdot 0) = P(-2),$$

so $\Theta_s \Delta(0) \simeq P(-2)$.

Similarly, $\Theta_s \Delta(-2) \simeq P(-2) \simeq \text{Sh}_s \nabla(0)$. The simple $L(0)$ is sent to 0 by Theorem 3.13 since $-1 \notin \widehat{C}$.

We postpone the computation of $\Theta_s P(-2)$ until Example 3.24.

As T_{on} and T_{off} are biadjoint, there are canonical natural isomorphisms

$$\text{Hom}_{\mathcal{O}_\lambda}(M, \Theta_s M) \simeq \text{Hom}_{\mathcal{O}_\mu}(T_{\text{on}} M, T_{\text{on}} M) \simeq \text{Hom}_{\mathcal{O}_\lambda}(\Theta_s M, M) \quad (3.3)$$

for each $M \in \mathcal{O}_\lambda$. Under the first isomorphism, the identity $\text{id}_{T_{\text{on}} M}$ corresponds to a map $\eta_M: M \rightarrow \Theta_s M$; under the second it corresponds to a map $\varepsilon_M: \Theta_s M \rightarrow M$. These maps assemble into natural transformations $\eta: \mathbb{1}_{\mathcal{O}_\lambda} \Rightarrow \Theta_s$ and $\varepsilon: \Theta_s \Rightarrow \mathbb{1}_{\mathcal{O}_\lambda}$, called the *unit* and *counit* of adjunction. By naturality, the components η_M and ε_M are functorial in M . (Compare with the proof of Theorem 3.16.)

In general, one cannot expect η_M and ε_M to be isomorphisms. For instance, η_M cannot be injective when $T_{\text{on}} M = 0$ but $M \neq 0$ (see Theorem 3.13). However, a useful special case follows immediately from (3.3).

Lemma 3.19. *Suppose $\dim \text{End}_{\mathcal{O}} T_\lambda^\mu M = 1$. Then η_M and ε_M are unique and non-zero.*

The following lemma uses the previous to show the effect of Θ_s on Verma modules.

Lemma 3.20. *For any $w \in W$ and $s \in S$, we have non-split short exact sequences*

$$\begin{aligned} 0 \longrightarrow \Delta(w \cdot 0) \longrightarrow \Theta_s \Delta(w \cdot 0) \longrightarrow \Delta(ws \cdot 0) \longrightarrow 0 & \quad \text{if } \ell(ws) > \ell(w), \\ 0 \longrightarrow \Delta(ws \cdot 0) \longrightarrow \Theta_s \Delta(w \cdot 0) \longrightarrow \Delta(w \cdot 0) \longrightarrow 0 & \quad \text{if } \ell(ws) < \ell(w). \end{aligned}$$

Proof. Suppose $\ell(ws) > \ell(w)$ and assume μ lies on the s -wall. Consider $\Theta_s \Delta(w \cdot 0)$. By Theorem 3.12, $\Theta_s \Delta(w \cdot 0) \simeq T_\mu^0 \Delta(w \cdot \mu)$. Since μ is on the s -wall, μ is fixed by s and e , so Lemma 3.8 tells that $\Delta(w \cdot 0)$ and $\Delta(w' \cdot 0)$ appear as subquotients in the standard filtration of $T_\mu^0(w \cdot \mu)$. As $\dim \text{End}_{\mathcal{O}} \Delta(w \cdot \mu) = 1$, it follows from Lemma 3.19 that $\Delta(w \cdot 0) \rightarrow \Theta_s \Delta(w \cdot 0)$ is a non-zero homomorphism. We therefore get the short exact sequence

$$0 \longrightarrow \Delta(w \cdot 0) \longrightarrow \Theta_s \Delta(w \cdot 0) \longrightarrow \Delta(ws \cdot 0) \longrightarrow 0.$$

The proof is analogous if $\ell(ws) < \ell(w)$. □

Wall-crossing functors play an important role in the braid group action on projective objects of $\mathcal{O}$. For an expression $\underline{w} = (s_1, s_2, \dots, s_k)$ where $s_i \in S$, set $P_{\underline{w}} := \Theta_{s_k} \cdots \Theta_{s_2} \Theta_{s_1} P(0)$.

Proposition 3.21. *For any $w \in W$, the standard filtration of $P(w \cdot 0)$ contains $\Delta(w \cdot 0)$ exactly once, and all other subquotient are of the form $\Delta(w' \cdot 0)$ with $w' \leq w$. Furthermore, if $\underline{w}$ is a reduced expression for w , then*

$$P_{\underline{w}} \simeq P(w \cdot 0) \oplus \bigoplus_{w' < w} P(w' \cdot 0)^{m_{w'}}$$

for certain multiplicities $m_{w'}$.

As a consequence $P(w \cdot 0)$ is the unique indecomposable direct summand of $P_{\underline{w}}$ not appearing as a direct summand of $P_{\underline{w'}}$ for any expression $\underline{w'}$ with $w' < w$.

3.4 (Co)shuffling functors

Although η_M and ε_M cannot be expected to be isomorphisms, there is always the cokernel and kernel to consider. We give the following definition which was first introduced by Carlin [Car86] and studied by Irving [Irv93] and Mazorchuk–Stroppel [MS05].

Definition 3.22. For $M \in \mathcal{O}_\lambda$, define the endofunctors on $\mathcal{O}_\lambda$

$$\mathrm{Sh}_s M := \mathrm{coker}(\eta_M: M \rightarrow \Theta_s M), \quad \mathrm{CSh}_s M := \ker(\varepsilon_M: \Theta_s M \rightarrow M),$$

which we call the *shuffling* and *coshuffling functors* respectively.

By the snake lemma, Sh_s is right-exact and CSh_s is left-exact. The two functors are connected in the following way.

Lemma 3.23 ([MS05, Lemma 5.1]). *Let s be a simple reflection. Then Sh_s is left adjoint to CSh_s .*

Proof. Let $M, N \in \mathcal{O}$. Both $\mathrm{Hom}(-, N)$ and $\mathrm{Hom}(M, -)$ are left-exact, while $\mathrm{Hom}(-, N)$ is contravariant. Applying them to the exact sequences

$$M \xrightarrow{\eta_M} \Theta_s M \rightarrow \mathrm{Sh}_s M \rightarrow 0 \quad \text{and} \quad 0 \rightarrow \mathrm{CSh}_s N \rightarrow \Theta_s N \xrightarrow{\varepsilon_N} N$$

gives the following diagram with exact rows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Hom}(\mathrm{Sh}_s M, N) & \longrightarrow & \mathrm{Hom}(\Theta_s M, N) & \xrightarrow{-\circ \eta_M} & \mathrm{Hom}(M, N) \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & \mathrm{Hom}(M, \mathrm{CSh}_s N) & \longrightarrow & \mathrm{Hom}(M, \Theta_s N) & \xrightarrow{\varepsilon_N \circ -} & \mathrm{Hom}(M, N). \end{array}$$

The middle arrow is an isomorphism coming from the self-adjunction of Θ_s . It takes $f \in \text{Hom}(\Theta_s M, N)$ to $g \in \text{Hom}(M, \Theta_s N)$ such that $\varepsilon_N \circ g = f \circ \eta_M$, thus making the right square commute. By five lemma, the dotted arrow is an isomorphism, proving the adjunction. $\square$

Example 3.24. Continuing on from the setup of Example 3.18, let us determine the images of indecomposable objects in $\mathcal{O}_0(\mathfrak{sl}_2)$ under the shuffling functor.

$L(0)$: Since $T_{\text{on}} L(0) = 0$ calculated in Example 3.9, $\text{Sh}_s L(0) = 0$.

$\Delta(0) \simeq P(0)$: We have $\Theta_s \Delta(0) \simeq P(-2)$. From Example 2.45, the projective $P(-2)$ has a standard filtration $0 \xrightarrow{\Delta(0)} \Delta(0) \xrightarrow{\Delta(-2)} P(-2)$. Therefore $\text{Sh}_s \Delta(0) \simeq \Delta(-2)$.

$\Delta(-2) \simeq L(-2)$: We have $\Theta_s \Delta(-2) \simeq P(-2)$, so $\text{Sh}_s \Delta(-2)$ fits into the short exact sequence

$$0 \longrightarrow \Delta(-2) \longrightarrow P(-2) \longrightarrow \text{Sh}_s \Delta(-2) \longrightarrow 0. \quad (3.4)$$

The composition series of $P(-2)$ is

$$0 \xrightarrow{L(-2)} L(-2) \xrightarrow{L(0)} \Delta(0) \xrightarrow{L(-2)} P(-2),$$

so $\text{Sh}_s \Delta(-2) \simeq P(-2)/L(-2)$ has composition series

$$0 \xrightarrow{L(0)} \Delta(0)/L(-2) \simeq L(0) \xrightarrow{L(-2)} P(-2)/L(-2).$$

Thus $\text{Sh}_s \Delta(-2) \simeq \nabla(0)$ with unique maximal submodule $L(0)$ and simple quotient $L(-2)$.

$\nabla(0)$: There is only one non-zero homomorphism from $\nabla(0)$ to $\Theta_s \nabla(0) \simeq P(-2)$ coming from the counit of adjunction. The composition series of $\nabla(0)$ is

$$0 \xrightarrow{L(0)} L(0) \xrightarrow{L(-2)} \nabla(0).$$

In terms of composition factors, the top of $\nabla(0)$ which is $L(-2)$ must be sent to the bottom $L(-2)$ factor of $P(-2)$. Then the cokernel of this map has top composition factor $L(-2)$ and bottom $L(0)$, which is the composition series for $\nabla(0)$. Thus $\text{Sh}_s \nabla(0) \simeq \nabla(0)$.

$P(-2)$: First, let us resolve the image of $P(-2)$ under Θ_s and Sh_s . Taking the short exact sequence (3.4), we apply Sh_s to each term factoring through Θ_s to

get the commutative diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & \Delta(-2) & \longrightarrow & P(-2) & \longrightarrow & \nabla(0) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & P(-2) & \longrightarrow & \Theta_s P(-2) & \longrightarrow & P(-2) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & P(-2) & \longrightarrow & \text{Sh}_s P(-2) & \longrightarrow & \nabla(0) \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & 0 & & 0 & & 0
\end{array}$$

using the results from Example 3.18. The second row is split as $P(-2)$ is projective, so $\Theta_s P(-2) \simeq P(-2)^{\oplus 2}$. Thus $\text{Sh}_s P(-2) \simeq P(-2)$, again due to the fact that $P(-2) \rightarrow \Theta_s P(-2) \simeq P(-2)^{\oplus 2}$ is non-zero coming from the counit.

The images of CSh_s can be determined similarly, which we have included for completeness.

	Θ_s	Sh_s	CSh_s
$L(0)$	0	0	0
$\Delta(0) \simeq P(0)$	$P(-2)$	$L(-2)$	$P(0)$
$\Delta(-2) \simeq L(-2)$	$P(-2)$	$\nabla(0)$	$P(0)$
$\nabla(0)$	$P(-2)$	$\nabla(0)$	$L(-2)$
$P(-2)$	$P(-2)^{\oplus 2}$	$P(-2)$	$P(-2)$

Figure 3.1: Images of indecomposable objects of $\mathcal{O}_0(\mathfrak{sl}_2)$ under Θ_s , Sh_s , and CSh_s .

This completes the classification for $\mathcal{O}_0(\mathfrak{sl}_2)$.

3.5 Braid group action on $D^b(\mathcal{O}_0)$

We introduce the definition of the braid group.

Definition 3.25. The *braid group* associated to the Coxeter system (W, S) is the group with presentation

$$B_W := \langle s \in S \mid \underbrace{sts \dots}_{m_{st} \text{ terms}} = \underbrace{tst \dots}_{m_{st} \text{ terms}} \rangle.$$

Here, W is the Weyl group associated to $\mathfrak{g}$ and S are the simple reflections of W . Since B_W forgets the square relation, W is a quotient of B_W . Every element $w \in B_W$ can be expressed as

$$w = s_1^{\varepsilon_1} \cdots s_n^{\varepsilon_n},$$

where $s_i \in S$ are simple reflections and $\varepsilon_i \in \{\pm 1\}$. We call $\underline{w} := (s_1^{\varepsilon_1}, \dots, s_n^{\varepsilon_n})$ a *braid word* expressing the element $w \in B_W$.

A (weak) *categorical group action* of a group G on a category $\mathcal{C}$ is a family $(\mathcal{F}_w)_{w \in G}$ of endofunctors on $\mathcal{C}$ such that $\mathcal{F}_e \simeq \mathbb{1}_{\mathcal{C}}$, $\mathcal{F}_{g_1} \mathcal{F}_{g_2} \simeq \mathcal{F}_{g_1 g_2}$ for all $g_1, g_2 \in G$. In particular, this means for $G = B_W$, the functors satisfy the natural isomorphisms

$$\underbrace{\mathcal{F}_s \mathcal{F}_t \mathcal{F}_s \dots}_{m_{st} \text{ terms}} \simeq \underbrace{\mathcal{F}_t \mathcal{F}_s \mathcal{F}_t \dots}_{m_{st} \text{ terms}}$$

for all $s, t \in S$.

Let us restrict to the principal block $\mathcal{O}_0$. If we want an action of B_W on $\mathcal{O}_0$, we must specify for each generator $s \in S$ an endofunctor giving the image of s and an endofunctor giving the image of s^{-1} . A natural first choice for the image of s are the wall-crossing functors Θ_s : Figure 3.1 shows that $\Theta_s^2 M \neq M$ for all indecomposables M . However, there is no obvious candidate for s^{-1} .

A second choice would be to let $s \mapsto \text{Sh}_s$ and $s^{-1} \mapsto \text{CSh}_s$. In [MS05], it was implicitly proven that Sh_s and CSh_s satisfy the braid relations, that is for distinct simple reflections $s, t \in S$,

$$\underbrace{\text{Sh}_s \text{Sh}_t \text{Sh}_s \dots}_{m_{st}} \simeq \underbrace{\text{Sh}_t \text{Sh}_s \text{Sh}_t \dots}_{m_{st}},$$

and similarly for CSh_s . However, $\text{CSh}_s \text{Sh}_s \not\simeq \text{id}$ on all of $\mathcal{O}_0$, evident in Figure 3.1, for example $\text{CSh}_s \text{Sh}_s \nabla(0) \simeq \text{CSh}_s \nabla(0) \simeq L(-2)$. Mazorchuk–Stroppel did prove that $\text{CSh}_s \text{Sh}_s \simeq \text{id}$ (and dually $\text{Sh}_s \text{CSh}_s \simeq \text{id}$) on certain objects.

Proposition 3.26 ([MS05, Corollary 4.4]). *Let s be a simple reflection. The adjunction morphism induces an isomorphism $\text{CSh}_s \text{Sh}_s \simeq \text{id}$ on the full subcategory of modules with a Verma flag.*

In particular, this implies that $\text{CSh}_s \text{Sh}_s \simeq \text{id}$ on $\mathcal{O}_0\text{-Proj}$, the full subcategory of projectives in $\mathcal{O}_0$. Because Sh_s is only right-exact and CSh_s is left-exact, one should pass to their derived functors. Working in $D^b(\mathcal{O}_0)$ by Theorem 2.47 should be the natural setting in which these functors have homotopy inverses and where the braid group action can be realised.

Let $\mathcal{L}^i\mathrm{Sh}_s$ be the i -th left derived functor of Sh_s and $\mathcal{R}^i\mathrm{CSh}_s$ be the i -th right derived functor of CSh_s . We have the following result.

Proposition 3.27 ([MS05, Proposition 5.3]). *For any simple reflection $s \in S$,*

- (a) $\mathcal{L}^i\mathrm{Sh}_s \Delta(w \cdot 0) = \mathcal{R}^i\mathrm{CSh}_s \Delta(w \cdot 0) = 0$ for $w \in W$, $i > 0$,
- (b) Sh_s and CSh_s are exact on projectives, and

$$(c) \quad \mathcal{L}^i\mathrm{Sh}_s = \begin{cases} \mathrm{Sh}_s & i = 0 \\ \ker(\eta) & i = 1, \text{ while } \mathcal{R}^i\mathrm{CSh}_s = \begin{cases} \mathrm{CSh}_s & i = 0 \\ \mathrm{coker}(\varepsilon) & i = 1. \\ 0 & i > 1 \end{cases} \\ 0 & i > 1 \end{cases}$$

Let $\underline{w} = (s_1, \dots, s_k)$ be a braid word for $w \in W$. Consider the compositions $\mathrm{Sh}_{\underline{w}} := \mathrm{Sh}_{s_k} \cdots \mathrm{Sh}_{s_1}$ and $\mathrm{CSh}_{\underline{w}} := \mathrm{CSh}_{s_k} \cdots \mathrm{CSh}_{s_1}$.

Theorem 3.28 ([MS05, Theorem 5.7, Lemma 5.10], [Rou06, Theorem 10.4]). *The assignment $s \mapsto \mathcal{R}\mathrm{CSh}_s$, $s^{-1} \mapsto \mathcal{L}\mathrm{Sh}_s$ defines a (weak) categorical braid group action of B_W on $D^b(\mathcal{O}_0)$.*

Mazorchuk—Stroppel established a weak braid action, that is relations hold up to unspecified isomorphisms. Rouquier upgrades this to a strict action via his 2-braid group, where he assigns a “canonical” object to each braid word which makes the isomorphisms explicit. We will come back to this point in Section 5.2.

3.6 $\mathcal{L}\mathrm{Sh}_s$ and $\mathcal{R}\mathrm{CSh}_s$ as functor complexes

We can also realise the derived functors $\mathcal{L}\mathrm{Sh}_s$ and $\mathcal{R}\mathrm{CSh}_s$ as *functor complexes*. For $s \in S$, consider the complexes of endofunctors

$$\begin{aligned} \mathcal{F}_s &:= (\cdots \longrightarrow 0 \longrightarrow 0 \longrightarrow \underline{\Theta}_s \xrightarrow{\varepsilon} \mathbb{1} \longrightarrow 0 \longrightarrow \cdots) \text{ and} \\ \mathcal{F}_{s^{-1}} &:= (\cdots \longrightarrow 0 \longrightarrow \mathbb{1} \xrightarrow{\eta} \underline{\Theta}_s \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots), \end{aligned} \tag{3.5}$$

where ε and η are the counit and unit of adjunction. These complexes were first studied by Rickard [Ric94] in the context of reductive groups in positive characteristic. Given a bounded complex $X^\bullet \in K^b(\mathcal{O}_0\text{-Proj})$, apply a functor complex termwise to form a double complex. For example, applying $\mathcal{F}_s$ yields the double complex

$$\begin{array}{ccccccc} \cdots & \longrightarrow & \Theta_s(X^{-1}) & \longrightarrow & \Theta_s(X^0) & \longrightarrow & \Theta_s(X^1) \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & X^{-1} & \longrightarrow & X^0 & \longrightarrow & X^1 \longrightarrow \cdots \end{array}$$

and its total complex is

$$\mathcal{F}_s(X^\bullet) = (\cdots \rightarrow \Theta_s(X^{-1}) \oplus X^{-2} \rightarrow \Theta_s(X^0) \oplus X^{-1} \rightarrow \Theta_s(X^1) \oplus X^0 \rightarrow \cdots).$$

The same applies to $\mathcal{F}_{s^{-1}}$. Since Θ_s and $\mathbb{1}$ are exact, $\mathcal{F}_s$ and $\mathcal{F}_{s^{-1}}$ define endofunctors on $K^b(\mathcal{O}_0\text{-Proj})$.

The following shows that $\mathcal{F}_s$ and $\mathcal{F}_{s^{-1}}$ are mutually-inverse equivalences.

Lemma 3.29. *The functor complexes $\mathcal{F}_s$ and $\mathcal{F}_{s^{-1}}$ are equivalent to $\mathcal{RCS}_s$ and $\mathcal{LS}_s$ respectively.*

Proof sketch. We will sketch a proof for $\mathcal{LS}_s$ and $\mathcal{F}_{s^{-1}}$ and illustrate their images on a one-term complex. The results hold for an arbitrary complex in $D^b(\mathcal{O}_0)$. Suppose $M \in D^b(\mathcal{O}_0)$ is the complex supported in degree 0. To compute $\mathcal{LS}_s M$, apply the exact functor Θ_s to a projective resolution $P^\bullet$ of A . Then $\text{Sh}_s P^\bullet \rightarrow \text{Sh}_s M$ fits into the following diagram:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & P^1 & \longrightarrow & P^0 & \longrightarrow & M \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \eta_M \\ \cdots & \longrightarrow & \Theta_s P^1 & \longrightarrow & \Theta_s P^0 & \longrightarrow & \Theta_s M \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \longrightarrow & \text{Sh}_s P^1 & \longrightarrow & \text{Sh}_s P^0 & \longrightarrow & \text{Sh}_s M \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

One can check that the total complex of $P^\bullet \rightarrow \Theta_s P^\bullet$ forms an invertible roof between $M \rightarrow \Theta_s M$ and $\text{Sh}_s P^\bullet$ using a spectral sequence argument. We omit the details. Alternatively, see [KS90, Proposition 1.8.8] and the discussion after for a general statement. $\square$

We introduce Soergel bimodules in the next chapter and develop the diagrammatic calculus needed for concrete computations. In Chapter 5 we reinterpret the functor complexes $\mathcal{F}_s$ and $\mathcal{F}_{s^{-1}}$ as complexes of Soergel bimodules, prove that the braid group acts on these complexes, and hence obtain the induced braid group action on $D^b(\mathcal{O})$ which proves Theorem 3.28.

Chapter 4

Soergel Bimodules

In 1992, Soergel [Soe92] defined a category SBim of graded bimodules over a polynomial ring which provided an intrinsic description of the Hecke category, discussed in the introduction. These bimodules were defined using a Coxeter system (W, S) and a chosen representation of W . When W is the Weyl group of a semisimple Lie algebra, there is a monoidal equivalence between the category of projective endofunctors in $\mathcal{O}_0$ and SBim , providing an algebraic model for $\mathcal{O}_0$. Although our applications to $\mathcal{O}$ involve only Weyl groups, we develop the theory of Soergel bimodules for any Coxeter system, both for completeness and because the general theory is of independent mathematical interest.

Despite the concrete algebraic definition of SBim , calculating morphisms between these bimodules become unwieldy very quickly. To overcome this, Elias, Khovanov, and Williamson [EK10, Eli10, Eli16, EW16] developed a diagrammatic calculus, in which morphisms are encoded by planar graphs, or so-called “Soergel graphs”. This provides a powerful combinatorial tool to compute morphisms between such bimodules. In Chapter 5, we recast Rouquier’s proof of Theorem 3.28 in this diagrammatic language, which makes his underlying arguments more transparent.

This chapter draws heavily from [EMTW20, §4,8,9]. The results in Section 4.4.3 can be found in [AMRW17].

4.1 Realisations of Coxeter systems

For a Weyl group W arising from a Lie algebra, there is a canonical root system $\Phi \subset E$ whose reflections generate W (see Section 1.2). For a general Coxeter system (W, S) , no such root system need exist so we introduce an analogue which

is a certain kind of representation of W .

Definition 4.1. A *realisation* $(W, \mathfrak{h})$ of (W, S) is a triple $(\mathfrak{h}, \{\alpha_s\}_{s \in S}, \{\alpha_s^\vee\}_{s \in S})$, where $\mathfrak{h}$ is a $\mathbb{C}$ -vector space, together with subsets

$$\{\alpha_s^\vee\}_{s \in S} \subseteq \mathfrak{h}, \quad \{\alpha_s\}_{s \in S} \subseteq \mathfrak{h}^* = \text{Hom}_{\mathbb{C}}(\mathfrak{h}, \mathbb{C})$$

called the *simple coroots* and *simple roots* of the realisation respectively. The *Cartan matrix* $(a_{s,t})_{s,t \in S}$ of a realisation is defined as $a_{s,t} := \langle \alpha_s, \alpha_t^\vee \rangle \in \mathbb{C}$, where $\langle -, - \rangle: \mathfrak{h}^* \times \mathfrak{h} \rightarrow \mathbb{C}$ is the natural pairing. This data satisfies the following conditions:

1. $a_{s,s} = 2$ for all $s \in S$;
2. $\mathfrak{h}$ is a W -representation via the assignment

$$s(v) := v - \langle \alpha_s, v \rangle \alpha_s^\vee \quad (4.1)$$

for all $s \in S$ and $v \in \mathfrak{h}$. We will refer to the action of $s \in S$ as *reflections*.

3. a minor technical condition that we defer to (4.17).

This definition can be generalised to any commutative ring (see [EW16, Definition 3.1]) so a realisation need not be faithful. It suffices for our purposes to consider realisations over $\mathbb{C}$.

Example 4.2. If $\mathfrak{g}$ is a Lie algebra, then any Cartan subalgebra $\mathfrak{h}$ and its dual $\mathfrak{h}^*$ yields a realisation of the Weyl group (W, S) of $\mathfrak{g}$, where the roots and coroots of the realisation are the usual Lie-theoretic roots and coroots. In this way, we obtain a realisation over $\mathbb{Z}$ of any finite crystallographic Coxeter system (see Section 1.2). We call this the *Cartan realisation*.

Example 4.3. Let V be the geometric representation of (W, S) over $\mathbb{C}$ (see Section 1.3). Let $\mathfrak{h} = V^*$. Define the elements $\alpha_s \in \mathfrak{h}^*$ by

$$\langle \alpha_s, \alpha_t^\vee \rangle := -2 \cos\left(\frac{\pi}{m_{st}}\right).$$

We call this the *geometric realisation*.

The reader should keep these two examples in mind for the rest of the thesis. For our purposes, as we are considering category $\mathcal{O}$, we will be specialising to the case when (W, S) is a Weyl group and its corresponding simple reflections in Chapter 6.

4.2 Algebraic definition

Fix a Coxeter system (W, S) and a realisation $(W, \mathfrak{h})$. In this section, we will define Soergel bimodules associated $(W, \mathfrak{h})$. Let

$$R := \text{Sym}(\mathfrak{h}^*) = \bigoplus_{i \in \mathbb{Z}} \text{Sym}^i(\mathfrak{h}^*)$$

denote the symmetric algebra of $\mathfrak{h}^*$, with $\deg \mathfrak{h}^* = 2$. The grading from R induces gradings on R -(bi)modules. We denote the grading shift by (1): if $M = \bigoplus_{i \geq 0} M^i$ is a graded (bi)module, then $M(1)^i = M^{i+1}$.

Remark 4.4. A classic result is that $\{\alpha_s\} \subseteq \mathfrak{h}^*$ is linearly independent if and only if W is finite (see [Hum92, §2]). Then $R \simeq \mathbb{C}[\alpha_s \mid s \in S]$, the polynomial algebra in $|S|$ variables.

Now W acts on $\mathfrak{h}^*$ via the contragredient representation $s(\gamma) = \gamma - \langle \gamma, \alpha_s^\vee \rangle \alpha_s$ for all $\gamma \in \mathfrak{h}^*$. This induces an action of W on the algebra R by $w \cdot \prod_{s \in S} \alpha_s^{k_s} = \prod_{s \in S} (w(\alpha_s))^{k_s}$ on monomials, and then extended linearly. For $s \in S$, write $R^s := \{f \in R \mid s(f) = f\}$ for the subring of s -invariants, and $R^{s,t} := R^s \cap R^t$. We call $f \in R$ *s-antiinvariant* when $s(f) = -f$. The next lemma records the decomposition of R as a graded R^s -bimodule.

Lemma 4.5. *Let $s \in S$. As R^s -bimodules, there is an isomorphism $R = R^s \oplus R^s \alpha_s \simeq R^s \oplus R^s(-2)$ so that R decomposes into s -invariants and s -antiinvariants. Therefore, R is a free R^s -module of rank 2.*

Proof. For any $f \in R$, we can write

$$f = \frac{f + s(f)}{2} + \frac{f - s(f)}{2}.$$

The first term is s -invariant, and the second is s -antiinvariant. The ideal (α_s) of R is invariant under the action of s , so $\{e, s\}$ acts on the quotient ring $R/(\alpha_s)$. In particular, the action of s on $R/(\alpha_s)$ is trivial as for any $v \in \mathfrak{h}^*$, $s(v) - v = -\langle v, \alpha_s^\vee \rangle \alpha_s \in (\alpha_s)$. If g is s -antiinvariant, then $g + (\alpha_s) = s(g + (\alpha_s)) = -g + (\alpha_s)$, which implies that $g \in (\alpha_s)$. This shows that $g = \alpha_s h$ for some unique $h \in R$. Furthermore, $-\alpha_s h = s(\alpha_s h) = -\alpha_s s(h)$ so $h \in R^s$. Therefore, the R^s -submodule of s -antiinvariants is exactly $R^s \alpha_s$. The decomposition of f above and the fact that $R^s \cap R^s \alpha_s = 0$ gives the decomposition

$$R = R^s \oplus R^s \alpha_s.$$

Finally, $\deg(\alpha_s) = 2$, so $R^s \alpha_s \simeq R^s(-2)$ as graded R^s -bimodules. $\square$

Remark 4.6. Lemma 4.5 is a special case of the Chevalley–Shephard–Todd theorem [Che55, ST54] which states that R is a graded free module of finite rank over R^I , where $I \subset S$ generates a finite subgroup of W .

Example 4.7. Consider $(S_2, \{s\})$ with the geometric realisation. If W is generated by the simple reflection s , then $R = \mathbb{R}[\alpha_s]$ and $R^s = \mathbb{R}[\alpha_s^2]$. In fact, for any arbitrary Coxeter system (W, S) , R^s is generated by α_s^2 and the elements $\alpha_t + \left(\cos \frac{\pi}{m_{st}}\right) \alpha_s$ for all $t \in S \setminus \{s\}$.

The following map, which will be useful in the following sections, allows us to project any element $f \in R$ to its s -antiinvariant component.

Definition 4.8. For $s \in S$, the *Demazure operator* ∂_s is the graded map

$$\partial_s: R \rightarrow R^s(-2), \quad f \mapsto \frac{f - s(f)}{\alpha_s}.$$

The Demazure operator is well-defined as the numerator is s -antiinvariant and hence in the ideal (α_s) . One could then describe the isomorphism $R \rightarrow R^s \oplus R^s(-2)$ explicitly by the maps $f \mapsto (\partial_s(f \frac{\alpha_s}{2}), \partial_s(f))$ with inverse $(g, h) \mapsto g + h \frac{\alpha_s}{2}$.

With the decomposition lemma in hand, we introduce the basic R -bimodules from which Soergel bimodules are built. Throughout, we abbreviate the tensor product over R by $MN := M \otimes_R N$.

Definition 4.9. For an expression $\underline{w} = (s_1, \dots, s_n)$, the *Bott–Samelson bimodule*, denoted by $\text{BS}(\underline{w})$, is the graded R -bimodule given by

$$\text{BS}(\underline{w}) := B_{s_1} \cdots B_{s_n},$$

where $B_{s_i} := R \otimes_{R^{s_i}} R(1)$.

Expanding each B_{s_i} , there is an isomorphism

$$\text{BS}(\underline{w}) \simeq R \otimes_{R^{s_1}} R \otimes_{R^{s_2}} \cdots \otimes_{R^{s_n}} R(n),$$

so $\text{BS}(\underline{w})$ is an R -bimodule under left and right multiplication. Using Lemma 4.5, we can decompose B_s as follows.

Lemma 4.10. *We have $B_s \simeq R(1) \oplus R(-1)$ as left R -modules.*

Proof. By Lemma 4.5 and distributivity of the tensor product with direct sums, we have

$$B_s = R \otimes_{R^s} R(1) \simeq R \otimes_{R^s} (R^s \oplus R^s(-2)) (1) \simeq R(1) \oplus R(-1).$$

Applying the decomposition of R to the second term allows us to only assert that $B_s \simeq R(1) \oplus R(-1)$ as left R -modules. $\square$

One can similarly analyse decompositions of $\text{BS}(\underline{w})$ for any expression $\underline{w}$. This motivates the following definition.

Definition 4.11. A *Soergel bimodule* is a graded bimodule isomorphic to a direct summand of a finite direct sum of grading shifts of Bott–Samelson bimodules. The category of Soergel bimodules $\mathbb{S}\text{Bim}$ is the smallest full subcategory of R -bimodules generated by B_s for each $s \in S$ that is closed under tensor products, grading shifts, direct sums, and direct summands.

The following lemma allows us to identify some indecomposable bimodules.

Lemma 4.12 ([EMTW20, Lemma 4.34]). *If a graded R -bimodule M is generated by a single homogeneous element m as an R -bimodules, that is $R \cdot m \cdot R = M$, then M is indecomposable.*

Proof. Let $d = \deg m$ and suppose $M = L \oplus N$. If M is generated by m as an R -bimodule, then $M^d = R^0 \cdot m \cdot R^0 = \mathbb{C}m$. Since $M^d \simeq L^d \oplus N^d$ as $\mathbb{C}$ -vector spaces, we may assume that $m \in L^d$ and $N^d = 0$ as M^d is one-dimensional. This means that $M = R \cdot m \cdot R \subseteq L$, so $N = 0$. $\square$

This lemma immediately gives that R is indecomposable. It is also easy to see that B_s is generated as an R -bimodule by the degree -1 homogeneous element $1 \otimes 1$, so it is also indecomposable.

Example 4.13. Consider $(S_2, \{s\})$. By the above discussion, R and B_s are indecomposable Soergel bimodules. Consider then the tensor product $B_s B_s$. We have that

$$\begin{aligned} B_s B_s &\simeq R \otimes_{R^s} R \otimes_{R^s} R(2) \\ &\simeq R \otimes_{R^s} (R^s \oplus R^s(-2)) \otimes_{R^s} R(2) && \text{(Lemma 4.5)} \\ &\simeq R \otimes_{R^s} R(2) \oplus R \otimes_{R^s} R \\ &= B_s(1) \oplus B_s(-1). \end{aligned}$$

Observe that, in contrast to Lemma 4.10 (which is only an isomorphism of left R -modules), the above identification is an isomorphism of R -bimodules since the middle R is regarded as an R^s -bimodule. Every tensor power of B_s splits as a direct sum of grading shifts of B_s , so R and B_s are the only indecomposable Soergel bimodules, up to shift and isomorphism.

Example 4.14. Consider $(S_3, \{s, t\})$. Then R , B_s , and B_t are indecomposable. Furthermore, $B_{st} := B_s B_t$ and $B_{ts} := B_t B_s$ are indecomposable as they are generated by the tensor $1 \otimes 1 \otimes 1$: it suffices to show that any element $1 \otimes f \otimes 1$ can be rewritten so that the middle factor is 1 by splitting f into s - and t -invariants via Demazure operators and sliding invariants across the tensor products. This implies that B_s and B_t are not isomorphic as we have seen that $B_s B_s$ is decomposable whilst $B_s B_t$ is not.

There is a further indecomposable $B_{sts} := R \otimes_{R^{s,t}} R(3)$ corresponding to the longest element. It appears in the decomposition $B_s B_t B_s \simeq B_s \oplus B_{sts}$ and $B_t B_s B_t \simeq B_t \oplus B_{sts}$. One way to prove this is to produce a degree 0 non-trivial idempotent in $\text{End}(B_s B_t B_s)$. Explicit inclusion and projection maps can be written down, though it becomes quite tedious to manipulate tensor products of polynomials (see [Tsv, Lemma 3.1] or [Dun, Proposition 10] for the full calculations). We will see another interpretation of this indecomposable in Example 4.23.

Using a similar method to the previous example, one can show that these are all the indecomposables.

As mentioned in the previous example, the method of tracking images of generators rapidly become unwieldy for larger Coxeter systems. This makes a direct, element-wise search for indecomposables an ineffective way to understand $\mathbb{S}\text{Bim}$. Remarkably, Soergel showed that the indecomposables are controlled by the Weyl group, which justifies the notation used in Example 4.14.

Theorem 4.15 (Soergel’s Categorification Theorem, [Soe07]). *Let $\underline{w}$ be a reduced expression of $w \in W$. Then there is a unique (up to isomorphism) indecomposable summand B_w of $\text{BS}(\underline{w})$ that does not occur in $\text{BS}(\underline{u})$, for any strictly shorter expression $\underline{u}$. The set $\{B_w \mid w \in W\}$ constitutes a complete set of non-isomorphic indecomposable Soergel bimodules, up to shift and isomorphism.*

The statement is non-constructive as it only identifies the indecomposables abstractly without describing their structure. In principle, one can recover B_w by finding all summands of $\text{BS}(\underline{w})$, removing those that also appear (up to shift) among Bott–Samelsons for shorter words, and taking the leftover summand. This amounts to calculating all idempotents in the degree-zero endomorphism algebra of $\text{BS}(\underline{w})$, which is hard using the methods we discussed. In the next sections, we turn to a combinatorial approach that makes these indecomposable summands explicit and computable. We begin by describing morphisms between Bott–Samelson bimodules involving just one simple reflection.

4.3 One-colour diagrammatics

The fundamental difficulty determining morphisms between Bott–Samelson bi-modules motivated the development of diagrammatic tools. In this section, we will introduce these tools and show how representing morphisms by planar diagrams provides a significant advantage to do computations. Let us first outline the diagrammatic setting for morphisms in $\mathbb{B}\mathbb{S}\mathbb{B}\mathbf{im}$ before setting up the dictionary between the two.

Fix a simple reflection $s \in S$, which will be drawn in **red**. A tensor product $B_s^{\otimes i}$ is depicted as i marked points on the x -axis at positions $1, \dots, i$ (the monoidal unit R has no marking). A morphism $B_s^{\otimes i}$ and $B_s^{\otimes j}$ is an $\mathbb{R}$ -linear combination of red planar graphs embedded in the strip $\mathbb{R} \times [0, 1]$, read bottom (domain) to top (codomain). More generally, horizontal juxtaposition of diagrams denotes the tensor product of the corresponding morphisms. These graphs have only two types of vertices, univalent vertices (dots) and trivalent vertices (merges and splits), and are decorated according to certain rules. We refer to this setting as *one-colour diagrammatics*.

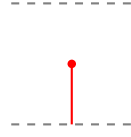
We begin setting up the diagram–morphism dictionary. The first morphism we present is the identity morphism id_{B_s} which is depicted as

$$\mathrm{id}_{B_s} := \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} . \quad (4.2)$$

This is a degree 0 morphism. Placing several such strands side by side depicts the identity on a tensor power $B_s^{\otimes i}$.

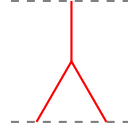
When the domain and target have different numbers of B_s factors, we can use *merges* and *splits* together with *dots*. We define some more diagrams and the morphisms they correspond to in (4.3). In particular, $\varepsilon: B_s \rightarrow R$ is drawn as a strand that originates from the bottom and ends in a dot. Dually, $\eta: R \rightarrow B_s$ begins with a dot and terminates at the top. Composing these simple diagrams by vertical stacking (in a compatible manner) allows us to describe more complicated morphisms.

Although the morphisms in (4.3) seem arbitrary, they equip B_s with the structure of a *Frobenius algebra object* in $\mathbb{B}\mathbb{S}\mathbb{B}\mathbf{im}$. Trivalent vertices encode multiplication μ and comultiplication δ , while univalent vertices encode unit η and counit ε (see [Koc04, §3.6.8]). The defining relations translate to the diagrammatic identities in (4.4).



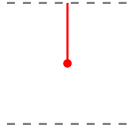
$$\begin{aligned} \varepsilon: B_s &\rightarrow R \\ f \otimes g &\mapsto fg \\ \deg \varepsilon &= -1 \end{aligned}$$

(4.3a)



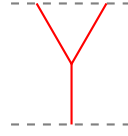
$$\begin{aligned} \mu: B_s B_s &\rightarrow B_s \\ f \otimes g \otimes h &\mapsto \partial_s(g) f \otimes h \\ \deg \mu &= -1 \end{aligned}$$

(4.3c)



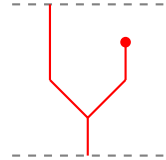
$$\begin{aligned} \eta: R &\rightarrow B_s \\ 1 &\mapsto \frac{1}{2}(1 \otimes \alpha_s + \alpha_s \otimes 1) \\ \deg \eta &= 1 \end{aligned}$$

(4.3b)



$$\begin{aligned} \delta: B_s &\rightarrow B_s B_s \\ f \otimes g &\mapsto f \otimes 1 \otimes g \\ \deg \delta &= 1 \end{aligned}$$

(4.3d)



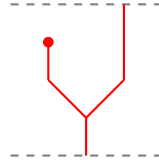
$$(\text{id}_{B_s} \otimes \varepsilon) \circ \delta$$

=



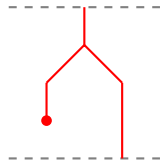
$$\text{id}_{B_s}$$

=



$$(\varepsilon \otimes \text{id}_{B_s}) \circ \delta$$

(4.4a)



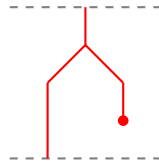
$$\mu \circ (\eta \otimes \text{id}_{B_s})$$

=



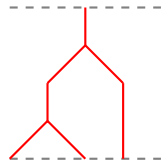
$$\text{id}$$

=



$$\mu \circ (\text{id}_{B_s} \otimes \eta)$$

(4.4b)



$$\mu \circ (\mu \otimes \text{id}_{B_s})$$

=



$$\mu \circ (\text{id}_{B_s} \otimes \mu)$$

(4.4c)

$$\begin{array}{ccc}
 \begin{array}{c} \text{---} \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ \text{---} \end{array} & = & \begin{array}{c} \text{---} \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ \text{---} \end{array} \\
 (\text{id}_{B_s} \otimes \delta) \circ \delta & & (\delta \otimes \text{id}_{B_s}) \circ \delta
 \end{array} \tag{4.4d}$$

$$\begin{array}{ccc}
 \begin{array}{c} \text{---} \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ \text{---} \end{array} & = & \begin{array}{c} \text{---} \\ \diagdown \quad \diagup \\ \diagup \quad \diagdown \\ \text{---} \end{array} \\
 (\text{id}_{B_s} \otimes \mu) \circ (\delta \otimes \text{id}_{B_s}) & & \delta \circ \mu \\
 & & \begin{array}{c} \text{---} \\ \diagup \quad \diagdown \\ \diagdown \quad \diagup \\ \text{---} \end{array} \\
 & & (\mu \otimes \text{id}_{B_s}) \circ (\text{id}_{B_s} \otimes \delta).
 \end{array} \tag{4.4e}$$

In the last relation, the dotted line is included to clearly show the composition. We will refer to (4.4e) as *Frobenius associativity*.

The last morphism to discuss is multiplication by a homogeneous polynomial $f \in R$. This will be represented by a box decoration:

$$\begin{array}{c}
 \text{---} \\
 \boxed{f} : 1 \mapsto f. \\
 \text{---}
 \end{array} \tag{4.5}$$

The degree of this map is simply $\deg f$. Addition and multiplication of homogeneous polynomials are represented by merging boxes. A box may be placed in any connected component of the planar graph, with its position indicating which tensor factor the operation acts on. Within a given component, boxes can be moved freely without altering the morphism. For example, it is easy to check that the diagrams represent the same morphism:

$$\begin{array}{ccc}
 \begin{array}{c} \text{---} \\ \bullet \\ \boxed{f} \\ \text{---} \end{array} & = & \begin{array}{c} \text{---} \\ \boxed{f} \\ \bullet \\ \text{---} \end{array} \\
 & & \begin{array}{c} \text{---} \\ \boxed{f} \\ \bullet \\ \text{---} \end{array}
 \end{array}$$

4.3.1 Isotopy

The true advantage of a diagrammatic presentation is that we can work with diagrams up to isotopy. Let us denote by *cups* and *caps* for the diagrams

$$\begin{array}{ccc}
 \begin{array}{c} \text{---} \\ \cup \\ \text{---} \end{array} & := & \begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \text{---} \end{array} \\
 & & \text{and} \\
 \begin{array}{c} \text{---} \\ \cap \\ \text{---} \end{array} & := & \begin{array}{c} \text{---} \\ \diagdown \quad \diagup \\ \bullet \end{array}
 \end{array} \tag{4.6}$$

$$(4.7)$$

In fact, the Frobenius algebra structure ensures that all isotopy relations hold:

$$\begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \diagup \\ \diagdown \end{array} = \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \diagup \\ \diagdown \end{array} = \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \diagup \\ \diagdown \end{array}, \quad \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \diagup \\ \diagdown \end{array} = \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \diagup \\ \diagdown \end{array} = \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \diagup \\ \diagdown \end{array} \quad (4.9)$$

This viewpoint allows us to simplify diagrams locally, via a collection of *local relations*. We indicate the portion of a morphism under consideration by enclosing it in a dotted circle. Everything inside the circle is the “local patch”, and everything outside is left unchanged. For example, the local versions of (4.4a) and (4.4b) can be drawn as

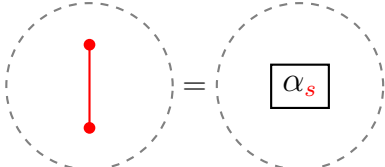
and (4.4e) becomes

These local relations allow us to zoom in on any patch of a Soergel diagram, manipulate the patch using local relations, and then glue the simplified piece back without affecting the rest of the diagram.

4.3.2 Further relations

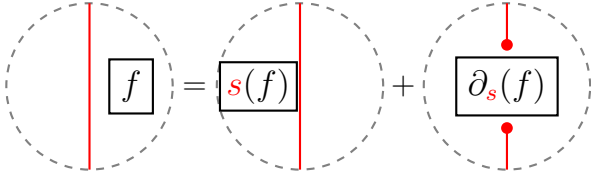
We list a few other relations that can be easily deduced which will be used frequently in the rest of this chapter and Chapter 5.

Barbell relation:



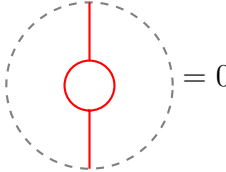
$$(4.12)$$

Sliding relation:



$$(4.13)$$

Keyhole relation:



$$(4.14)$$

Relation (4.13) implies that an s -invariant f can slide across s -strands without other changes. One also gets a sliding relation with f on the left and $s(f)$ on the right by applying (4.13) to $s(f)$, and noting that $\partial_s(s(f)) = -\partial_s(f)$.

Remark 4.16. The category of Soergel bimodules we have defined is one incarnation of the Hecke category introduced in the introduction. One can define a *diagrammatic Hecke category* [EW16], rather than just using diagrams to encode morphisms between Soergel bimodules. This diagrammatic Hecke category is defined where objects are finite sequences of coloured points on a line, and whose morphisms are diagrams between these sequences, taken up to isotopy and the usual local relations. The diagrammatic category is defined over $\mathbb{Z}$ and may be base-changed to any ring, in particular to fields of prime characteristic, so it provides a combinatorial model of the Hecke category even in settings where classical Soergel bimodules break down (see for example [Wil18]).

4.3.3 A simple diagrammatic calculation

Recall from Example 4.13 that $B_s B_s \simeq B_s(-1) \oplus B_s(1)$ as R -modules. We now give a diagrammatic description of this splitting. The inclusion and projection maps will be displayed as matrices whose entries are the corresponding Soergel diagrams, so that composing these matrices yields the identity.

Example 4.17. Consider the maps

$$f := \left[\begin{array}{c} \text{diagram 1} \\ \text{diagram 2} \end{array} \right] : B_s B_s \rightarrow B_s(-1) \oplus B_s(1),$$

$$g := \left[\begin{array}{cc} \text{diagram 3} & \text{diagram 4} \end{array} \right] : B_s(-1) \oplus B_s(1) \rightarrow B_s B_s.$$

We will show diagrammatically that they are mutual inverses to each other.

The composition $f \circ g$ is given by

$$f \circ g = \left[\begin{array}{c} \text{diagram 1} \\ \text{diagram 2} \end{array} \right] \left[\begin{array}{cc} \text{diagram 3} & \text{diagram 4} \end{array} \right] = \left[\begin{array}{cc} \text{diagram 5} & \text{diagram 6} \\ \text{diagram 7} & \text{diagram 8} \end{array} \right],$$

where the morphisms in the entries of g are at the bottom as they are applied first.

The top-right entry is zero by the keyhole relation (4.14). The top-left and bottom-right entries can be diagrammatically simplified by the computation

$$\left[\begin{array}{c} \text{diagram 5} \\ \text{diagram 7} \end{array} \right] \stackrel{(4.13)}{=} \left[\begin{array}{c} \text{diagram 9} \\ \text{diagram 10} \end{array} \right] + \left[\begin{array}{c} \text{diagram 11} \\ \text{diagram 12} \end{array} \right] = \left[\begin{array}{c} \text{diagram 13} \end{array} \right],$$

where the keyhole relation gives zero for the first term, and the calculation

$$\partial_s \left(\frac{\alpha_s}{2} \right) = \frac{\alpha_s - s(\alpha_s)}{2\alpha_s} = \frac{\alpha_s - (\alpha_s - \langle \alpha_s, \alpha_s^\vee \rangle \alpha_s)}{2\alpha_s} = 1 \quad (4.15)$$

along with (4.10) simplifies to the identity.

A similar computation shows that the bottom left entry is equal to the zero map. This can be also seen from the fact that α_s^2 is s -invariant, and thus can be pulled across a red strand, after which the keyhole relation applies.

The composition $g \circ f$ can be seen to equal the identity on $B_s B_s$ by

$$\begin{aligned}
g \circ f &= \left[\begin{array}{c} \text{Diagram 1: A box labeled } \frac{\alpha_s}{2} \text{ with two lines entering from the bottom and two exiting from the top.} \\ \text{Diagram 2: A box labeled } \frac{\alpha_s}{2} \text{ with two lines entering from the top and two exiting from the bottom.} \end{array} \right] \left[\begin{array}{c} \text{Diagram 3: A box labeled } \frac{\alpha_s}{2} \text{ with two lines entering from the top and two exiting from the bottom.} \\ \text{Diagram 4: A box labeled } \frac{\alpha_s}{2} \text{ with two lines entering from the bottom and two exiting from the top.} \end{array} \right] = \frac{1}{2} \left(\begin{array}{c} \text{Diagram 5: A box labeled } \alpha_s \text{ with two lines entering from the bottom and two exiting from the top.} \\ \text{Diagram 6: A box labeled } \alpha_s \text{ with two lines entering from the top and two exiting from the bottom.} \end{array} \right) \\
&\stackrel{(4.11)}{=} \frac{1}{2} \left(\begin{array}{c} \text{Diagram 7: A box labeled } \alpha_s \text{ with two lines entering from the bottom and two exiting from the top.} \\ \text{Diagram 8: A box labeled } s(\alpha_s) \text{ with two lines entering from the top and two exiting from the bottom.} \end{array} \right) \stackrel{(4.13)}{=} \frac{1}{2} \left(\begin{array}{c} \text{Diagram 9: A box labeled } \partial_s(\alpha_s) \text{ with two lines entering from the bottom and two exiting from the top.} \\ \text{Diagram 10: A box labeled } \partial_s(\alpha_s) \text{ with two lines entering from the top and two exiting from the bottom.} \end{array} \right) \\
&= \begin{array}{c} \text{Diagram 11: A box labeled } \partial_s(\alpha_s) \text{ with two lines entering from the bottom and two exiting from the top.} \\ \text{Diagram 12: A box labeled } \partial_s(\alpha_s) \text{ with two lines entering from the top and two exiting from the bottom.} \end{array} = \begin{array}{c} \text{Diagram 13: A box labeled } \partial_s(\alpha_s) \text{ with two lines entering from the bottom and two exiting from the top.} \\ \text{Diagram 14: A box labeled } \partial_s(\alpha_s) \text{ with two lines entering from the top and two exiting from the bottom.} \end{array}.
\end{aligned}$$

This gives the desired isomorphism.

Whilst the diagrammatic calculation is more involved than the abstract algebraic decomposition for this simple example, it makes the isomorphism completely explicit. The morphism f is the projection onto the direct summands of B_s , and g the inclusion. For $B_s B_s \simeq B_s(-1) \oplus B_s(1)$ we require $f \circ g = \text{id}_{B_s(-1) \oplus B_s(1)}$ and $g \circ f = \text{id}_{B_s B_s}$. The components of f and g are the inclusions and projections between $B_s B_s$ and $B_s(-1) \oplus B_s(1)$. The above calculation is an example of a result in arbitrary additive categories, whose form will be convenient to use later.

Proposition 4.18. *Let X, M, N be objects in an additive category. Then $X \simeq M \oplus N$ if and only if there are morphisms*

$$i_M: M \rightarrow X, \quad i_N: N \rightarrow X, \quad p_M: X \rightarrow M, \quad p_N: X \rightarrow N$$

which satisfy

$$\begin{aligned}
p_M \circ i_N &= 0, \quad p_N \circ i_M = 0, \quad p_M \circ i_M = \text{id}_M, \quad p_N \circ i_N = \text{id}_N, \\
i_M \circ p_M + i_N \circ p_N &= \text{id}_X.
\end{aligned}$$

4.4 Two-colour diagrammatics

In the one-colour case, every indecomposable summand of $B_s^{\otimes k}$ is B_s itself (up to shift) by Example 4.13. In particular, Example 4.17 gives an explicit diagrammatic idempotent $B_s B_s \rightarrow B_s B_s$ realising the decomposition $B_s B_s \simeq$

$B_s(1) \oplus B_s(-1)$, so the summands are exactly the images of the projections $B_s B_s \rightarrow B_s(1)$ and $B_s B_s \rightarrow B_s(-1)$.

Fix another simple reflection $t \in S$, and colour it **blue**. In Example 4.14, the bimodule $B_s B_t B_s$ contains the indecomposable B_{sts} , which is not Bott–Samelson, so there is no obvious diagrammatic realisation. In this section, we introduce the Jones–Wenzl morphisms for alternating tensor products of B_s and B_t which are projections to precisely the indecomposable Soergel bimodules.

4.4.1 Quantum numbers

To define diagrammatics for two colours in full generality over an arbitrary realisation, let us make the following technical definition.

Definition 4.19 ([EW16, Definition 3.5]). The *2-coloured quantum numbers* $[n]_s$ and $[n]_t$ are defined by $[0]_s = [0]_t = 0$, $[1]_s = [1]_t = 1$, $[2]_s = -\langle \alpha_t, \alpha_s^\vee \rangle$, $[2]_t = -\langle \alpha_s, \alpha_t^\vee \rangle$, and the recursive formula

$$[n]_s = [2]_s [n-1]_t - [n-2]_s, \quad (4.16a)$$

$$[n]_t = [2]_t [n-1]_s - [n-2]_t. \quad (4.16b)$$

For a realisation to admit a diagrammatic presentation, we require that

$$[m_{st}]_s = [m_{st}]_t = 0, \quad (4.17)$$

which is the missing condition from Definition 4.1. We will not go into detail about this condition or the technicalities of quantum numbers. See Remark 4.21 or refer to [Eli16, Appendix A] for motivation and details.

4.4.2 Jones–Wenzl projectors

We give the following definition.

Definition 4.20. Consider the Bott–Samelson bimodule $BS_k^s := \underbrace{B_s B_t B_s \dots}_{k \text{ terms}}$

which is the tensor product of alternating B_s and B_t of length k starting with B_s . Then the *Jones–Wenzl projector* $JW_k^s \in \text{End}_{\mathbb{B}\text{SBim}}(M)$ is defined recursively,

with $\text{JW}_0^s := \text{id}$, $\text{JW}_1^s := \text{id}_{B_s}$, and

$$\text{JW}_{k+1}^s := \text{JW}_k^t + \frac{[k-1]_t}{[k]_s} \left(\text{JW}_k^t \text{ JW}_{k-1}^s \text{ JW}_k^t \right)$$

for $k \geq 2$ even. We will represent the right-most strand by **purple** as the actual colour depends on the parity of k . We define BS_k^t and JW_k^t similarly.

For the remainder of this section, we present results in one colouring, noting that they also hold with the colours swapped.

Remark 4.21. The appearance of quantum numbers used to define the Jones–Wenzl projector originates from the study of the Temperley–Lieb algebra. The recursive formula given in Definition 4.20, which we take to be the definition, was first discovered in [Wen87], and the diagrammatic version for Soergel bimodules was given in [Eli16]. The two-coloured quantum number was introduced to account for realisations with non-symmetric Cartan matrices, such as those corresponding to certain Weyl groups.

When the meaning is clear, we omit the superscripts and subscripts on Jones–Wenzl projectors. We first list their key properties, then give an example.

Proposition 4.22. *Jones–Wenzl projectors satisfy the following properties:*

(JW1) JW_k^s and JW_k^t are idempotents for each k .

(JW2) There is a diagram $\underline{\text{JW}}_k^s$ for every k such that

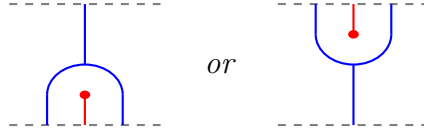
$$\text{JW}_k^s = \left[\text{JW}_k^s \right],$$

and similarly for JW_k^t .

To clarify, the box decoration of JW_k^s is a formal decoration with k lines of alternating colour going into the bottom of the rectangle and out the top, whereas the dotted circle $\underline{\text{JW}}_k^s$ represents a local diagram (or linear

combinations of diagrams) with $2k - 2$ strands around it with alternating colours and two trivalent vertices making it compatible with the top and bottom sequences.

(JW3) The diagram obtained by placing the morphisms



to the top or bottom respectively of a JW projector equals zero.

(JW4) JW is invariant under horizontal reflection.

Example 4.23. Note that $\text{JW}_2^s = \text{id}_{B_s B_t}$. The first non-identity Jones–Wenzl projector is when $k = 3$. Applying the recurrence relation once gives

$$\begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} = \boxed{\text{JW}_3^s} - \frac{1}{[2]_s} \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array}.$$

By (JW1), JW_3^s is an idempotent. Using (4.13) and Definition 4.19, one can check that the second map is idempotent also. (JW3) kills the cross terms, so the images of the two maps are orthogonal. The second idempotent factors through B_s , so its image is isomorphic to B_s . Consequently, $B_s B_t B_s \simeq B_s \oplus \text{Im}(\text{JW}_3^s)$, so by Example 4.14, we can identify $\text{Im}(\text{JW}_3^s) \simeq B_{sts}$.

Although indecomposables do not generally have a diagrammatic presentation, they do appear as the images of certain idempotents. Denote by $B_k^s := \underbrace{B_{sts} \dots}_{k \text{ terms}}$ the indecomposable summand from Theorem 4.15. The following theorem specifies the idempotent whose image is B_k^s .

Theorem 4.24 ([Eli16, Theorem 5.29]). *The image of the Jones–Wenzl projector JW_k^s is the indecomposable summand B_k^s .*

We therefore identify the indecomposable bimodule B_k^s with the idempotent morphism JW_k^s . Accordingly, for BS_k^s and BS_k^t , determining indecomposables reduces to manipulating the corresponding JW diagrams and reading off the resulting idempotent. This is still quite involved, but yields an explicit decomposition.

4.4.3 Further properties of Jones–Wenzl projectors

We finish this chapter with some useful lemmas needed in the next.

Lemma 4.25 ([AMRW17, Lemma 8.3.1]). *We have*

Proof. This follows from an application of (JW2), after which the dot can be absorbed, and a reapplication of (JW2) gives the desired result. $\square$

Lemma 4.26 ([AMRW17, Lemma 8.3.2]). *We have*

Proof. Apply (JW2), after which the keyhole relation turns the diagram into the zero morphism. $\square$

Lemma 4.27 ([AMRW17, Lemma 8.3.3]). *We have*

Proof. Looking at just the portion of the morphism above the Jones–Wenzl projector, we have the following equalities:

$$\begin{aligned}
 \begin{array}{c} \text{red dot} \\ | \\ \text{blue dot} \end{array} &= \partial_s \left(\frac{\alpha_s}{2} \right) \begin{array}{c} \text{red line} \\ | \\ \text{blue line} \end{array} \stackrel{(4.13)}{=} \begin{array}{c} \frac{\alpha_s}{2} \\ | \\ \text{blue line} \end{array} - s \left(\frac{\alpha_s}{2} \right) \begin{array}{c} \text{red line} \\ | \\ \text{blue line} \end{array} \\
 &\stackrel{(4.13)}{=} \begin{array}{c} \text{red line} \\ | \\ \text{blue line} \end{array} s \left(\frac{\alpha_s}{2} \right) + \partial_s \left(\frac{\alpha_s}{2} \right) \begin{array}{c} \text{red line} \\ | \\ \text{blue line} \end{array} - s \left(\frac{\alpha_s}{2} \right) \begin{array}{c} \text{red line} \\ | \\ \text{blue line} \end{array} \stackrel{(\text{JW3})}{=} \begin{array}{c} \text{red line} \\ | \\ \text{blue line} \end{array} .
 \end{aligned}$$

We have used the identity $\partial_s(\frac{\alpha_s}{2}) = 1$ (see (4.15)). Since each diagram is pre-composed with a Jones–Wenzl projector, (JW3) kills the first and third terms on the second line, leaving only the middle term.

Similarly, a t -dot juxtaposed with an s -dot slides across a t -strand, becoming an s -dot followed by a t -dot. Iterating yields the desired equality. $\square$

Lemma 4.28 ([AMRW17, Lemma 8.3.5]). *We have*

and similarly with the small JW boxes on top.

Proof. Omitted. $\square$

The following result is [AMRW17, Lemma 8.4.1], which generalises [Rou06, Lemma 9.1], and gives an inductive description of how Soergel bimodules decompose.

Lemma 4.29. *We have $B_s B_k^s = B_k^s(-1) \oplus B_k^s(1)$ for $1 \leq k \leq m_{st}$, and*

$$B_s B_k^t = \begin{cases} B_2^s & k = 1 \\ B_{k+1}^s \oplus B_{k-1}^s & 2 \leq k \leq m_{st} - 1 \\ B_k^t(-1) \oplus B_k^t(1) & k = m_{st}. \end{cases}$$

Proof. Consider $B_s B_k^t$ first. The case $k = 1$ was worked out in Example 4.14.

Suppose $2 \leq k \leq m_{st} - 1$. We claim that the maps

$$\begin{aligned} p_{+1} &= \begin{array}{c} \text{JW} \\ \vdots \\ \text{JW} \end{array} : B_s B_k^t \rightarrow B_{k+1}^s & p_{-1} &= -\frac{[k-1]_t}{[k]_s} \begin{array}{c} \text{JW} \\ \vdots \\ \text{JW} \end{array} : B_s B_k^t \rightarrow B_{k-1}^s \\ i_{+1} &= \begin{array}{c} \text{JW} \\ \vdots \\ \text{JW} \end{array} : B_{k+1}^s \rightarrow B_s B_k^t & i_{-1} &= \begin{array}{c} \text{JW} \\ \vdots \\ \text{JW} \end{array} : B_{k-1}^s \rightarrow B_s B_k^t \end{aligned}$$

(4.18)

satisfy the relations

$$\begin{aligned} p_{-1} \circ i_{+1} &= 0, & p_{+1} \circ i_{-1} &= 0, & p_{+1} \circ i_{+1} &= \text{id}_{B_{k+1}^s}, & p_{-1} \circ i_{-1} &= \text{id}_{B_{k-1}^s}, \\ i_{+1} \circ p_{+1} + i_{-1} \circ p_{-1} &= \text{id}_{B_s B_k^t}, \end{aligned}$$

from where the claim follows by Proposition 4.18. We can write $\text{id}_{B_{k+1}^s}$ as just JW_{k+1}^s , and similarly for $\text{id}_{B_{k-1}^s}$, so $\text{id}_{B_s B_k^t} = \text{id}_{B_s} \otimes \text{JW}_k^t$.

The first two relations are satisfied by Lemma 4.28, where JW_k^t is absorbed into JW_{k+1}^s , after which (JW3) is applied. The third relation holds by (JW1) and Lemma 4.28. The fourth relation holds by the following calculation:

$$\begin{aligned} -\frac{[k-1]_t}{[k]_s} \left(\text{JW} \right) &= -\frac{[k-1]_t}{[k]_s} \left(\left(\text{JW} \right) + \frac{[k-2]_s}{[k-1]_t} \left(\text{JW} \right) \right) \\ &= -\frac{[k-1]_t}{[k]_s} \left(s(\alpha_t) \left(\text{JW} \right) + \partial_s(\alpha_t) \left(\text{JW} \right) - \frac{[k-2]_s}{[k-1]_t} \left(\text{JW} \right) \right) \\ &= \frac{[k-1]_t [2]_s - [k-2]_s}{[k]_s} \text{JW}_{k-1}^s = \text{JW}_{k-1}^s. \end{aligned}$$

The equality in the first line follows by applying the recursive definition of JW. The first term is simplified using the sliding relation (4.13), and the second using Lemma 4.25 after the blue dots are “pulled in”, yielding the equality in the second line. In the parentheses of the second line Lemma 4.26 kills the first term, the last term becomes $-\frac{[k-2]_s}{[k-1]_t} \text{JW}_{k-1}^s$ by Lemma 4.28, and the middle term becomes

$$\begin{aligned} \partial_s(\alpha_t) \text{JW}_{k-1}^s &= \frac{\alpha_t - s(\alpha_t)}{\alpha_s} \text{JW}_{k-1}^s = \frac{\alpha_t - (\alpha_t - \langle \alpha_t, \alpha_s^\vee \rangle \alpha_s)}{\alpha_s} \text{JW}_{k-1}^s \\ &= -[2]_s \text{JW}_{k-1}^s. \end{aligned}$$

Then using (4.16a), the coefficient of JW_{k-1}^s on the third line is 1, yielding the fourth relation. The last relation holds by Lemma 4.28 and the recursive definition for Jones–Wenzl projectors in Definition 4.20.

Suppose $k = m_{st}$. As before, we show that the morphisms

$$\begin{aligned}
 p'_{+1} &= \begin{array}{c} \cdots \\ \text{JW} \\ \cdots \\ \frac{\alpha_s}{2} \\ \text{JW} \\ \cdots \end{array} : B_s B_k^t \rightarrow B_k^t(-1) & p'_{-1} &= \begin{array}{c} \cdots \\ \text{JW} \\ \cdots \\ \text{JW} \\ \cdots \end{array} : B_s B_k^t \rightarrow B_k^t(1) \\
 i'_{+1} &= \begin{array}{c} \cdots \\ \text{JW} \\ \cdots \\ \text{JW} \\ \cdots \end{array} : B_k^t(-1) \rightarrow B_s B_k^t & i'_{-1} &= \begin{array}{c} \cdots \\ \frac{\alpha_s}{2} \\ \text{JW} \\ \cdots \\ \text{JW} \\ \cdots \end{array} : B_k^t(1) \rightarrow B_s B_k^t
 \end{aligned}$$

satisfy the relations

$$\begin{aligned}
 p'_{-1} \circ i'_{+1} &= 0, & p'_{+1} \circ i'_{-1} &= 0, & p'_{+1} \circ i'_{+1} &= \text{JW}_k^t, & p'_{-1} \circ i'_{-1} &= \text{JW}_k^t, \\
 i'_{+1} \circ p'_{+1} + i'_{-1} \circ p'_{-1} &= \text{id}_{B_s} \otimes \text{JW}_k^s.
 \end{aligned}$$

The first relation is satisfied by (JW2) and Lemma 4.26. The second relation holds similarly after pulling the s -invariant α_s^2 through the s -strand and applying Lemma 4.26. The third and fourth relation holds by pulling out $\frac{\alpha_s}{2}$ using the sliding relation 4.13 and the keyhole relation. The computation $\partial_s(\frac{\alpha_s}{2}) = 1$ gives the desired equality. Using (JW2), the last relation holds from the computation

$$\begin{aligned}
 & \begin{array}{c} \text{---} \\ \text{---} \\ \frac{\alpha_s}{2} \\ \text{---} \end{array} \text{JW} + \begin{array}{c} \frac{\alpha_s}{2} \\ \text{---} \\ \text{---} \end{array} \text{JW} \stackrel{(4.11)}{=} \begin{array}{c} \frac{\alpha_s}{2} \\ \text{---} \\ \text{---} \end{array} \text{JW} + \begin{array}{c} \frac{\alpha_s}{2} \\ \text{---} \\ \text{---} \end{array} \text{JW} \\
 & \stackrel{(4.13)}{=} \begin{array}{c} \frac{-\alpha_s}{2} \\ \text{---} \\ \text{---} \end{array} \text{JW} + \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \text{JW} + \begin{array}{c} \frac{\alpha_s}{2} \\ \text{---} \\ \text{---} \end{array} \text{JW}
 \end{aligned}$$

after which (JW1) is applied to the top and bottom JW boxes.

Finally, the decomposition for $B_s B_k^s$ is completely analogous to the previous case. $\square$

Chapter 5

Rouquier Complexes

Let $\mathbb{S}\text{Bim}$ be the associated category of Soergel bimodules for a fixed realisation $(W, \mathfrak{h})$. In this chapter, we will study a certain family of complexes within $K^b(\mathbb{S}\text{Bim})$, called Rouquier complexes. These complexes are the Soergel-theoretic analogues of the two-term functor complexes $\mathcal{F}_s$ and $\mathcal{F}_s^{-1}$ corresponding to the (co)shuffling functors introduced in Section 3.6. Rouquier [Rou06] showed that these complexes satisfy the braid relations up to homotopy, so they provide a (weak) categorification of the braid group. Because the morphisms between these complexes carry extra structure (not just the induced self-equivalences), Rouquier calls the resulting structure the *2-braid group*.

This section will be dedicated to explaining the proof that the Rouquier complexes braid using Soergel calculus. Our discussion of Rouquier complexes follows [EMTW20, §19.3]. We discuss the idea of Rouquier’s original argument, which was originally presented in terms of coherent sheaves on $\mathfrak{h}^* \times \mathfrak{h}^*$ using the geometrical realisation $\cdot$. Then we give the diagrammatic version of the proof following [AMRW17, §9.1]. We also point out a minor inaccuracy from Rouquier’s paper in Remark 5.7.

5.1 Definition of Rouquier complexes

Recall the definition of the braid group from Section 3.5. To lift the structure of the braid group, we use the monoidal structure on $K^b(\mathbb{B}\text{SBim})$. The tensor product $\otimes_R$ on $\mathbb{B}\text{SBim}$ extends to $K^b(\mathbb{B}\text{SBim})$ in the following way: if $A, B \in K^b(\mathbb{B}\text{SBim})$, then $A \otimes_R B$ has terms

$$(A \otimes_R B)^n := \bigoplus_{i+j=n} A^i \otimes_R B^j,$$

with differential

$$d(a \otimes_R b) = d(a) \otimes_R b + (-1)^{\deg a} a \otimes_R d(b).$$

In fact, the homotopy category of any monoidal category is monoidal via this definition. Here, R is the monoidal unit viewed as the complex with R in degree 0. From now on, we omit $\otimes_R$ and denote tensor product of complexes simply by concatenation.

Fix a realisation $(W, \mathfrak{h})$. Mimicking (3.5), we make the following definition.

Definition 5.1. For a simple reflection $s \in S$, let $F_s, F_{s^{-1}} \in K^b(\mathbb{S}\text{Bim})$ be the complexes

$$\begin{array}{ccccccc} F_s = \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \underline{B}_s \xrightarrow{\uparrow} R(1) \longrightarrow 0 \longrightarrow \cdots \\ & & & & \downarrow & & \\ F_{s^{-1}} = \cdots & \longrightarrow & 0 & \longrightarrow & R(-1) & \longrightarrow & \underline{B}_s \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots \end{array}$$

Then the *Rouquier complex* $F_{\underline{w}} \in K^b(\mathbb{S}\text{Bim})$ associated to a braid word $\underline{w} = (s_1^{\varepsilon_1}, \dots, s_n^{\varepsilon_n})$ is the complex

$$F_{\underline{w}} := F_{s_1^{\varepsilon_1}} \cdots F_{s_n^{\varepsilon_n}}.$$

Each term of a Rouquier complex may then be decomposed into a direct sum of Soergel bimodules. Gaussian elimination of complexes allows us to simplify complexes with contractible summands (see Appendix A). Since all Soergel bimodules have finitely many direct summands, repeated applications of Gaussian elimination on a complex in $K^b(\mathbb{S}\text{Bim})$ will terminate. We call the resulting complex *minimal* if it does not have any contractible direct summands.

For the remainder of this chapter, fix distinct simple reflections $s, t \in S$. Let us first see some examples of Rouquier complexes.

Example 5.2. Consider the complex $F_s F_s$. Then we have that

$$\begin{aligned} F_s F_s &= (\underline{B}_s \longrightarrow R(1)) \otimes (\underline{B}_s \longrightarrow R(1)) \\ &= \underline{B}_s \underline{B}_s \begin{array}{ccc} \xrightarrow{\uparrow} & B_s \otimes R(1) & \xrightarrow{\uparrow} \\ & \oplus & \\ \xrightarrow{\uparrow} & R(1) \otimes B_s & \xrightarrow{-\uparrow} \end{array} R(1) \otimes R(1). \end{aligned}$$

Write this complex as

$$\underline{B_s B_s} \xrightarrow{\begin{bmatrix} | & | \\ | & | \\ | & | \end{bmatrix}} B_s(1) \oplus B_s(1) \xrightarrow{\begin{bmatrix} | & -| \\ | & -| \end{bmatrix}} R(2)$$

after simplifying the monoidal units R . We saw in Example 4.17 that $B_s B_s \simeq B_s(-1) \oplus B_s(1)$. This gives an isomorphism of complexes

$$\begin{array}{ccccc} \underline{B_s B_s} & \longrightarrow & B_s(1) \oplus B_s(1) & \longrightarrow & R(2) \\ f \downarrow & \uparrow g & \downarrow \text{id} & & \downarrow \text{id} \\ \underline{B_s(-1) \oplus B_s(1)} & \longrightarrow & B_s(1) \oplus B_s(1) & \longrightarrow & R(2) \end{array}$$

where f and g were the explicit inclusion and projection morphisms calculated. The first differential on the bottom row is the composition of g and the first differential of the top row, which is given by

$$\begin{bmatrix} | & | \\ | & | \\ | & | \end{bmatrix} \begin{bmatrix} \boxed{\frac{\alpha_s}{2}} & \\ & \end{bmatrix} = \begin{bmatrix} \boxed{\frac{\alpha_s}{2}} & \\ & \boxed{\frac{\alpha_s}{2}} \end{bmatrix} = \begin{bmatrix} | & | \\ | & | \\ | & | \end{bmatrix}.$$

The component $B_s(1) \rightarrow B_s(1)$ appearing in the lower-right entry of the differential matrix is an isomorphism (the identity), so removing this contractible summand by Gaussian elimination yields the complex

$$\underline{B_s(-1)} \xrightarrow{\frac{1}{2} \begin{bmatrix} | & | \\ | & | \end{bmatrix}} B_s(1) \xrightarrow{\begin{bmatrix} | \\ | \end{bmatrix}} R(2),$$

which is homotopy equivalent to $F_s F_s$. This resulting complex is minimal as there are no remaining contractible direct summands. Therefore, $F_s F_s \not\simeq R$.

Example 5.3. As $B_s B_t = B_{st}$ is indecomposable, $F_s F_t$ is already minimal.

Example 5.4. Using a similar method to Example 5.2, one can show that $F_s F_{s-1} \simeq R \simeq F_{s-1} F_s$. See [AMRW17, Lemma 4.2.4] for details.

5.2 The 2-braid group

The existence of homotopy inverses and the failure of the Coxeter square relation $F_s F_s \not\simeq R$ are necessary preliminary checks, but to obtain a (weak) categorification of the braid group, one must verify the braid relations. We have the following theorem of Rouquier.

Theorem 5.5 ([Rou06, Proposition 9.2]). *Let $(W, \mathfrak{h})$ be a realisation of a Coxeter system (W, S) . For distinct simple reflections $s, t \in S$ with $m_{st} < \infty$, the complexes*

$$\underbrace{F_s F_t F_s \cdots}_{m_{st} \text{ terms}} \quad \text{and} \quad \underbrace{F_t F_s F_t \cdots}_{m_{st} \text{ terms}}$$

are homotopy equivalent.

Remark 5.6. In [Rou06], it was conjectured that the map f above is injective: if $F_{\underline{w}} \simeq F_{\underline{u}}$, can we say that w and u are the same element in B_W ? Then the quotient of B_W of isomorphism classes would then just be B_W . This was proven in all finite types by Khovanov–Seidel [KS02], Brav–Thomas [BT11], and Jensen [Jen17].

Remark 5.7. Rouquier proved Theorem 5.5 for Soergel bimodules in the geometric realisation (Example 4.3), identifying R -bimodules with quasi-coherent sheaves on $V \times V$, where V is the geometric representation. This is not the realisation best suited for lifting the braid action to category $\mathcal{O}$. Rather, one should work in the Cartan realisation (Example 4.2) coming from $\mathfrak{h}$, which is the setting compatible with translation functors on $\mathcal{O}$ and the $\mathbb{V}$ functor, to be defined in Chapter 6. Fortunately, the braid relations are independent of the chosen realisation.

Theorem 5.5 shows that for any two expressions $\underline{w}_1, \underline{w}_2$ for the element $w \in B_W$, there is an isomorphism $F_{\underline{w}_1} \simeq F_{\underline{w}_2}$ in $K^b(\text{SBim})$, but does not specify what the isomorphism is. Rouquier proved a stronger result [Rou06, Theorem 9.7] “rigidifying” $K^b(\text{SBim})$: he constructs a monoidal functor from the strict monoidal category objects are elements of B_W (only identity morphisms, tensor product given by multiplication) to the strict monoidal category of endofunctors of $K^b(\text{SBim})$. Thus, Rouquier complexes attached to the same braid element are canonically isomorphic independent of the chosen word. Concretely, any sequence of braid moves taking $\underline{w}_1$ to $\underline{w}_2$ yields a homotopy equivalence by composing the ones corresponding to the braid relations. Rouquier’s result says different such sequences produce homotopic chain maps. This rigidification forms the *2-braid group*.

5.3 Proving Rouquier complexes braid

We first make a reduction following [Rou06, Proposition 9.2] and prove Theorem 5.5 in a simpler case.

Lemma 5.8. *Theorem 5.5 follows if the braid relations hold in the case that W is finite dihedral with $S = \{s, t\}$.*

Proof. Let (W, S) be a Coxeter system and fix a realisation $(W, \mathfrak{h})$. Then decompose $\mathfrak{h}^*$ as $\mathfrak{h}^* = \mathfrak{h}_1^* \oplus \mathfrak{h}_2^*$, where $\mathfrak{h}_2^* = \mathbb{C}\alpha_s \oplus \mathbb{C}\alpha_t$ and $\mathfrak{h}_1^*$ its complement, so that $\mathfrak{h}_1^*$ is precisely fixed by $\langle s, t \rangle$. This gives an algebra decomposition $R \simeq \mathbb{C}[\mathfrak{h}_1^*] \otimes_{\mathbb{C}} \mathbb{C}[\mathfrak{h}_2^*]$ (recall that we assume R is the polynomial algebra $\mathbb{C}[\alpha_s \mid s \in S]$ from Section 4.2). Denote both tensor factors by $R = R_1 \otimes R_2$, and note that $R_1 = R^{s,t}$.

Recall that Bott–Samelson bimodules are defined as $B_s = R \otimes_{R^s} R(1)$ for any $s \in S$, tensored over R . We claim there are natural isomorphisms of functors

$$R \otimes_{R^s} - \simeq R_2 \otimes_{R_2^s} - \quad \text{and} \quad R \otimes_R - \simeq R_2 \otimes_{R_2} -,$$

and similarly with t , so that the proof reduces to the finite dihedral case. Since the left and right actions of R_1 commute across all tensor products $\otimes_{R^s}, \otimes_{R^t}$, we obtain the following natural transformation of functors on $R\text{-gmod-}R$:

$$\begin{aligned} \text{Ind}_{R_2}^R \circ (R_2 \otimes_{R_2^s} -) \circ \text{Res}_{R_2}^R &\simeq R_1 \otimes_{\mathbb{C}} (R_2 \otimes_{R_2^s} -) \simeq (R_1 \otimes_{\mathbb{C}} R_2) \otimes_{R_2^s} - \\ &\simeq R \otimes_{R^s} - \end{aligned}$$

$$\begin{aligned} \text{Ind}_{R_2}^R \circ (R_2 \otimes_{R_2} -) \circ \text{Res}_{R_2}^R &\simeq R_1 \otimes_{\mathbb{C}} (R_2 \otimes_{R_2} -) \simeq (R_1 \otimes_{\mathbb{C}} R_2) \otimes_{R_1 \otimes_{\mathbb{C}} R_2} - \\ &\simeq R \otimes_R - \end{aligned}$$

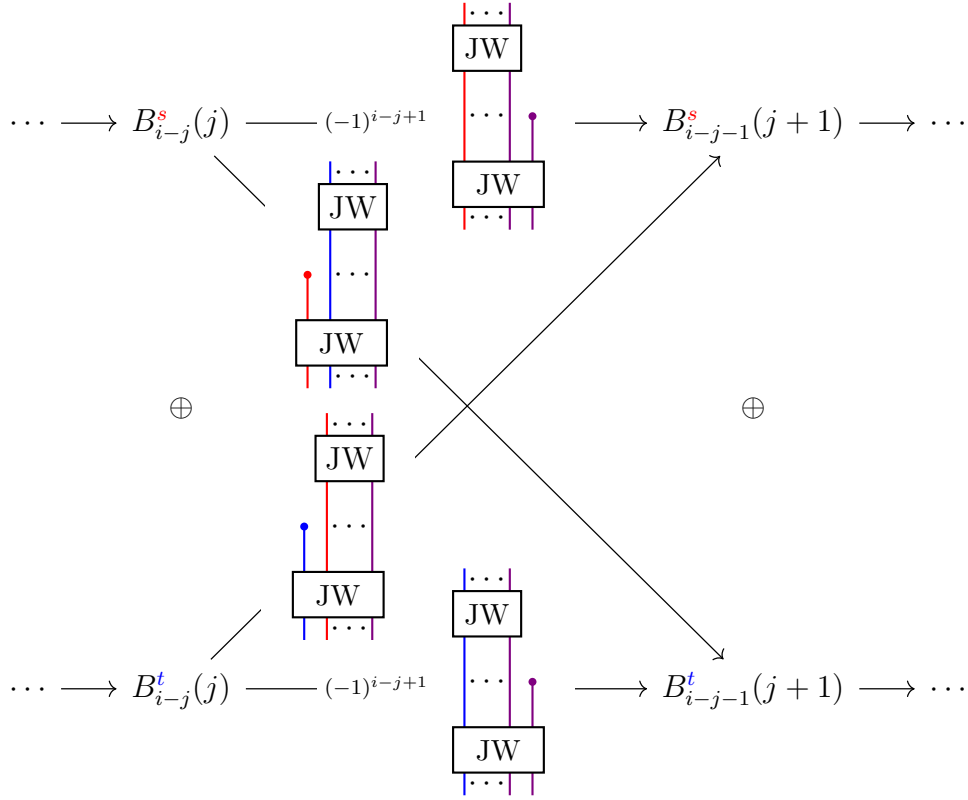
Therefore, it suffices to prove Theorem 5.5 for $W = \langle s, t \rangle$. □

Thanks to the previous lemma, let $W = \langle s, t \rangle$. To prove Theorem 5.5 in this case, we will construct a nice minimal complex that is homotopy equivalent to both $\underbrace{F_s F_t F_s \dots}_{m_{st} \text{ times}}$ and $\underbrace{F_t F_s F_t \dots}_{m_{st} \text{ times}}$.

Recall from Theorem 4.24 the notation $B_i^s := \underbrace{B_{st}^s \dots}_{i \text{ terms}}$, and similarly for B_i^t .

Define the complex

$$\begin{array}{ccccccc} & & B_{i-1}^s(1) & \xrightarrow{(-1)^i} & B_{i-2}^s(2) & \xrightarrow{(-1)^{i-1}} & \dots \xrightarrow{-1} B_1^s(i-1) \\ & \nearrow^{(-1)^{i+1}} & & \searrow & & & \searrow \\ F_{\min}^{s,i} := \underline{B_i^s} & & \oplus & & \oplus & & \oplus \\ & \searrow & & \nearrow & & & \nearrow \\ & & B_{i-1}^t(1) & \xrightarrow{(-1)^i} & B_{i-2}^t(2) & \xrightarrow{(-1)^{i-1}} & \dots \xrightarrow{-1} B_1^t(i-1) \end{array} \rightarrow R(i),$$

Figure 5.1: Differential of $F_{\min}^{s,i}$.

where the components of the differential are given by the maps in Figure 5.1. Note the negative sign on the horizontal components whenever the source is indexed by a term whose length is even. Together with Lemma 4.27, this ensures that the differentials of the complex square to 0, so $F_{\min}^{s,i}$ is indeed a well-defined complex. Furthermore, no component of the differential is an isomorphism, hence the complex is minimal. Moreover $F_{\min}^{s,i}$ is indecomposable, that is not homotopy equivalent to a non-trivial direct sum $A \oplus B$ of non-zero complexes.

Observe that the top and bottom row are symmetric in s and t . When $i = m_{st}$, the complexes $F_{\min}^{s,m_{st}}$ and $F_{\min}^{t,m_{st}}$ are isomorphic (as complexes) since $B_{m_{st}}^s \simeq B_{m_{st}}^t$. We will show using induction that there is a homotopy equivalence $\underbrace{F_s F_t F_s \dots}_{i \text{ terms}} \simeq F_{\min}^{s,i}$ (and similarly $\underbrace{F_t F_s F_t \dots}_{i \text{ terms}} \simeq F_{\min}^{t,i}$), which would then show that Rouquier complexes satisfy the braid relations when $i = m_{st}$.

To show that $F_s F_{\min}^{t,i} \simeq F_{\min}^{s,i+1}$, observe that

$$(F_s F_{\min}^{t,i})[1] = \left((B_s \rightarrow R(1)) F_{\min}^{t,i} \right)[1] \simeq \text{cone} \left(B_s F_{\min}^{t,i} \xrightarrow{\lambda} F_{\min}^{t,i}(1) \right)$$

for some non-zero map λ of complexes. Here, $[1]$ denotes cohomological shift by 1 and cone is the mapping cone for complexes. We will first show that $\underline{B_s}F_{\min}^{t,i}$ simplifies to a two-term complex via repeated applications of Gaussian elimination. This is Proposition 5.9. Next we identify the resulting cone with $F_{\min}^{s,i+1}[1]$. This is Corollary 5.10. All results hold with s and t swapped. Together, these steps complete the proof of Theorem 5.5.

Proposition 5.9 ([AMRW17, Lemma 9.1.2]). *Let $1 \leq i \leq m_{st} - 1$. Then there is a homotopy equivalence*

$$B_s F_{\min}^{t,i} \simeq (\underline{B_{i+1}^s} \rightarrow B_i^s(1)),$$

where the morphism of the resulting complex is given by the top horizontal arrow of Figure 5.1.

Proof. The complex $B_s F_{\min}^{t,i}$ is given by

$$\begin{array}{ccccccc} & & B_s B_{i-1}^t(1) & \xrightarrow{(-1)^i} & B_s B_{i-2}^t(2) & \xrightarrow{(-1)^{i-1}} \cdots \xrightarrow{-1} & B_s B_1^s(i-1) \\ & \nearrow^{(-1)^{i+1}} & & & & & \searrow \\ \underline{B_s B_i^t} & & \oplus & & \oplus & & \oplus \\ & \searrow & & & & & \nearrow \\ & & B_s B_{i-1}^s(1) & \xrightarrow{(-1)^i} & B_s B_{i-2}^s(2) & \xrightarrow{(-1)^{i-1}} \cdots \xrightarrow{-1} & B_s B_1^t(i-1) \end{array}$$

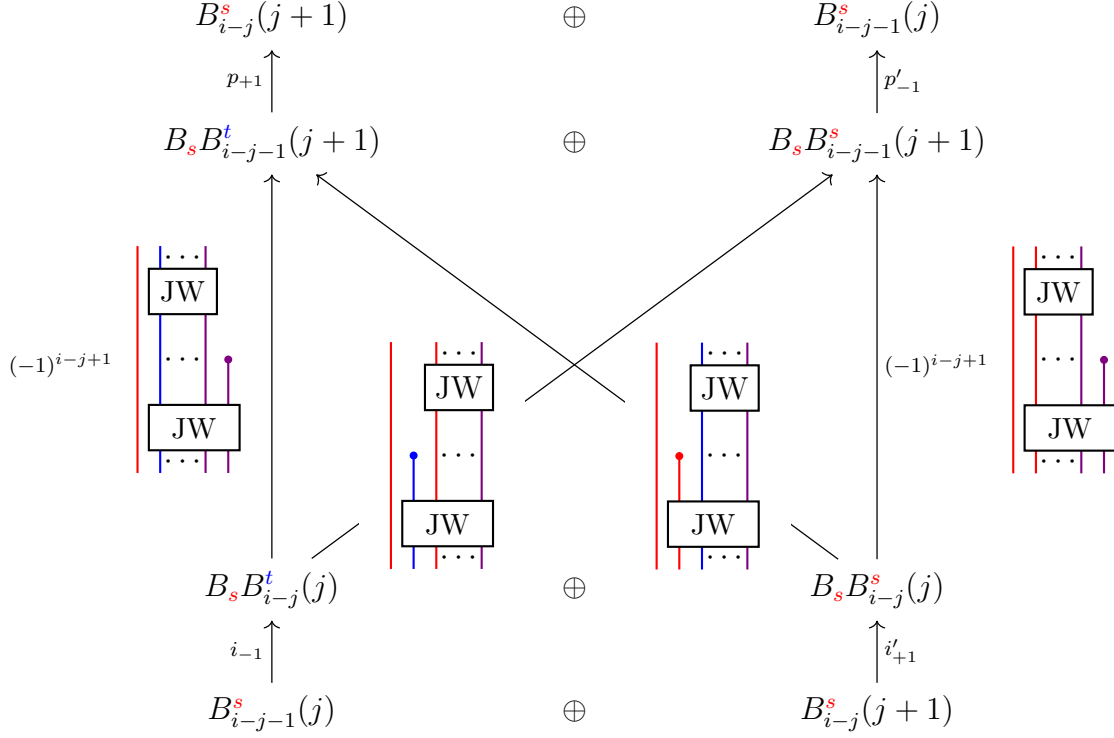
$B_s(i).$

The differentials of this complex have the same components as $F_{\min}^{t,i}$ with the s -strand id_{B_s} juxtaposed on the left. Using Lemma 4.29, we can decompose the terms as follows:

$$\begin{array}{ccccccc} & & B_i^s(1) \oplus B_{i-2}^s(1) & \xrightarrow{(-1)^i} & B_{i-1}^s(2) \oplus B_{i-3}^s(2) & \xrightarrow{(-1)^{i-1}} & \\ & \nearrow^{(-1)^{i+1}} & & & & & \\ \underline{B_{i+1}^s \oplus B_{i-1}^s} & & \oplus & & \oplus & & \\ & \searrow & & & & & \\ & & B_{i-1}^s \oplus B_{i-1}^s(2) & \xrightarrow{(-1)^i} & B_{i-2}^s(1) \oplus B_{i-2}^s(3) & \xrightarrow{(-1)^{i-1}} & \\ & & \cdots \xrightarrow{-1} & B_1^s(i-2) \oplus B_1^s(i) & & & \\ & & & & \oplus & & \\ & & \cdots \xrightarrow{-1} & B_2^s(i-1) & & & \end{array}$$

$B_s(i)$

(5.1)

Figure 5.2: Differential of decomposed $B_s F_{\min}^{t,i}$.

Recall from the proof of Lemma 4.29 that we have the morphisms

$$i_{-1} = \begin{array}{c} \text{JW} \\ \vdots \\ \text{JW} \end{array}, \quad i'_{+1} = \begin{array}{c} \text{JW} \\ \vdots \\ \text{JW} \end{array}, \quad p_{+1} = \begin{array}{c} \text{JW} \\ \vdots \\ \text{JW} \end{array}, \quad p'_{-1} = \begin{array}{c} \text{JW} \\ \vdots \\ \text{JW} \end{array},$$

which are the inclusion and projection morphisms to and from their respective indecomposables. Then the differentials of (5.1) are shown in Figure 5.2. Each differential is given by restricting to the chosen direct summand via the inclusions, apply the usual differential (Figure 5.1), and then project back, so that the resulting maps define the differential of the complex on the indecomposable summand.

We will show the following isomorphisms:

- (1) $\underline{B_{i-1}^s} \simeq B_{i-1}^s$ in degree 0 to 1;
- (2) $B_{i-j-1}^s(j) \oplus B_{i-j}^s(j+1) \simeq B_{i-j}^s(j+1) \oplus B_{i-j-1}^s(j)$ in degree j to $j+1$ for $1 \leq j \leq i-1$;

(3) $B_1^s(i-2) \simeq B_s(i)$ in degree i to $i+1$.

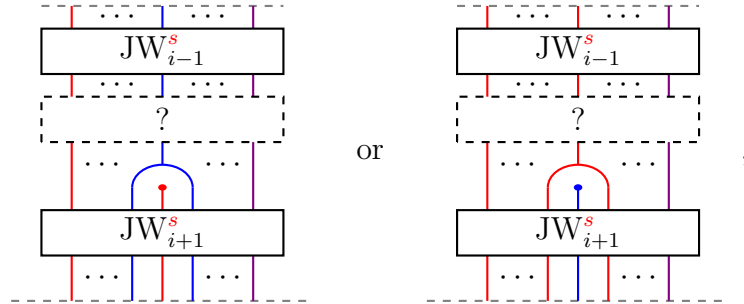
Using Gaussian elimination, first cancel the summands in (3). Then proceed downward in cohomological degree, iteratively cancelling the summands in (2) until degree 0 is reached where (1) can also be eliminated. At each stage, the cancellation only involves a contractible pair within the current degree, so the map to be reduced at the next step is unchanged. What remains is the two-term complex $\underline{B}_{i+1}^s \rightarrow B_i^s(1)$.

We now show that the morphism of the resulting complex is the original component of the degree 0 differential. After cancelling the summands in (2) and (3), we obtain

$$\underline{B}_{i+1}^s \oplus \underline{B}_{i-1}^s \longrightarrow B_i^s(1) \oplus B_{i-1}^s$$

and the component $\underline{B}_{i+1}^s \rightarrow B_i^s(1)$ is unchanged. It therefore suffices to show that no non-zero map $\underline{B}_{i+1}^s \rightarrow B_{i-1}^s$ can appear, in which case the $\underline{B}_{i+1}^s \rightarrow B_i^s(1)$ component remains unaffected by the eliminations (see Appendix A for the explicit form of the resulting morphism after eliminating).

The vanishing $\text{Hom}(B_{i+1}^s, B_{i-1}^s) = 0$ follows for degree reasons, which we will not go into full details. The idea is that any degree 0 map from B_{i+1}^s to B_{i-1}^s must be of the form



which is the zero morphism by (JW3). One can make this precise by using the *light-leaves basis* for graded hom-spaces between Soergel bimodules. We refer to the proof in [AMRW17, Lemma 9.1.2] for more details, and [EW16, §6] for a discussion on the light-leaves basis.

Let us now show that the isomorphism between summands in (2) is given by the matrix

$$\begin{bmatrix} * & \text{id}_{B_{i-j}^s(j+1)} \\ -\frac{[i-j]_s}{[i-j-1]_t} \text{id}_{B_{i-j-1}^s(j)} & 0 \end{bmatrix}$$

which is invertible, and thus the terms can be eliminated away. The starred entry $*$ is irrelevant for invertibility. The maps in (2) can be studied via the composition shown in Figure 5.2.

The composition $B_{i-j}^s(j+1) \rightarrow B_{i-j}^s(j)$ on the right vertical is zero by a similar argument to Lemma 4.26. The composition $B_{i-j}^s(j+1) \rightarrow B_{i-j}^s(j+1)$ gives the identity by Lemma 4.25. The composition $B_{i-j-1}^s(j) \rightarrow B_{i-j-1}^s(j)$ equals $-\frac{[i-j]_s}{[i-j-1]_t} \text{id}_{B_{i-j-1}^s(j)}$ by a very similar computation to $p_{-1} \circ i_{-1} = \text{id}_{B_{k-1}^s}$ in the proof of Lemma 4.29.

Similarly, components (1) and (3) can be calculated as they are special cases of (2). $\square$

Corollary 5.10 ([AMRW17, Lemma 9.1.3]). *Let $1 \leq i \leq m_{st} - 1$. Then there is a homotopy equivalence*

$$F_s F_{\min}^{t,i} \simeq F_{\min}^{s,i+1}$$

Proof. We will proceed by induction on i . The case for $i = 1$ is clear since $F_s F_{\min}^{t,1} = F_s F_t \simeq F_{\min}^{s,2}$.

Let $i > 1$. Recall the discussion preceding Proposition 5.9. Observe that

$$F_{\min}^{s,i+1}[1] \simeq \text{cone} \left(\left(\underline{B_{i+1}^s} \longrightarrow B_i^s(1) \right) \xrightarrow{\lambda} F_{\min}^{t,i}(1) \right).$$

The map λ is just the components of the differential in the zeroth and first degree of $F_{\min}^{s,i+1}$. Similarly,

$$\begin{aligned} (F_s F_{\min}^{t,i})[1] &\simeq \text{cone} (B_s F_{\min}^{t,i} \rightarrow F_{\min}^{t,i}(1)) \\ &\stackrel{\text{Proposition 5.9}}{\simeq} \text{cone} \left(\left(\underline{B_{i+1}^s} \longrightarrow B_i^s(1) \right) \xrightarrow{\lambda'} F_{\min}^{t,i}(1) \right) \end{aligned}$$

for some λ' . By induction, $F_{\min}^{t,i}$ is homotopy equivalent to a product of alternating copies of F_t and F_s . Using the fact that F_s is inverse to $F_{s^{-1}}$ by Example 5.4, $F_s F_{\min}^{t,i}$ must be indecomposable, otherwise $F_{\min}^{t,i} \simeq F_{s^{-1}} F_s F_{\min}^{t,i} \simeq F_{s^{-1}} A \oplus F_{s^{-1}} B$ which contradicts the indecomposability of $F_{\min}^{t,i}$. Therefore, $\lambda' \neq 0$. Moreover, the components of λ' have to be a non-zero multiple of the components of λ , again by degree reasons, so we get the desired isomorphism. $\square$

Chapter 6

From Category $\mathcal{O}$ to Bimodules

In this chapter, we establish the connection between Soergel bimodules and category $\mathcal{O}$. We will introduce Soergel's $\mathbb{V}$ functor [Soe90], and explain results by Soergel from the same paper. In particular, the $\mathbb{V}$ functor is fully faithful on the subcategory of projectives of the principal block $\mathcal{O}_0$, the image of which resemble a module version of Soergel bimodules. This can be summarised in the following diagram:

$$\begin{array}{ccccc}
 \mathcal{O}_0 & \xrightarrow{\mathbb{V}_0} & \text{mod-}C & \xrightarrow{\text{inflate}} & \text{mod-}R \\
 \cup & & \cup & & \cup \\
 \mathcal{O}_0\text{-Proj} & \xrightarrow{\mathbb{V}_0} & \text{Smod-}C & \xrightarrow{\text{inflate}} & \text{Smod-}R. \\
 \downarrow \Theta_s & & \downarrow (-\otimes_{C^s} C) & & \downarrow (-\otimes_{R^s} R)
 \end{array} \tag{6.1}$$

Because the $\mathbb{V}$ functor is fully faithful on projectives, the categorical braid action on $K^b(\text{SBim})$ given by the Rouquier complexes $F_s, F_{s^{-1}}$ transports along $\mathbb{V}$ to a categorical braid action on $K^b(\mathcal{O}_0\text{-Proj}) \simeq D^b(\mathcal{O}_0)$ via (co)shuffling functors, proving Theorem 3.28.

This chapter gives a detailed exposition of Rouquier's proof of the braid action (following [Rou06, §10.2]), based on [EMTW20, §15.4–15.5] and drawing on results from Soergel's original paper [Soe90].

6.1 Soergel's theorems

Recall that $R = \text{Sym}(\mathfrak{h}) \simeq \mathbb{C}[\mathfrak{h}^*]$. We first give the following definition.

Definition 6.1. Let $R_+ \subset R$ denote the subspace of homogeneous elements of positive degree, and write I_W for the ideal generated by R_+^W the W -invariants in

R_+ . The *coinvariant algebra* of R is defined as the quotient

$$C := R/I_W.$$

By construction, C inherits the grading from R and is finite-dimensional (see [Hum92, §3.5-3.6]). The first remarkable theorem of Soergel allows us to pass from the setting of category $\mathcal{O}$ to a simpler category of (right) C -modules by describing the endomorphism algebra of a certain indecomposable projective module.

Theorem 6.2 ([Soe90, Endomorphismensatz]). *Let $\mu \in \mathfrak{h}^*$ be dominant integral with stabiliser $W_\mu \subseteq W$ under the dot-action. Then there is a algebra isomorphism*

$$\mathrm{End}_{\mathcal{O}} P(w_\circ \cdot \mu) \simeq C^{W_\mu}.$$

In particular, if μ is regular, then $\mathrm{End}_{\mathcal{O}} P(w_\circ \cdot \mu) \simeq C$ by noting that $W_\mu = \{\mathrm{id}\}$ for μ regular, so $C^{W_\mu} = C$.

To prove the Endomorphismensatz, Soergel makes heavy use of translation and wall-crossing functors to compare the two algebra homomorphisms

$$q \circ \gamma: Z(\mathfrak{g}) \rightarrow C \quad \text{and} \quad Z(\mathfrak{g}) \rightarrow \mathrm{End}_{\mathfrak{g}}(P(w_\circ \cdot \mu)).$$

The homomorphism on the left is the composition of the Harish-Chandra homomorphism γ (Definition 2.31) with the quotient $q: R \rightarrow C$. The homomorphism on the right is given by the natural action of $Z(\mathfrak{g})$ on $P(w_\circ \cdot \mu)$. He shows that both maps are surjective with the same kernel, which shows $\mathrm{End}_{\mathcal{O}} P(w_\circ \cdot \mu) \simeq C$.

Remark 6.3. The proof of the Endomorphismensatz was simplified considerably through Bernstein’s “trace formula” for $Z(\mathfrak{g})$ [Ber90]. Ginzburg [Gin91] gave a geometric proof using perverse sheaves, which leaves almost all of the representation theory aside. In [BG99], Beilinson and Ginzburg gave a proof via the Beilinson–Bernstein localisation correspondence between $\mathfrak{g}$ -modules and $\mathcal{D}$ -modules, where the geometric analogues of translation and wall-crossing functors behave in a greater generality.

Observe that $P(w_\circ \cdot \mu)$ is a $(\mathfrak{g}, C^{W_\mu})$ -bimodule: it is a left $\mathfrak{g}$ -module in the natural way and a right module over its endomorphism algebra. Using this bimodule structure, Soergel defined the following functor.

Definition 6.4. Let $\mathbb{V}_\mu: \mathcal{O}_\mu \rightarrow \mathrm{mod}\text{-}\mathrm{End}_{\mathcal{O}_\mu}(P(w_\circ \cdot \mu))$ be the functor defined by

$$\mathbb{V}_\mu := \mathrm{Hom}_{\mathfrak{g}}(P(w_\circ \cdot \mu), -).$$

We will refer to this as the $\mathbb{V}$ functor when μ is clear. This functor is well-defined since for any $M \in \mathcal{O}_\mu$, $\text{Hom}_{\mathcal{O}_\mu}(P(w_\circ \cdot \mu), -)$ is a right $\text{End}_{\mathcal{O}_\mu}(P(w_\circ \cdot \mu))$ -module under the action of precomposition. By definition of $P(w_\circ \cdot \mu)$ being projective, $\mathbb{V}_\mu$ is also an exact functor.

At first glance, the $\mathbb{V}$ functor seems like a promising tool that allows us to pass from the “difficult” $\mathcal{O}_\mu$ to the “easier” finite-dimensional coinvariant algebra. However, one quickly sees that a large amount of information is lost. On simple objects $L(w \cdot \mu)$,

$$\dim_{\mathbb{C}} \mathbb{V}_\mu(L(w \cdot \mu)) = \dim_{\mathbb{C}} \text{Hom}(P(w_\circ \cdot \mu), L(w \cdot \mu)) = \begin{cases} 1 & \text{if } w = w_\circ \\ 0 & \text{else,} \end{cases}$$

since by definition, $P(w_\circ \cdot \mu)$ is the projective cover of $L(w_\circ \cdot \mu)$, so it admits no non-zero morphisms to other simple modules (recall from the proof of Lemma 2.44). For any $M \in \mathcal{O}_\mu$, taking a composition series and using exactness of $\mathbb{V}_\mu$ implies

$$\dim_{\mathbb{C}} \mathbb{V}_\mu(M) = \sum_{w \in W} [M : L(w \cdot \mu)] \dim_{\mathbb{C}} \mathbb{V}_\mu(L(w \cdot \mu)) = [M : L(w_\circ \cdot \mu)], \quad (6.2)$$

where the sum runs over W , counted with multiplicity. Consequently, for Verma modules $\Delta(w \cdot \mu)$, Theorem 2.14 and the formula above gives

$$\dim_{\mathbb{C}} \mathbb{V}_\mu(\Delta(w \cdot \mu)) = 1 \quad \text{for all } w \in W. \quad (6.3)$$

There is one family of objects on which the $\mathbb{V}$ functor behaves well.

Theorem 6.5 ([Soe90, Struktursatz]). *The $\mathbb{V}$ functor is fully faithful on the subcategory $\mathcal{O}_\mu\text{-Proj}$ of projective modules in $\mathcal{O}_\mu$, that is the map induced by the functor is an isomorphism*

$$\text{Hom}_{\mathfrak{g}}(P, Q) \cong \text{Hom}_{\mathbb{C}^{W_\mu}}(\mathbb{V}_\mu(P), \mathbb{V}_\mu(Q))$$

for $P, Q \in \mathcal{O}_\mu\text{-Proj}$.

Remark 6.6. The $\mathbb{V}$ functor is in fact quite natural from the perspective of Morita theory (see [Bas68, Chapter II] for more details). Indeed, one has an equivalence $\mathcal{O}_\mu \simeq \text{mod-End}_{\mathcal{O}_\mu}(P)$ for $P := \bigoplus_{w \in W} P(w \cdot \mu)$, realised by the functor $\text{Hom}_{\mathcal{O}_\mu}(P, -)$. What is remarkable is that even if one restricts attention to the single projective cover $P(w_\circ \cdot \mu)$, the resulting $\mathbb{V}$ functor still encodes a large amount of information seen from the Endomorphismensatz and the Struktursatz. In this sense, $P(w_\circ \cdot \mu)$ deserves the name *big projective module*. It is the largest and most informative of the indecomposable projectives, despite the fact that the $\mathbb{V}$ functor loses lots of information in the sense discussed above.

Whilst the proof of the Endomorphismensatz and the Struktursatz are very technical, they can be verified directly for $\mathfrak{g} = \mathfrak{sl}_2$.

Example 6.7. Let $\mathfrak{g} = \mathfrak{sl}_2$. Here, $P(w_\circ \cdot 0) = P(-2)$. To check the Endomorphismensatz, notice that $\dim \text{End } P(-2) = 2$. This is because $L(-2) \simeq \Delta(-2)$ appears in both the top and socle (see Example 2.45). From this, the algebra $\text{End } P(-2)$ is local of dimension two. Therefore, it is isomorphic to $\mathbb{C}[x]/(x^2)$, which is the coinvariant algebra of $W \simeq S_2$.

Next, we check the Struktursatz, that is $\mathbb{V}_0$ is fully faithful on projectives. Let $\varphi: M \rightarrow N$ be a morphism in $\mathcal{O}_0$, which factors through the maps $\pi: M \rightarrow \text{im } \varphi$ and $\iota: \text{im } \varphi \rightarrow N$. If $\mathbb{V}_0(\varphi) = 0$, then $\mathbb{V}_0(\varphi) = \mathbb{V}_0(\iota) \circ \mathbb{V}_0(\pi) = 0$. Now π is surjective so $\mathbb{V}_0(\pi)$ is also surjective by exactness of $\mathbb{V}_0$, forcing $\mathbb{V}_0(\iota) = 0$. Recall that every proper submodule of $\Delta(0)$ or $\Delta(-2)$ contains $L(-2)$, and $\mathbb{V}_0(L(-2)) = \text{Hom}(P(-2), L(-2)) \neq 0$. Therefore, any non-zero submodule of a module filtered by $\Delta(0)$ and $\Delta(-2)$ has a non-zero image under $\mathbb{V}_0$. Since the indecomposable projectives are filtered by these Vermas, $\text{im } \varphi \neq 0$ means that $\mathbb{V}_0(\varphi) \neq 0$. Hence $\mathbb{V}_0$ is faithful on projectives. Noting that $\dim \text{Hom}(P(-2), P(0)) = 1$ which corresponds to sending the top $L(-2)$ of $P(-2)$ to the socle $L(-2)$ of $P(0)$, a straightforward check shows that the dimension of the hom-sets of $P(-2)$ and $P(0)$ under $\mathbb{V}_0$ are equal. Indeed $\dim \mathbb{V}_0(P(0)) = \dim \mathbb{V}_0(\Delta(0)) = 1$ by (6.3). This shows that $\mathbb{V}_0$ is fully faithful on the projective objects of $\mathcal{O}_0$.

Let us restrict to the principal block $\mathcal{O}_0$. The functor $\mathbb{V}_0$ is an exact functor from $\mathcal{O}_0$ into graded C -modules. Wall-crossing functors Θ_s act on $\mathcal{O}$ and preserve projectives, so it is natural to ask how $\mathbb{V}_0$ interacts with Θ_s . The theorem below answers this by identifying the effect of Θ_s under $\mathbb{V}_0$ with a simple algebraic operation on C -modules.

Theorem 6.8 ([Soe90, Theorem 10, Korollar 1]). *For every $s \in S$, there is a natural isomorphism*

$$\mathbb{V}_0 \circ \Theta_s \simeq (- \otimes_{C^s} C) \circ \mathbb{V}_0.$$

This statement says that $\mathbb{V}_0$ intertwines the action of applying the wall-crossing functor Θ_s with tensoring on the right with $- \otimes_{C^s} C$. Equivalently, we can write the tensor action as $- \otimes_C (C \otimes_{C^s} C)$, which resembles the familiar definition of the Bott–Samelson bimodule B_s defined in Section 4.2 but over C instead of R . We will reconcile this in the next section.

Proof of Theorem 6.8. Let $\mu \in \Lambda$ be such that μ is on the s -wall in the closure of the fundamental chamber, and consider the translation functors $T_{\text{on}}: \mathcal{O}_0 \rightarrow \mathcal{O}_\mu$

and $T_{\text{off}}: \mathcal{O}_\mu \rightarrow \mathcal{O}_0$. By Theorem 3.14, $T_{\text{off}}P(w_\circ \cdot \mu) \simeq P(w_\circ \cdot 0)$. This gives us a commutative diagram

$$\begin{array}{ccc} C^{W_\mu} & \xrightarrow{\quad} & C \\ \downarrow \simeq & & \downarrow \simeq \\ \text{End}_{\mathcal{O}_\mu} P(w_\circ \cdot \mu) & \longrightarrow & \text{End}_{\mathcal{O}_0} P(w_\circ \cdot 0), \end{array}$$

where the vertical isomorphisms come from Theorem 6.2, and the bottom horizontal arrow is the map induced by the naturality of the diagram

$$\begin{array}{ccc} P(w_\circ \cdot \mu) & \xrightarrow{\psi} & P(w_\circ \cdot \mu) \\ \downarrow T_{\text{off}} & & \downarrow T_{\text{off}} \\ P(w_\circ \cdot 0) & \xrightarrow{T_{\text{off}}(\psi)} & P(w_\circ \cdot 0). \end{array} \quad (6.4)$$

As restriction of scalars is a contravariant functor, we get the following diagram:

$$\begin{array}{ccc} \text{mod-}C & \xrightarrow{\text{Res}_{C^{W_\mu}}^C} & \text{mod-}C^{W_\mu} \\ \downarrow \simeq & & \downarrow \simeq \\ \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot 0) & \xrightarrow{T_{\text{off}}^*} & \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot \mu). \end{array} \quad (6.5)$$

Here, the functor T_{off}^* is defined as follows. If $M \in \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot 0)$, then T_{off}^*M is a right $\text{End}_{\mathcal{O}_0} P(w_\circ \cdot \mu)$ -module, where $\psi \in \text{End}_{\mathcal{O}_0} P(w_\circ \cdot \mu)$ acts by $T_{\text{off}}^*m \cdot \psi := m \cdot T_{\text{off}}(\psi)$ for $m \in M$.

Next, we claim that the diagram

$$\begin{array}{ccc} \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot 0) & \xrightarrow{T_{\text{off}}^*} & \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot \mu) \\ \mathbb{V}_0 \uparrow & & \mathbb{V}_\mu \uparrow \\ \mathcal{O}_0 & \xrightarrow{T_{\text{on}}} & \mathcal{O}_\mu \end{array} \quad (6.6)$$

commutes. Indeed,

$$T_{\text{off}}^* \circ \mathbb{V}_0 \simeq \text{Hom}_{\mathcal{O}_0}(T_{\text{off}}P(w_\circ \cdot \mu), -) \simeq \text{Hom}_{\mathcal{O}_\mu}(P(w_\circ \cdot \mu), T_{\text{on}}(-)) = \mathbb{V}_\mu \circ T_{\text{on}}.$$

Dual to the diagrams (6.5) and (6.6), we have the commutative diagram

$$\begin{array}{ccc} \text{mod-}C^{W_\mu} & \xrightarrow{\text{Ind}_{C^{W_\mu}}^C} & \text{mod-}C \\ \downarrow \psi & & \downarrow \varphi \\ \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot \mu) & \xrightarrow{T_{\text{on}}^*} & \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot 0) \\ \mathbb{V}_\mu \uparrow & & \mathbb{V}_0 \uparrow \\ \mathcal{O}_\mu & \xrightarrow{T_{\text{off}}} & \mathcal{O}_0. \end{array} \quad (6.7)$$

where ψ and φ are isomorphisms. The top square commutes since $\text{Ind}_{C^{W_\mu}}^C$ is left-adjoint to $\text{Res}_{C^{W_\mu}}^C$ (see Section 1.4) and T_{on} is left-adjoint to T_{off} , so

$$\begin{aligned} \text{Hom}(\text{Ind}_{C^{W_\mu}}^C \psi^{-1} M, N) &\simeq \text{Hom}(M, \psi \text{Res}_{C^{W_\mu}}^C N) \\ &\simeq \text{Hom}(M, T_{\text{off}}^* \varphi N) \\ &\simeq \text{Hom}(\varphi^{-1} T_{\text{on}}^* M, N) \end{aligned}$$

so $\varphi \circ \text{Ind}_{C^{W_\mu}}^C = T_{\text{on}}^* \circ \psi$. The bottom square can be shown to commute similarly to (6.6).

Combining the four squares, we have the following commutative diagram:

$$\begin{array}{ccccc} \text{mod-}C & \xrightarrow{\text{Res}_{C^{W_\mu}}^C} & \text{mod-}C^{W_\mu} & \xrightarrow{\text{Ind}_{C^{W_\mu}}^C} & \text{mod-}C \\ \downarrow \simeq & & \downarrow \simeq & & \downarrow \simeq \\ \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot 0) & \xrightarrow{T_{\text{off}}^*} & \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot \mu) & \xrightarrow{T_{\text{on}}^*} & \text{mod-End}_{\mathcal{O}_0} P(w_\circ \cdot 0) \\ \mathbb{V}_0 \uparrow & & \mathbb{V}_\mu \uparrow & & \mathbb{V}_0 \uparrow \\ \mathcal{O}_0 & \xrightarrow{T_{\text{on}}} & \mathcal{O}_\mu & \xrightarrow{T_{\text{off}}} & \mathcal{O}_0. \end{array}$$

In particular,

$$\mathbb{V}_0 \circ \Theta_s = \mathbb{V}_0 \circ T_{\text{off}} \circ T_{\text{on}} \simeq \text{Ind}_{C^{W_\mu}}^C \circ \text{Res}_{C^{W_\mu}}^C \circ \mathbb{V}_0 = (- \otimes_{C^s} C) \circ \mathbb{V}_0$$

proving the claim. $\square$

6.2 Soergel modules

The $\mathbb{V}$ functor allows us to bridge between the setting of category $\mathcal{O}$ and that of Soergel bimodules and Rouquier complexes. To do so, we fix the realisation $(W, \mathfrak{h}^*)$ coming from the Cartan subalgebra and the root system, and consider Soergel (bi)modules over $R = \text{Sym}(\mathfrak{h})$. In the following, we view $\mathbb{C}$ as an R -module via $\mathbb{C} \simeq R/R_+$.

Definition 6.9. For an expression $\underline{w} = (s_1, \dots, s_n)$, the *Bott–Samelson module*, denoted by $\overline{\text{BS}}(\underline{w})$, is the graded right R -module

$$\overline{\text{BS}}(\underline{w}) := \mathbb{C} \otimes_R \text{BS}(\underline{w}).$$

A *Soergel module* is a graded right R -module isomorphic to a direct summand of $\mathbb{C} \otimes_R B$, where B is a Soergel bimodule. Denote the category of Soergel modules over R as $\text{Smod-}R$.

Analogous to the classification of bimodules from Theorem 4.15, we have the following theorem.

Theorem 6.10 ([EMTW20, Corollary 15.28]). *Let B be a Soergel bimodule. Then $\mathbb{C} \otimes_R B$ is an indecomposable Soergel module if and only if B is indecomposable. Therefore, the elements of $\{\mathbb{C} \otimes_R B_w \mid w \in W\}$ are all indecomposable Soergel modules up to shift and isomorphism.*

Since the coinvariant algebra arises in both the setting of category $\mathcal{O}$ via the Harish-Chandra isomorphism and as a quotient of R , passing from C -modules to Soergel modules is straightforward.

Proposition 6.11. *Bott–Samelson and Soergel modules can be defined using the coinvariant algebra, that is, we have a natural identification*

$$\overline{\text{BS}}(s_1, \dots, s_m) \simeq \mathbb{C} \otimes_{C^{s_1}} C \otimes_{C^{s_2}} \cdots \otimes_{C^{s_m}} C(m). \quad (6.8)$$

Proof. Any graded right C -module M is a graded right R -module by extension of scalars, that is, R_+^W acts by zero on the right. Recall that $C = R/I_W$ where I_W is the ideal generated by R_+^W which are all positive-degree W -invariant polynomials. Then I_W is closed under the action of W . In particular, the quotient map $R \rightarrow C$ is s -equivariant for every $s \in S$, and so restricts to a map $R^s \rightarrow C^s$. For any $M \in \text{gmod-}C$, there is a natural map

$$M \otimes_{R^s} R(1) \rightarrow M \otimes_{C^s} C(1). \quad (6.9)$$

The decomposition $R \cong R^s \oplus \alpha_s R^s$ from Lemma 4.5 of R^s -modules thus descends to a decomposition $C \simeq C^s \oplus \alpha_s C^s$ as C^s -modules. Since α_s has degree 2, we have an isomorphism

$$\begin{aligned} M \otimes_{R^s} R(1) &\simeq M \otimes_{R^s} (R^s(1) \oplus R^s(-1)) \\ &\simeq M(1) \oplus M(-1), \end{aligned}$$

and similarly for $M \otimes_{C^s} C(1)$. Therefore (6.9) is an isomorphism.

Recall that we view $\mathbb{C}$ as an R -module via $\mathbb{C} \simeq R/R_+$. This means that on any Bott–Samelson module, elements of R_+^W act by zero on the left. Furthermore, any element of R_+^W is in R^s , and so passes through all tensor products $\otimes_{R^s}$ for all $s \in S$, and so also acts by zero on the right. As direct summands of Bott–Samelson modules, Soergel modules also behave the same. This proves the proposition. $\square$

Let $\text{Smod-}C$ be the right Soergel modules over C from (6.8). There is an inflation functor $\text{mod-}C \rightarrow \text{mod-}R$ along $R \twoheadrightarrow C$. The motivation is that Soergel modules are precisely the images of projectives in $\mathcal{O}_0$ under the functor $\mathbb{V}_0$. The action on $\text{Smod-}R$ of tensoring with B_s on the right corresponds to the action of the wall-crossing functor Θ_s . We make this precise in the next theorem, thereby explaining (6.1). Recall from Section 3.3 that for an expression $\underline{w} = (s_{i_1}, s_{i_2}, \dots, s_{i_k})$, we set $P_{\underline{w}} := \Theta_{s_{i_k}} \cdots \Theta_{s_{i_2}} \Theta_{s_{i_1}} P(0)$. There is a forgetful functor $\text{For}: \text{gmod-}R \rightarrow \text{mod-}R$ which forgets the grading.

Theorem 6.12 ([EMTW20, Corollary 15.37]). *The $\mathbb{V}$ functor is an equivalence of $\mathbb{C}$ -linear additive categories*

$$\mathcal{O}_0\text{-Proj} \simeq \text{For}(\text{Smod-}R).$$

For every $s \in S$, we have $\mathbb{V}_0 \circ \Theta_s \simeq (- \otimes_R B_s) \circ \mathbb{V}_0$. Moreover,

$$\mathbb{V}_0(P_{\underline{w}}) \simeq \mathbb{C} \otimes_R \text{For}(\text{BS}(\underline{w})) \quad \text{and} \quad \mathbb{V}_0(P(w \cdot 0)) \simeq \mathbb{C} \otimes_R B_w$$

for any expression $\underline{w}$ of any $w \in W$.

Proof. To show $\mathbb{V}_0$ is an equivalence of categories, we need to show that $\mathbb{V}_0$ is fully faithful and essentially surjective. The Struktursatz gives the first condition. To show that $\mathbb{V}_0$ is essential surjective, it suffices to show that $\mathbb{V}_0(P(w \cdot 0)) \simeq \mathbb{C} \otimes_R B_w$, after which Theorem 6.10 applies.

Choose a reduced expression $\underline{w} = (s_{i_1}, \dots, s_{i_k})$ for $w \in W$. Firstly, the action of Θ_s and $(- \otimes_R B_s)$ are intertwined by $\mathbb{V}_0$ as a result of Theorem 6.8 and Proposition 6.11. Since $\Delta(0) \simeq P(0)$ by Theorem 2.37, we have that $\mathbb{V}_0(P(0)) \simeq \mathbb{V}_0(\Delta(0)) \simeq \mathbb{C}$ by (6.3). By repeated applications of Theorem 6.8, we have that

$$\begin{aligned} \mathbb{V}_0(P_{\underline{w}}) &= \mathbb{V}_0 \circ \Theta_{s_{i_k}} \circ \cdots \circ \Theta_{s_{i_1}} P(0) \\ &\simeq \mathbb{V}_0(P(0)) \otimes_{C^{s_1}} C \otimes_{C^{s_{i_2}}} \cdots \otimes_{C^{s_{i_k}}} C \\ &\simeq \mathbb{C} \otimes_{C^{s_1}} C \otimes_{C^{s_{i_2}}} \cdots \otimes_{C^{s_{i_k}}} C \\ &\simeq \mathbb{C} \otimes_R \text{BS}(\underline{w}), \end{aligned}$$

where the last isomorphism comes from Proposition 6.11.

As $P(w \cdot 0) \subset P_{\underline{w}}$ by Proposition 3.21, we have that $\mathbb{V}_0(P(w \cdot 0)) \subset \mathbb{C} \otimes_R \text{BS}(\underline{w})$. By Theorem 6.10, direct summands of $\mathbb{C} \otimes_R \text{BS}(\underline{w})$ are of the form $\mathbb{C} \otimes_R B$ where B is a direct summand of $\text{BS}(\underline{w})$. Now $P(w \cdot 0)$ is the indecomposable summand of $P_{\underline{w}}$ that is not a summand of $P_{\underline{w}'}$ for any strict subexpression w' of w by the discussion after Proposition 3.21. Therefore, the same is true for $\mathbb{V}_0(P(w \cdot 0))$ as $\mathbb{V}_0$ is fully faithful on projectives, so $\mathbb{V}_0(P(w \cdot 0)) \simeq \mathbb{C} \otimes_R B_w$. $\square$

Remark 6.13. Note that $\mathbb{V}_0$ is an equivalence between $\mathcal{O}_0\text{-Proj}$ and *ungraded* Soergel modules. One can lift objects in $\mathcal{O}_0$ to a graded version by assigning to it its corresponding grading in C . Continuing the discussion from Remark 6.6, we can induce the grading from $\text{End}_{\mathcal{O}_0} P(w_\circ \cdot 0) \simeq C$ to a $\mathbb{Z}$ -grading on $\text{End}_{\mathcal{O}_0}(P)$ (see [Str03, Theorem 2.1]). This allows us to define the graded category $\mathcal{O}_0^{\mathbb{Z}}$ as the category of graded $\text{End}_{\mathcal{O}_0}(P)$ -modules. Another way to realise a graded lift of $\mathcal{O}$ is to define a grading on $K^b(\mathbb{S}\text{mod-}R)$ and use the equivalence of the $\mathbb{V}$ functor. The *Soergel-theoretic graded category* $\mathcal{O}_0$ is defined as the heart of a “perverse” t-structure on $K^b(\mathbb{S}\text{mod-}R)$ (see [Mak19]). We will not be concerned with the graded equivalence as it is not within the scope of this thesis.

6.3 Braid group action

We are now finally able to reap the rewards of all the technical machinery that we have sown.

Pass to the bounded homotopy categories $K^b(\mathcal{O}_0)$ and $K^b(C)$. A functor on the abelian level acts on chain complexes by applying it to each term, so it extends canonically to the homotopy level. In particular the $\mathbb{V}$ functor extends to a functor $K^b(\mathcal{O}_0) \rightarrow K^b(C)$. Likewise each wall-crossing functor Θ_s preserves $\mathcal{O}_0\text{-Proj}$ and therefore induces an endofunctor of $K^b(\mathcal{O}_0\text{-Proj})$.

Recall from Lemma 3.29 that $\mathcal{F}_s$ and $\mathcal{F}_s^{-1}$ corresponds to $\mathcal{R}\text{CSh}_s$ and $\mathcal{L}\text{Sh}_s$. In the homotopy category, Theorem 6.12 then shows that

$$\mathbb{V}_0 \circ \text{CSh}_s \simeq \mathbb{V}_0 \circ (\underline{\Theta}_s \rightarrow \mathbb{1}) \simeq (\underline{\text{For}}(B_s) \rightarrow R) \circ \mathbb{V}_0 = \text{For}(F_s) \circ \mathbb{V}_0,$$

and similarly

$$\mathbb{V}_0 \circ \text{Sh}_s \simeq \mathbb{V}_0 \circ (\mathbb{1} \rightarrow \underline{\Theta}_s) \simeq (R \rightarrow \underline{\text{For}}(B_s)) \circ \mathbb{V}_0 = \text{For}(F_s^{-1}) \circ \mathbb{V}_0.$$

Finally, via Theorem 6.5, $\mathbb{V}_0$ is fully faithful on $\mathcal{O}_0\text{-Proj}$. Therefore the natural isomorphisms coming from the categorical braid action that ensure the braid relations for the Rouquier complexes F_s and F_s^{-1} lift via $\mathbb{V}_0$ to the natural isomorphisms for CSh_s and Sh_s as endofunctors of $K^b(\mathcal{O}_0\text{-Proj})$. By Theorem 2.47, $K^b(\mathcal{O}_0\text{-Proj}) \simeq D^b(\mathcal{O}_0)$. Hence, there is a categorical braid group action on $D^b(\mathcal{O}_0)$.

Appendix A

Gaussian Elimination for Complexes

Lemma A.1 ([BN07, Lemma 4.2]). *Let $\mathcal{A}$ be an additive category. Suppose that $C \in C(\mathcal{A})$ has the form*

$$\dots \longrightarrow C^{i-1} \xrightarrow{\begin{pmatrix} a \\ b \end{pmatrix}} \tilde{C}^i \oplus X \xrightarrow{\begin{pmatrix} c & d \\ e & \varphi \end{pmatrix}} \tilde{C}^{i+1} \oplus Y \xrightarrow{\begin{pmatrix} f & g \end{pmatrix}} C^{i+2} \longrightarrow \dots$$

where $\varphi: X \rightarrow Y$ is an isomorphism. Then the inclusion $\tilde{C} \hookrightarrow C$ is an isomorphism in $K(\mathcal{A})$, where $\tilde{C}$ is the complex

$$\dots \longrightarrow C^{i-1} \xrightarrow{a} \tilde{C}^i \xrightarrow{c-d\varphi^{-1}e} \tilde{C}^{i+1} \xrightarrow{f} C^{i+2} \longrightarrow \dots$$

Proof. First, we show that C is isomorphic to C' , the complex

$$\dots C^{i-1} \xrightarrow{\begin{pmatrix} a \\ 0 \end{pmatrix}} \tilde{C}^i \oplus X \xrightarrow{\begin{pmatrix} c-d\varphi^{-1}e & 0 \\ 0 & \varphi \end{pmatrix}} \tilde{C}^{i+1} \oplus Y \xrightarrow{\begin{pmatrix} f & 0 \end{pmatrix}} C^{i+2} \dots$$

We see that the middle differential of C' differ by an invertible column and row operation. The “change of basis” matrix at $\tilde{C}^i \oplus X$ from C to C' is given by

$$\begin{pmatrix} 1 & 0 \\ \varphi^{-1}e & 1 \end{pmatrix} \text{ with inverse } \begin{pmatrix} 1 & 0 \\ -\varphi^{-1}e & 1 \end{pmatrix}.$$

The change of basis matrix at $\tilde{C}^{i+1} \oplus Y$ from C to C' is

$$\begin{pmatrix} 1 & -d\varphi^{-1} \\ 0 & 1 \end{pmatrix} \text{ with inverse } \begin{pmatrix} 1 & d\varphi^{-1} \\ 0 & 1 \end{pmatrix}.$$

Therefore, the middle differential of C under base change becomes

$$\begin{pmatrix} 1 & -d\varphi^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c & d \\ e & \varphi \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\varphi^{-1}e & 1 \end{pmatrix} = \begin{pmatrix} c - d\varphi^{-1}e & 0 \\ 0 & \varphi \end{pmatrix}.$$

Apart from the change of basis maps for the middle two terms, all other terms are related by the identity map, so the side differentials become

$$\begin{pmatrix} 1 & 0 \\ \varphi^{-1}e & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ \varphi^{-1}ea + b \end{pmatrix}.$$

Now $ea + \varphi b = 0$ as the differential squares to zero in C . This means that $\varphi^{-1}ea + b = 0$, so the resulting differential is just $\begin{pmatrix} a \\ 0 \end{pmatrix}$. Similarly, the last differential becomes

$$\begin{pmatrix} f & g \end{pmatrix} \begin{pmatrix} 1 & -d\varphi^{-1} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} f & -fd\varphi^{-1} + g \end{pmatrix} = \begin{pmatrix} f & 0 \end{pmatrix}.$$

Thus $C \cong C'$ in $C(\mathcal{A})$.

Lastly, the complex

$$\cdots \longrightarrow 0 \longrightarrow X \xrightarrow{\varphi} Y \longrightarrow 0 \longrightarrow \cdots$$

is contractible as φ is an isomorphism. Hence $C \cong C' \simeq \tilde{C}$ in $C(\mathcal{A})$, so $\tilde{C} \cong C$ in $K(\mathcal{A})$. $\square$

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